

Capital Flows and Foreign Exchange Intervention

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Abstract

I consider a small open economy model where international financial markets are imperfect and the exchange rate is determined by capital flows. I use this framework to study the effects of portfolio flow shocks, derive the optimal foreign exchange intervention policy, and characterize its interaction with monetary policy. I derive the optimal intervention rule in closed form as a function of three implicit targets. Finally, using Swiss data I provide evidence of the inefficiencies generated by capital flow shocks and quantify them by estimating the financial market friction that determines deviations from uncovered interest rate parity.

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The volatility of international capital flows has increased substantially since the global financial crisis of 2008-2010. To lean against the wind of capital inflows, many policymakers around the world have relied on heterodox policies such as non-conventional monetary policies, capital controls and foreign exchange interventions. According to the IMF¹, the last decade has seen an increasing number of countries actively managing their exchange rates. Brazil, Chile, Colombia, Turkey, and other emerging markets with announced inflation targeting regimes have increased both the frequency and the size of their operations in the foreign exchange market.² Along them, also developed economies such as Israel and Switzerland have engaged in considerable intervention in the foreign exchange market and have accumulated substantial reserves.³

Exchange rate interventions are typically attributed to precautionary or mercantilist motives. Accumulating foreign reserves might provide a cushion of liquidity if the flow of short-term debt is interrupted. Alternatively, currency interventions can be used to undervalue the domestic currency and boost the competitiveness of the export sector. Recent evidence (see for example Adler and Tovar Mora (2011) and Daude et al. (2014), among others) documents the prevalence of an alternative reason for intervening: that of stabilizing the exchange rate and reducing its volatility. The decision to intervene is increasingly driven by the goal of limiting what the policymaker perceives as unwarranted deviations of the exchange rates from their fundamental levels. Interventions correlate negatively with exchange rate pressures, that is they “lean against the wind”, and positively with foreign financial conditions and capital flows. Central bankers appear to be particularly worried about the negative effects of a strong currency on the competitiveness of their exports and ultimately on domestic output. Hence their attempt to fight non-fundamental appreciating pressures.⁴

However popular, the logic behind such interventions has not been formally tested. The goal of this paper is to characterize the optimal use of sterilized foreign exchange intervention in response to exchange rate fluctuations driven by capital flows. I consider a New Keynesian model of a small open economy, as originally formulated by Gali and Monacelli (2005) and later extended to continuous time by Farhi and Werning (2012), in which I depart from the assumption of frictionless financial markets and assume that international financial markets are imperfect. Following the model developed by Gabaix and Maggiori (2015), I assume that agents in each country have limited access to international capital market. International financiers absorb any imbalance between demand and supply of assets denominated in different currencies and clear the market. To do so, however, they require premium and the currency expected return must adjust accordingly. I use this model to study the effects of shocks to foreign investors demand for domestic assets. When international asset markets are imperfect, portfolio flow shocks generate boom-bust cycles

¹International Monetary Fund (2012), Annual Report on Exchange Arrangements and Exchange Restrictions, Washington, DC.

²The Central Bank of Brazil (BC) raised foreign reserves from \$54 billion at the end of 2005 to \$350 billion at the end of 2011. Source: Central Bank of Brazil

³Between March 2008 and August 2011 the Bank of Israel (BOI) accumulated \$28.1 billion in foreign reserves, through daily purchases initially of \$25 million and later raised to \$100 million. From August 2009 to July 2011 the BOI continued to occasionally purchase foreign exchange to bring total reserves to \$77.9 billion, an increase of almost 170% with respect to their level in March 2008. Source: Flug and Shpitzer (2013)

⁴Former Israel central bank governor Stanley Fisher remarked: “I have no doubt that the massive purchases [of foreign exchange] we made between July 2008 and into 2010 [...] had a serious effect on the exchange rate which I think is part of the reason that we succeeded in having a relatively short recession”. Source: Levinson (2010)

in the domestic economy. An increase in the foreign demand for domestic assets appreciates the domestic currency and reduces the domestic real interest rate. Capital flows in the domestic economy and fuels a consumption boom, while output falls. The dynamics of consumption and output are reversed during the bust phase of the cycle. When foreign demand for domestic assets subsides, their return increases and the external debt accumulated by domestic households must be reabsorbed. Consumption falls below its steady state level and output rises until the Home country pays off its debt.

I will show that, in response to such a shock, the optimal foreign exchange intervention leans against the wind and stabilizes the path of the exchange rate. Following an increase in the demand for domestic assets, the central bank increases their net supply and accumulates foreign reserves. By doing so, the central bank absorbs part of the capital inflow and reduces its impact on the exchange rate. By leaning against the wind the central bank achieves two purposes. First, it stabilizes the domestic real interest rate and smooths out the consumption boom. Smoothing out consumption fluctuations increases welfare as it reduces the output gap caused by the wealth effect on labor supply and maximizes the present discounted value of domestic consumption. Second, if domestic prices are sticky, by reducing the initial appreciation of the domestic currency the central bank helps sustain foreign demand, therefore reducing the output gap. However, it is never optimal for the central bank to fully stabilize domestic consumption. The reason is that, by creating a wedge between the return of domestic and foreign assets, portfolio flow shocks are akin to positive wealth shocks for the Home country. Foreign exchange interventions are costly since they reduce this wedge and its benefits. In determining the optimal amount of stabilization, the central bank trades off the benefits that arise from the positive wealth shock and the inefficiencies caused by the exchange rate fluctuations.

The optimal policy mix required to deal with portfolio flow shocks includes both foreign exchange intervention and monetary policy. Consumption fluctuations induced by portfolio flow shocks shift the Phillips curve to an inefficient position, away from first best. While monetary policy select the welfare loss-minimizing allocation along the shifted Phillips curve, foreign exchange intervention smooths out consumption fluctuations and reduce the magnitude of the shift. The two tools are complements rather than substitutes. This conclusion is in sharp contrast with the standard view of foreign exchange intervention which emphasizes its monetary aspect. The use of continuous-time techniques allows me to solve the planning problem in closed form and characterize the optimal foreign exchange intervention policy rule. I will show that the optimal intervention policy can be implemented by a simple rule that is a function of three implicit targets: a wedge in the Backus-Smith condition, domestic net foreign assets, and the level of foreign reserves.

Finally, I will provide empirical evidence of the inefficiencies generated by capital flow shocks quantify them, by estimating the key friction of the model. I will first estimate a reduced form VAR model using Swiss and EU data over the period 1999Q1 to 2013Q3 and show that, consistently with the theoretical model, in response to an increase in the foreign demand for Swiss assets, net capital inflow into Switzerland rises and the franc appreciates, while Swiss GDP falls. I will then use the empirical impulse responses to estimate the friction that governs the imperfect substitutability between domestic and foreign assets which is at the core of the model. The estimated friction implies that an increase in foreign demand for Swiss assets equal to 16 percent of Swiss GDP, which corresponds to a one-standard deviation increase in the VSTOXX index,

causes the Swiss franc to appreciate by 6 percentage points in nominal term, and around 3 percentage points in real terms. The optimal foreign exchange intervention is large as the domestic central bank accumulates reserves equal to more than 10 percent of GDP.

Conceptually, the literature distinguishes between three channels through which foreign exchange intervention can have an impact on the exchange rate: a portfolio balance channel which arises when domestic and foreign assets are imperfect substitutes and uncovered interest rate parity does not hold; a signaling channel where the central bank, with its intervention, provides information to the private sector shifts expectations about the future path of the monetary base; a coordination channel whereby the central bank's intervention influences trading decisions of currency traders, which in turn move the exchange rate through their order flow. The mechanism developed in this paper belongs to the first channel. Since the early literature on portfolio balance models, started by Kouri (1976) and Driskill and McCafferty (1980), we know that when domestic and foreign assets are imperfect substitutes, foreign exchange intervention is an additional tool available to the central bank. However, that literature is mostly positive as the lack of solid micro-foundations prevents a rigorous normative analysis. Therefore, it cannot say anything about the optimal use of foreign exchange intervention and the relationship with monetary policy. The goal of this paper is to fill this gap.

The focus of this paper is the desirability of foreign exchange intervention in response to currency misalignments arising in financial markets. Questions regarding its effectiveness are the focus of a vast and lively literature. Most of the studies from the late 1980s found that the coordinated interventions, conducted after the Plaza and Louvre agreements, were effective. Dominguez and Frankel (1993) analyze interventions using data from U.S. dollar, German mark and Swiss franc between 1982 and 1988. Their study points to the presence of a significant portfolio channel in the U.S. Dollar and German mark markets. More recently, Sarno and Taylor (2001) survey central bank interventions conducted in the 1990s and conclude that foreign exchange intervention can be effective especially if it is publicly announced and provided that it is consistent with the underlying stance of monetary and fiscal policy. Menkhoff (2010) and Menkhoff (2013) survey the literature focusing on studies that use high-frequency data for both developed and emerging economies. The author finds that the evidence corroborates the hypothesis that interventions move the exchange rate level in the desired direction, especially in emerging market economies where higher reserves and shallow markets give central banks more leverage. Adler and Tovar Mora (2011) and Adler et al. (2015) provide evidence that interventions can affect the level of the exchange rate and the pace of appreciation. Finally, using daily cross-country data, Fratzscher et al. (2015) show that foreign exchange intervention works well in smoothing the path of exchange rates, and in stabilizing the exchange rate in countries with narrow band regimes. Although the evidence of its effectiveness is far from conclusive, it suggests that foreign exchange intervention can be a useful tool for central banks.

A recent strand of literature has focused on testing the relationship between capital flows, asset demands, and exchange rates movement. Froot and Ramadorai (2005) show that medium-term variations in expected currency returns are associated with capital flows, while long-term variations are correlated with macroeconomic fundamentals. Hau et al. (2010), in the context of a natural experiment, provide direct evidence that exogenous capital flows affect the exchange rate. They show that countries that experienced capital inflows

as a result of an increase in their weight in the MSCI World Equity Index, saw their currencies appreciate. More recently, Della Corte et al. (2016) find evidence of a global imbalance risk factor in currency excess returns. They show that the currencies of net debtor countries have higher returns, tend to be on the receiving end of carry trade flows, and depreciate when global financial conditions worsen. Menkhoff et al. (2016) study order flow data from a very large FX dealer and conclude that order flows are highly informative about future exchange rates, and in the case of short-term demand-side investment managers' flows their predictive content seems to be driven by liquidity effects associated with downward-sloping demand curves.

The rest of the paper is organized as follows: Section 1 reviews the relevant literature; Section 2 describes the model and its equilibrium; Section 3 describes the log-linearized dynamics around the steady state and the planner problem; Section 4 characterizes the optimal foreign exchange intervention under flexible and rigid prices; Section 5 studies the interaction between foreign exchange intervention and monetary policy; Section 6 characterizes the optimal foreign exchange intervention rule; Section 7 describes the empirical evidence and the estimation of the model; finally, Section 8 concludes.

1 Literature Review

This paper is related to three broad streams of literature: the international macroeconomic literature on the volatility of capital flows, the literature on international financial intermediation and the literature on optimal policy and welfare in open economies.

The volatility of capital flows and their effects on open economies are at the center of a large literature. Starting with the seminal work of Calvo (1998) most of the literature has focused on “sudden stops”, episodes in which the inflow of foreign capitals suddenly reverses. See for example Caballero and Krishnamurthy (2004), Mendoza (2010), Bianchi (2009), Bianchi and Mendoza (2010), Jeanne and Korinek (2010), Korinek (2011), and Korinek and Sandri (2016). These papers emphasize the role of domestic borrowing constraints and pecuniary externalities in amplifying shocks to the external financing conditions. More recently Farhi and Werning (2014a) study the effects of foreign risk premia shocks in a small open economy with nominal rigidities. These papers focus on capital controls as a tool to manage capital flows and stabilize the domestic economy. I contribute to this literature by studying the optimal use of an alternative tool, namely sterilized foreign exchange intervention, in response to capital flow shocks. A related strand of literature has focused on the role of capital flows in the transmission of financial conditions across countries. Rey (2015), Agrippino and Rey (2013), Shin (2014) and Blanchard et al. (2015) provide evidence of a global financial cycle driving capital flows, asset prices and credit growth across countries. The cycle might impose constraints and might render domestic monetary policy ineffective if the capital account is not managed. Consistent with this conjecture, I find that monetary policy independence increases when foreign exchange intervention is used to deal with currency misalignments caused by capital flows.

This paper also relates to the literature on international financial intermediation and exchange rate determination in the presence of frictions. The earlier and most prominent strand of the literature on asset demand and exchange rate determination in general equilibrium has focused on complete markets. See for example Lucas Jr (1982), Backus, Kehoe and Kydland (1992), Backus and Smith (1993), Pavlova and

Rigobon (2007) Verdelhan (2010), Colacito and Croce (2011), Stathopoulos (2015) and Farhi and Gabaix (2016). An other stream of literature has analyzed the role of market incompleteness in the determination of international portfolios. For recent examples see Chari, Kehoe and McGrattan (2002), Corsetti et al. (2008) and Rigobon et al. (2011). As shown by Backus and Kehoe (1989), sterilized intervention is not an extra policy instrument available to central banks in these classes of models. When portfolio decisions are frictionless, the imperfect substitutability between assets postulated by the portfolio balance channel is not enough for sterilized intervention to have an effect on prices and quantities. Considering this result, this paper follows a different approach based on frictions in the intermediation process of international capital flows. Important contributions in the study of international financial frictions include Caballero and Krishnamurthy (2001), Jeanne and Rose (2002), Evans and Lyons (2002), Hau and Rey (2006) and Bruno and Shin (2015). A related stream of literature has focused on the effects of domestic financial frictions and market segmentation on the dynamics of portfolio flows and exchange rates. See for example Alvarez et al. (2009), Bacchetta and Van Wincoop (2010) and Maggiori (2013). Bruno and Shin (2014) develop a model of the international banking system where global and local funding constraints interact in the transmission of financial conditions across borders. Gabaix and Maggiori (2015) build an analytically tractable 2-period general equilibrium model where constrained international financiers intermediate capital flows across countries. They provide a novel micro-foundation for the portfolio balance channel and analyze the welfare effects of heterodox policies such as foreign intervention and capital controls. I build on their analysis and incorporate the intermediation friction in a standard New Keynesian small open economy. This allows me to provide a complete characterization of the optimal foreign exchange intervention policy and its interaction with standard monetary policy.

Finally, this paper relates to the literature on optimal policy and welfare in New Keynesian open economies. Aguiar et al. (2009), Schmitt-Grohé and Uribe (2012), Farhi and Werning (2012; 2014b; 2015) and Farhi et al. (2014) provide innovative analyses of the effects of conventional policies such as capital controls, fiscal transfers and fiscal devaluations on the welfare of a small open economy. This paper is most closely related to the literature on the desirability of stabilizing exchange rates. Much of this literature is concerned with investigating the conditions under which it is optimal to include the exchange rate in the reaction function of monetary policy. See for example Sutherland (2005), Corsetti and Pesenti (2005), Benigno (2009), De Paoli (2009) and the comprehensive review of Engel (2014). Engel (2011) characterizes the optimal response of monetary policy to currency misalignments arising from differences between domestic and foreign prices due to Local Currency Pricing (LCP). Cavallino and Sandri (2017) develop a model in which the presence of currency mismatches and domestic borrowing constraints limit the expansionary power of monetary policy. The focus of this paper is on currency misalignments arising in financial markets. There are few existing papers that allow for a non-trivial role for sterilized intervention. Some notable exceptions are Benes et al. (2015), Montoro and Ortiz (2013), Devereux and Yetman (2014) and Ghosh et al. (2016). These papers analyze the welfare effects of different exogenously specified intervention policies aimed at stabilizing the exchange rate. My contribution to this literature is to derive the optimal foreign exchange intervention policy in a utility-maximizing framework and characterize its implementation.

2 A Small Open Economy Model

Consider a continuous-time model with infinite horizon. The world economy is composed of a continuum of measure one of countries indexed by $i \in [0, 1]$. Every country is inhabited by a continuum of measure one of identical households, and each household is composed of a measure one of family members with identical preferences over consumption paths. By assumption, there is perfect consumption insurance within the household because all members pay out their earnings to be shared equally across the entire household. In each country there is also a measure one of monopolistically competitive firms that produce a continuum of differentiated tradable goods. International financial markets are incomplete and segmented. The only assets available in the world economy are a continuum of riskless nominal bonds denominated in different currencies. Each bond pays one unit of the currency of a specific country. Households can freely trade domestic assets, i.e. bonds denominated in domestic currency, but they are constrained in their holdings of foreign assets. Hence, imbalances in financial markets might arise. The excess demand or supply of assets is absorbed, at some premium, by financial firms. Financial firms are allowed to invest in bonds denominated in different currencies and clear markets.

Uncertainty is represented by a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}(t)\}, \mathbb{P})$ on which is defined a standard Brownian motion $Z(t)$ for $t \in [0, \infty)$. All stochastic processes are assumed to be adapted to $\{\mathcal{F}(t)\}_{t=0}^T$, the augmented filtration generated by Z . In what follows, I assume all regularity conditions which ensure that all processes introduced are well defined.

Since the focus of this paper is on the policy of a single economy, I will describe the model from the point of view of one country, which I call “Home” or “the domestic country”, and can be thought of as a particular value of $H \in [0, 1]$. To simplify the analysis, I will also assume that all foreign countries are at their symmetric equilibrium at all points in time.⁵ I refer to them as rest of the world and denote their variables with a star superscript.

In the next sections, I detail the problem faced by households and firms located in Home. Unless noted otherwise, the problems faced by foreign agents are symmetric. I will then describe the decision of financial intermediaries and the instruments available to the domestic policymaker.

Households

Home is inhabited by a continuum of measure one of identical households. Each household chooses consumption and labor for all its family members. The representative household maximizes

$$\mathbb{E} \left[\int_0^\infty e^{-\rho t} \left(\ln C(t) - \frac{L(t)^{1+\varphi}}{1+\varphi} \right) dt \right] \quad (1)$$

where C is consumption and L is the amount of labor supplied. The parameter $\rho > 0$ is the time discount factor, and φ the inverse of the Frisch elasticity of labor supply. The consumption index C is a composite of

⁵This assumption allows me to keep track of only one set of international prices rather than a continuum of bilateral prices.

Home and imported goods, given by

$$C(t) \equiv \frac{C_H(t)^{1-\alpha} C_F(t)^\alpha}{(1-\alpha)^{1-\alpha} \alpha^\alpha}$$

where $\alpha \in [0, 1]$ is the degree of Home bias in consumption and measures the degree of openness of the Home economy. The imported goods index C_F is itself an aggregator of goods produced in different countries and it is defined by $C_F(t) \equiv \exp \int_0^1 \ln C_i(t) di$. Each country produces a continuum of varieties of the domestic good. Therefore each C_i is an index of consumption of all varieties defined by

$$C_i(t) \equiv \left[\int_0^1 C_{i,j}(t)^{\frac{\epsilon-1}{\epsilon}} dj \right]^{\frac{\epsilon}{\epsilon-1}} \quad i \in [0, 1]$$

where the parameter $\epsilon > 1$ measures the elasticity of substitution across different varieties produced in a single country.

International financial markets are incomplete and segmented. The representative Home household can only invest in two assets: a Home bond paying one unit of the domestic currency, and a continuum of foreign bonds each paying one unit of a foreign currency. However, while it can freely trade domestic assets, its holding of foreign bonds is constrained. Let D denote the value of Home bonds held by the representative household and F the foreign-currency value of its portfolio of foreign bonds.⁶ For simplicity, I will model the demand for foreign assets as an exogenous process.⁷ The portfolio flow F can be thought of as an inelastic demand coming from noise family members. It can also be motivated as a liquidity shock or as the result of time-varying portfolio constraints. Understanding the determinants of international portfolios is a key research question in international finance, and the focus of a developing literature⁸, but it goes beyond the scope of this paper. Similarly, I assume that foreign households can freely hold foreign bonds but are constrained in their holding of Home bonds. With a slight abuse of notation, I denote with F_H^* the aggregate Home-currency value of Home bonds held by foreign households. As for F , I model foreign demand for domestic assets as an exogenous process. Since each country has measure zero, Home assets represent a negligible share of foreign households' portfolios. Thus, shocks to the foreign demand for Home bonds do not affect foreign prices nor aggregate quantities.

Define the Home terms of trade as the Home price of imported goods divided by the price of Home goods, $\mathcal{S} \equiv P_F/P_H$, such that an increase in $\mathcal{S}$ represents a deterioration in the Home terms of trade. Let $\mathcal{E}$ be the nominal exchange rate between the Home country and the rest of the world, defined as the Home currency price of one unit of any foreign currency, such that a decrease in $\mathcal{E}$ is an appreciation of the domestic currency. Let $A \equiv D + \mathcal{E}F$ be the Home-currency value of the assets held by the representative

⁶Formally $F \equiv \int_{i \neq H} F_i di$, but since foreign countries are identical at any point in time, the actual composition of the portfolio is indeterminate.

⁷Alternatively, I could allow domestic agent to choose F subject to a portfolio constraint of the form $F \in [\underline{F}, \overline{F}]$. The main idea is that, in order for international financial intermediaries to play a role, domestic household must be unable to fully absorb foreign demand for domestic assets and arbitrage away any excess return. Both approaches yield similar results.

⁸See for example Pavlova and Rigobon (2010), Rigobon et al. (2011) and Maggiori (2013), among others.

household. Its dynamic budget constraint is

$$\begin{aligned} dA(t) = & A(t) i(t) dt + \mathcal{E}(t) F(t) [(i^*(t) - i(t) + \mu_{\mathcal{E}}(t)) dt + \sigma_{\mathcal{E}}(t) dZ(t)] \\ & + [W(t) L(t) - P(t) C(t)] dt + dM(t) + dR(t) + dT(t) \end{aligned} \quad (2)$$

where i is the domestic nominal interest rate, i^* is the nominal interest rate in the rest of the world, $\mu_{\mathcal{E}}$ and $\sigma_{\mathcal{E}}$ are drift and volatility of the nominal exchange rate⁹, W is the nominal wage, P is the domestic Consumer Price Index (CPI), M represents nominal profits received from domestic firms, R profits received from domestic financial intermediaries, and T is a nominal lump-sum component of income which includes all taxes paid to and transfer received from the government.

The optimal allocation of expenditure across different goods yields the following demand functions

$$\begin{aligned} C_H(t) &= (1 - \alpha) \left(\frac{P_H(t)}{P(t)} \right)^{-1} C(t) & C_F(t) &= \alpha \left(\frac{P_F(t)}{P(t)} \right)^{-1} C(t) \\ C_i(t) &= \left(\frac{P_i(t)}{P_F(t)} \right)^{-1} C_F(t) & C_{i,j}(t) &= \left(\frac{P_{i,j}(t)}{P_i(t)} \right)^{-\epsilon} C_i(t) \end{aligned}$$

$\forall i \in [0, 1]$ and $\forall j \in [0, 1]$, where the Home's CPI is defined as $P \equiv P_H^{1-\alpha} P_F^{\alpha}$, the price of imported goods is given by $P_F \equiv \exp \int_0^1 \ln P_i dj$, and the price of the good produced in country i is $P_i \equiv \left(\int_0^1 P_{i,j}^{1-\epsilon} dj \right)^{\frac{1}{1-\epsilon}}$.

The problem of the representative household is to choose consumption, labor and investment to maximize (1) subject to the budget constraint (2), given policies and the exogenous demand for foreign assets F . The optimal consumption/saving policy is described by the Euler equation

$$\frac{dC(t)}{C(t)} = \left[i(t) - \pi(t) - \rho + \sigma_P(t)^2 + \sigma_C(t) \sigma_P(t) + \sigma_C(t)^2 \right] dt + \sigma_C(t) dZ(t)$$

where π and σ_P are the drift and the volatility of the CPI, and will be determined in equilibrium. Finally, the labor supply schedule is $L(t)^\varphi C(t) = \frac{W(t)}{P(t)}$.

Foreign households have identical preferences and solve a symmetric problem. Hence, their demand function for Home goods is $C_H^*(t) = \alpha \left(\frac{P_H^*(t)}{P^*(t)} \right)^{-1} C^*(t)$, where C^* satisfies the Euler equation

$$\frac{dC^*(t)}{C^*(t)} = \left[i^*(t) - \pi^*(t) - \rho + \sigma_{P^*}(t)^2 + \sigma_{C^*}(t) \sigma_{P^*}(t) + \sigma_{C^*}(t)^2 \right] dt + \sigma_{C^*}(t) dZ(t)$$

⁹Since the exogenous processes in the model are driven by a Brownian motion, all endogenous variables follow Ito processes whose coefficients will be determined in equilibrium. For example, the nominal exchange rate $\mathcal{E}$ follows the Ito process

$$\frac{d\mathcal{E}(t)}{\mathcal{E}(t)} = \mu_{\mathcal{E}}(t) dt + \sigma_{\mathcal{E}}(t) dZ(t)$$

whose drift, $\mu_{\mathcal{E}}$, and volatility, $\sigma_{\mathcal{E}}$, will be determined endogenously in equilibrium.

Firms

A continuum of monopolistically competitive firms, indexed by $j \in [0, 1]$, produce different varieties of the domestic good. All firms use the same technology, described by the production function

$$Y_j(t) = L_j(t) \quad (3)$$

for $j \in [0, 1]$. The profits generated by a generic firm j are given by

$$dM_j(t) = [P_{H,j}(t) C_{H,j}(t) + \mathcal{E}(t) P_{H,j}^*(t) C_{H,j}^*(t) - (1 - \tau) W(t) L_j(t)] dt$$

where $P_{H,j}$ is the Home-currency price of good j when it is sold in the Home country while $P_{H,j}^*$ is the foreign-currency price of the good when it is sold abroad. $C_{H,j}$ represents sales of the good in the Home country while $C_{H,j}^*$ represents its export. Aggregate profits are given by $dM = \int_0^1 dM_j dj$. The Home policymaker is assumed to have only limited fiscal instruments, a constant labor subsidy τ which is chosen to maximize domestic welfare in steady state.

Firms face an identical isoelastic demand schedule for their own good and set prices infrequently à la Calvo (1983). Each firm is allowed to reset its prices only at stochastic dates determined by a Poisson process with intensity θ . I assume that firms set their prices in domestic currency and the law of one price holds:

$$P_{H,j}^*(t) \equiv \frac{P_{H,j}(t)}{\mathcal{E}(t)}$$

Under this assumption there is perfect exchange rate pass-through as exchange rate shocks are transmitted one-to-one to import prices.¹⁰ The firm's objective is to maximize the present discounted value of its stream of profits, subject to the sequence of domestic and foreign demand schedules.

In what follows, I will consider three different values for θ . First, I will consider the case of flexible prices ($\theta \uparrow \infty$). When firms are allowed to reset their prices continuously, then the optimal pricing setting decision simplifies to $P_{H,j}(t) = \frac{\epsilon}{\epsilon-1} (1 - \tau) W(t)$. Second I will consider the case of rigid prices ($\theta = 0$). Under this assumption firms are not able to change their prices at any point in time, therefore $P_{H,j}(t) = P_{H,j}(0)$, $\forall t \in [0, \infty)$. Finally, I will consider the intermediate case of staggered pricing ($\theta \in (0, \infty)$). A firm that is allowed to reset its prices at time t solves the following problem

$$\max_{\hat{P}_{H,j}(t)} \mathbb{E} \left[\int_t^\infty \theta e^{-(\rho+\theta)(k-t)} \frac{C(t) P(t)}{C(k) P(k)} \left\{ \hat{P}_{H,j}(t) Y_j(k|t) - (1 - \tau) W(k) L_j(k|t) \right\} dk \right]$$

subject to (3), where $Y_j(k|t) = \left(\frac{\hat{P}_{H,j}(t)}{P_H(k)} \right)^{-\epsilon} Y(k)$. In the interest of space, the complete solution to the firms' price-setting problem under sticky prices, together with the implied laws of motion of domestic inflation and price dispersion, are reported in the appendix.

¹⁰This price-setting assumption is also known as Producer Currency Pricing (PCP) in contrast with Local Currency Pricing (LCP). Under the latter, firms choose their domestic and foreign prices independently and the law of one price does not hold.

International Financial Intermediaries

Households can freely trade domestic bonds, but they are constrained in their holdings of foreign assets. Hence, imbalances in financial markets might arise. A mass one of financial firms operates in the global financial market. They can invest in assets denominated in different currencies and are therefore able to absorb any excess demand or supply. To do so, however, they require a premium in the form of expected currency appreciation/depreciation. Formally, in each country each household owns a continuum of measure one of financial intermediaries, each managed by one of its family members. Each intermediary is specialized in trading assets issued by a specific country. That is, the financial intermediary specialized in country- i assets is restricted to hold a portfolio composed only of such assets and domestic bonds.¹¹ Therefore, in the market for Home bonds operate a measure one of intermediaries owned by Home households and a measure one of intermediaries owned by foreign households.

To keep the analysis tractable, I assume that the management of financial firms is a short-term job, such that there is no capital accumulation. At time t , family members are randomly assigned to financial intermediaries. Each manager then chooses a self-financed portfolio of domestic assets and assets issued by the country in which her intermediary is specialized in. At time $t + dt$, with $dt \downarrow 0$, they collect profits and pay them back to the household, where they will be shared equally among all family members. Managers are then randomly reassigned to new intermediaries and the cycle starts over.

Consider the problem facing the representative foreign financial intermediary.¹² At time t , the financier chooses a portfolio of assets $\{Q_H^*(t), Q^*(t)\}$ subject to the balance sheet constraint

$$Q_H^*(t) + \mathcal{E}(t) Q^*(t) = 0$$

where Q_H^* is the Home-currency value of Home assets and Q^* is the foreign-currency value of foreign assets. At time $t + dt$, this portfolio generates the net return

$$dR^*(t) = Q^*(t) (i^*(t) - i(t) + \mu_{\mathcal{E}}(t)) dt + Q^*(t) \sigma_{\mathcal{E}}(t) dZ(t)$$

Following Gabaix and Maggiori (2015), I will assume that the borrowing process is subject to an agency friction that limits the size of the balance sheet of the intermediary and prevents perfect arbitrage between assets denominated in different currencies. At time t , after taking positions $\{Q_H^*(t), Q^*(t)\}$, the financier can divert borrowed funds at rate $\Gamma^* \sigma_{\mathcal{E}}(t)^v |Q^*(t)|$ per unit of time, where $\Gamma^*, v \geq 0$.¹³ Hence, the total

¹¹This arrangement seems overly complex but is necessary to guarantee that in the market for Home bonds households and investors have comparable masses, while maintaining the representative household construct in each country. Since all foreign countries are identical, one can think of the model as being composed of one big country and one small country, where by small I mean of negligible size with respect to the big country. If all agents in the big country were allowed to demand assets of the small country, demand and supply would not be comparable measures because they would belong to two different dimensions (like area and length in geometry). Hence I need to restrict the mass of agents in the big country that are allowed to trade assets of the small country. Notice that this arrangement also prevents financial intermediary from diversifying their portfolios. This restriction is without consequences in the current model since all foreign countries are identical, and therefore there are only two bonds.

¹²The construction of Home and foreign representative financial intermediaries is made possible by the assumption that all foreign countries are identical. Hence their bonds also are.

¹³Since the financial intermediary has no capital, the foreign-currency value of the funds borrowed is always equal to $|Q^*(t)|$, regardless of the position the manager is taking.

amount of divertible funds is $\Gamma^* \sigma_{\mathcal{E}}(t)^v Q^*(t)^2 dt$.¹⁴ Notice that the manager, and implicitly the household owners of the intermediary, has limited commitment before returns are realized but full commitment after that. At $t + dt$ the financier is not allowed to default and any loss is absorbed by the household. Since being a financier is a short-term job, and there is perfect consumption insurance within the household, managers only care about the instantaneous expected return of their portfolio. Hence, they choose not to divert funds if the following incentive compatibility constraint holds

$$\mathbb{E}[dR^*(t)] \geq \Gamma^* \sigma_{\mathcal{E}}(t)^v Q^*(t)^2 dt \quad (4)$$

Creditors correctly anticipate the incentives of the financier to divert funds and are willing to lend as long as their incentive compatibility condition holds. Therefore, financiers solve the following problem:

$$\max_{Q^*(t)} \mathbb{E}[dR^*(t)] = Q^*(t) (i^*(t) - i(t) + \mu_{\mathcal{E}}(t)) dt$$

subject to 4. Since the value function of the financier is linear in Q^* , while the incentive compatibility condition is convex, the constraint always binds and the solution to the Home financier's problem yields the following demand for Home bonds

$$\frac{Q_H^*(t)}{\mathcal{E}(t)} = \frac{i(t) - i^*(t) - \mu_{\mathcal{E}}(t)}{\Gamma^* \sigma_{\mathcal{E}}(t)^v}$$

Foreign financier's demand for Home assets is increasing in the excess return of Home bonds vis-à-vis foreign bonds, while it is decreasing in the volatility of the exchange rate. The parameter Γ^* determines the size of the balance sheet of the financier and is therefore an inverse measure of her risk-bearing capacity. The higher is Γ^* , the lower the financier ability to sustain the currency risk of their portfolio and the higher the required compensation per unit of risk. As $\Gamma^* \uparrow \infty$ then $Q^* \rightarrow 0$ and the financier is unable to absorb any imbalance. Vice versa, as $\Gamma^* \downarrow 0$ then $i^* - i + \mu_{\mathcal{E}} \rightarrow 0$ and all bonds have the same expected rate of return, once measured in the same currency. The financier's risk-bearing capacity is so high that any excess return is immediately arbitrated away. The parameter v governs the sensitivity of the demand of the financier to the volatility of the exchange rate. As $v \downarrow 0$ this sensitivity decreases and the financiers only cares about the excess return.

Home financiers are subject to a similar constraint which yields the domestic demand for Home bonds:

$$\frac{Q_H(t)}{\mathcal{E}(t)} = \frac{i(t) - i^*(t) - \mu_{\mathcal{E}}(t)}{\Gamma \sigma_{\mathcal{E}}(t)^v}$$

In order to map the model to the data, I assume the following functional forms for Γ^* and Γ

$$\Gamma^* = \frac{\gamma / (1 - \beta)}{P_H^*(t) Y(t)}$$

¹⁴For the constraint to make economic sense it must be the case that $\Gamma^* \sigma_{\mathcal{E}}(t)^v Q^*(t)^2 \leq 1$. That is, the financier cannot steal more than 100% of the funds borrowed. In what follows I assume that the parameters of the model are such that this condition always holds.

$$\Gamma = \frac{\gamma/\beta}{P_H^*(t) Y(t)}$$

The parameter γ determines the aggregate size of financiers balance sheets and is an inverse measures of the aggregate risk-bearing capacity of the financial sector. It is the key parameter of the model, and will be the main object of interest in the estimation. The parameter β represents the relative size of the Home financiers balance sheets with respect to foreign financiers, and therefore governs how revenue from intermediation are distributed between domestic and foreign households. If $\beta = 0$ then domestic financiers are unable to intermediate any flow and all revenues accrue to foreign households. Finally, I assume that both domestic and foreign balance sheets are directly proportional to the foreign currency value of Home output. It seems reasonable to assume that financial intermediaries dedicate larger fraction of their balance sheets to assets issued by larger economies. This assumption allows me to focus on the scaled value of assets and facilitate the mapping of the model to the data.

These functional forms imply that the aggregate financiers demand for Home assets is given by

$$\hat{Q}_H^*(t) + \hat{Q}_H(t) = \frac{i(t) - i^*(t) - \mu_{\mathcal{E}}(t)}{\gamma \sigma_{\mathcal{E}}(t)^v} \quad (5)$$

where hats denote variables scaled by the domestic value of Home production.

Central Bank

The Home central bank has two instruments to stabilize the domestic economy: monetary policy and foreign exchange intervention. Throughout the paper, I specify monetary policy in terms of an interest rate rule (directly or indirectly), therefore I do not need to introduce money explicitly in the model.¹⁵

The second tool, currency intervention, is the focus of this paper. The Home central bank can intervene in the asset market by buying and selling domestic and foreign bonds. Unlike households, the central bank is not constrained in his holdings of foreign assets. Let X denote the foreign-currency value of the portfolio of foreign bonds held by the central bank, i.e. its foreign reserves. The central bank funds its holding of foreign reserves by issuing domestic bonds. Let X_H denote the amount of Home bonds held by the central bank. Without loss of generality, since Ricardian equivalence holds, assume that the central bank has no capital and rebates all profits and losses generated by its portfolio of assets to domestic households. Its balance sheet equation is

$$X_H(t) + \mathcal{E}(t) X(t) = 0 \quad (6)$$

A foreign exchange intervention is any purchase/sale of foreign assets that alters the relative supply of Home bonds. Through foreign exchange purchases the central bank increases foreign reserves ($X(t) \uparrow$) and increases the net supply of assets denominated in Home currency ($X_H(t) \downarrow$). Vice versa, foreign exchange sales reduce the amount of foreign reserves held by the central bank ($X(t) \downarrow$) and decrease the net supply of Home-currency bonds ($X_H(t) \uparrow$). Notice that, by focusing on a cashless economy, I am ignoring the

¹⁵Money can be explicitly introduced in the model by using a cash-in-advance constraint or by augmenting the utility function to include real money balances.

currency component of the balance sheet of the central bank. The central bank cannot increase (decrease) its holding of foreign assets by issuing (purchasing) domestic currency. Thus, foreign exchange interventions considered in this paper are sterilized interventions. Non-sterilized interventions can be modeled as a combination of sterilized intervention and interest rate policy.

The net transfer to domestic households includes central bank's revenues/losses and taxes raised to finance the labor subsidy. It is given by

$$dT(t) = [X_H(t)(i(t) - i^*(t) - \mu_{\mathcal{E}}(t)) - \tau W(t)L(t)]dt - X_H(t)\sigma_{\mathcal{E}}(t)dZ(t)$$

Uncovered Interest Parity and Shock

This section characterizes the link between capital flows and exchange rates. The time- t market clearing condition for Home bonds is:

$$D(t) + F_H^*(t) + Q_H(t) + Q_H^*(t) + X_H(t) = 0$$

By using the aggregate demand from financial intermediaries, equation (5), and the balance sheet equation of Home households, I obtain the arbitrage condition between Home and foreign assets. Following Sarno et al. (2012), I will call it the risk-adjusted Uncovered Interest Parity (UIP) condition of the model.

Lemma 1. *The risk-adjusted UIP condition of the model is*

$$i(t) - i^*(t) - \mu_{\mathcal{E}}(t) + \gamma\sigma_{\mathcal{E}}(t)^v \left(\underbrace{\hat{A}(t)}_{CA\ Flow} + \underbrace{\hat{F}_H^*(t) - \hat{F}(t)}_{Portfolio\ Flow} + \underbrace{\hat{X}_H(t)}_{Intervention} \right) = 0 \quad (7)$$

Lemma 1 shows that the expected Home currency excess return includes a risk premium component that is proportional to the risk-bearing capacity of the financial sector, the volatility of the exchange rate, and capital flows. If $\gamma > 0$, capital flows are a key determinant of the expected return and therefore of the path of the exchange rate. When international asset markets are imbalanced, the relative return of Home and foreign bonds must adjust to induce financial intermediaries to absorb the imbalance. The nominal exchange rate is the relative price between assets denominated in different currencies and provides the adjustments necessary to clear the markets. Lemma 1 shows that deviations from uncovered interest rate parity are endogenous and proportional to the imbalances in demand and supply of assets denominated in different currencies. When an excess demand for Home assets arises, that is the terms in parenthesis is positive, the Home currency expected risk premium must fall to induce international financiers to take short positions denominated in domestic currency. Hence, the exchange rate appreciates on the spot and depreciates in expectation, that is $\mu_{\mathcal{E}}$ rises. Vice versa, when the supply of Home bonds exceeds their demand, the Home currency expected risk premium must rise. The exchange rate depreciates on the spot and appreciates in expectation. That is, $\mu_{\mathcal{E}}$ falls.

The decomposition in equation 7 highlights three components of capital flows that drive the expected currency excess return. The first component is associated with capital account flows. These flows are

generated by the desire of the Home households to reallocate their consumption intertemporally and they are accompanied by symmetric trade flows. The second component of the demand for Home assets is associated with portfolio flows. Portfolio flows are induced by the desire of domestic or foreign households to alter the composition of their portfolio of assets, and therefore alter the country's gross external position. Without any form of asset markets segmentation, portfolio flows don't have any impact on the exchange rate as households in other countries would be willing to take the other side of the portfolio as long as uncovered interest parity holds. However, when asset markets are segmented the relative demand for domestic and foreign assets matters, and changes in countries' gross external positions affect the path of the exchange rates. The last component of the demand for Home assets is given by the central bank intervention. By expanding and contracting its balance sheet, the Home central bank can alter the relative supply of assets and therefore the size of the imbalance that financiers must absorb. This affects the Home currency risk premium and therefore the path of the exchange rate. Formally, foreign exchange intervention is central bank intermediation of international flows that coexists and complements intermediation provided by financial intermediaries. Foreign exchange intervention affects the exchange rate to the extent there exists limits to arbitrage in private intermediation, that is $\gamma > 0$.¹⁶

Shocks to portfolio flows, and the optimal use of foreign exchange intervention, will be the focus of this paper. Without loss of generality, I assume $\hat{F} = 0$ and consider a shock to the foreign demand for domestic assets, $\hat{F}_H^*$. Although I model these shocks as exogenous, shifts in foreign investors demand for Home bonds capture a broad range of situations: they might be triggered by shocks to the perceived risk of investing in the home country, or by shocks to the overall riskiness of the investor's portfolio which induce yield-searching or safe-haven type of flows, or they may represent investor's preferences for a particular country's bonds. Importantly, these shocks induce exchange rate fluctuations that would not arise in standard models with frictionless asset markets. In this sense, they capture exchange rate movements that are not driven by standard macroeconomic fundamentals.

Equilibrium

Following the literature on open economies with incomplete markets, I define the consumption wedge as the wedge between the marginal rate of substitution between Home and foreign consumption, that is the ratio of their marginal utilities, and their marginal rate of transformation, the real exchange rate. Formally

$$\Lambda(t) \equiv \frac{1}{Q(t)} \left(\frac{C^*(t)}{C(t)} \right)^{-1} \quad (8)$$

where $Q \equiv \mathcal{E}P^*/P$ is the real exchange rate, defined as the relative price of one unit of foreign consumption in terms of domestic consumption, such that a decrease in Q is an increase in the purchasing power of the Home currency. Using the Euler equations of domestic and foreign agents, we can derive the law of motion of the consumption wedge:

¹⁶In a closed economy, similar forms of intervention have been studied in the context of large scale asset purchases and unconventional monetary policy. See for example Gertler and Karadi (2011), Gertler et al. (2012) and Gertler et al. (2013)

$$\frac{d\Lambda(t)}{\Lambda(t)} = \left(i(t) - i^* - \mu_{\mathcal{E}}(t) + \sigma_{\mathcal{E}}(t)^2 + \sigma_{\Lambda}(t) \sigma_{\mathcal{E}}(t) + \sigma_{\Lambda}^2(t) \right) dt + \sigma_{\Lambda}(t) dZ(t) \quad (9)$$

where $\sigma_{\Lambda} \equiv \sigma_P + \sigma_C - \sigma_{\mathcal{E}}$. Notice that, if asset markets were complete and frictionless ($\gamma = 0$), marginal utilities at Home and in the rest of the world would grow at the same rate once converted in the same units. Thus, Λ would be constant for all t and the real exchange rate would be proportional to their ratio: $\mathcal{Q}(t) \propto \left(\frac{C^*(t)}{C(t)} \right)^{-1}$. This is the well-known Backus-Smith condition (Backus and Smith (1993)) when there is perfect risk sharing across countries. This condition fails in the present model since households must trade with constrained financiers. Equation 9 establishes a link between the risk-adjusted UIP condition of the model, equation 7, and the dynamics of the consumption wedge. A fall in the Home currency expected risk premium increases the consumption wedge, since it tends to reduce its drift. Either domestic consumption increases relative to foreign consumption or the real exchange rate appreciates, or both. The domestic currency is overvalued as the real exchange rate falls below its efficient level, defined as the value that would arise under complete and frictionless markets given the current relative consumption. Vice versa, an increase in the Home currency expected risk premium reduces the consumption wedge and the domestic currency is undervalued, as the real exchange rate rises above the level that would arise under complete and frictionless markets given the current relative consumption.¹⁷ It must be stressed that the term 'efficient' here is used to denote the globally efficient allocation, i.e. the one that arises under complete markets. My focus however will be on the Home country. The Home policymaker might prefer allocations in which the real exchange rate is undervalued/overvalued and thus differ from the global optimum.

Let $Y \equiv \left[\int_0^1 Y_j(t)^{\frac{\epsilon-1}{\epsilon}} dj \right]^{\frac{\epsilon}{\epsilon-1}}$ be aggregate domestic output. Market clearing in the goods markets requires $Y(t) = C_H(t) + C_H^*(t)$, therefore

$$Y(t) = C^*(t) \mathcal{S}(t) [\alpha + (1 - \alpha) \Lambda(t)] \quad (10)$$

The labor market clearing condition is

$$L(t) = \int_0^1 L_j(t) dj = \Delta(t) Y(t) \quad (11)$$

where $\Delta(t) \equiv \int_0^1 \left(\frac{P_{H,j}(t)}{P_H(t)} \right)^{-\epsilon} dj$ is an index of price dispersion. The nominal wage level is determined by the households labor supply schedule: $W(t) = L(t)^{\varphi} C(t) P(t)$. The law of motion of Home net foreign assets is

$$dA(t) = [A(t) i(t) + P_H(t) Y(t) - P(t) C(t) + (X_H(t) + Q_H(t)) (i(t) - i^*(t) - \mu_{\mathcal{E}}(t))] dt \quad (12)$$

¹⁷A stream of the international finance literature, pioneered by Colacito and Croce (2013), studies the optimal risk-sharing arrangement when agents have Epstein and Zin (1989) preferences. In a two-countries complete market model when agents have recursive preferences, the optimal risk-sharing condition features a time-varying relative component that indexes the relative share of world consumption. This component, which arises from the agents' desire to trade-off higher expected future utility level for lower expected future utility risk in response to permanent shocks to their endowments (also referred to as long-run shocks), behaves like Λ . Therefore, capital flow shocks can be thought of as long-run shocks. The equivalence, however, is limited to the positive part of the analysis, as the difference in preferences could affect the normative part dramatically.

$$- (X_H(t) + Q_H(t)) \sigma_{\mathcal{E}}(t) dZ(t)$$

where $Q_H(t) = P_H(t) Y(t) \beta^{\frac{i(t)-i^*(t)-\mu_{\mathcal{E}}(t)}{\gamma \sigma_{\mathcal{E}}(t)v}}$. This is the Home aggregate budget constraint, subject to the initial condition $A(0) = 0$ and a no-Ponzi terminal condition. The real exchange rate is given by $\mathcal{Q}(t) = \mathcal{S}(t)^{1-\alpha}$. Finally, market clearing in the Home bond market yields equation (7). The model is closed by the firms' pricing policies which determine the laws of motion of prices and their dispersions.

The next lemma characterizes the deterministic symmetric steady state of the economy and derives the optimal static labor subsidy.

Proposition 1. *Suppose prices are flexible, and there are no portfolio flows, $\hat{F}_H^* = 0$. Then, in steady state $\hat{X}_H = \hat{A}$. If the steady state is symmetric, that is $\hat{X}_H = \hat{A} = 0$, then $\Lambda = 1$ and the optimal labor subsidy is $\tau = 1 - \frac{\epsilon-1}{\epsilon} \frac{1}{1-\alpha}$.*

Proposition 1 proves that in steady state the central bank of the Home country cannot permanently manipulate the level of the exchange rate. Assume Home households have positive assets in steady state, $\hat{A} > 0$, then the central bank simply act as a financial intermediary. It borrows from domestic households in domestic currency and buys an equivalent amount of foreign reserves, such that $\hat{X}_H = \hat{A}$. By doing so, the central bank converts the Home net foreign assets position in foreign currency and circumvents the financial intermediaries. Hence, the financial market imperfection is muted in equilibrium. This result highlights the fact that the use of foreign exchange intervention in this model is not driven by mercantilist motives. The policymaker does not have any incentive to sustain a permanently undervalued currency with the goal of permanently increasing foreign demand for domestic goods.

3 The Log-Linearized Model

To study the effects of portfolio flow shocks on the small open economy, I use a log-linearized version of the equilibrium conditions described in the previous section.¹⁸ I then characterize optimal policies by considering a second order approximation of the welfare function around the deterministic steady state. The optimal policies obtained from the solution of the linear-quadratic problem are linear and in many cases an analytical solution can be derived. In what follows, lowercase letters denote log-deviations from steady state, if the variable has a non-zero steady state value. Otherwise, they denote simple deviations from steady state.

The equations that approximate the local dynamics of the equilibrium around its steady state are the log-linearized analogues of the stochastic differential equations described in the previous section. The consumption wedge and scaled foreign assets evolve as follows:

$$d\lambda(t) = -\gamma \left(\hat{a}(t) + \hat{f}_H^*(t) - \hat{x}(t) \right) dt \quad (13)$$

$$d\hat{a}(t) = (\rho \hat{a}(t) - \alpha \lambda(t)) dt \quad (14)$$

¹⁸I also take the limit for $v \downarrow 0$, to preserve the financial friction that is at the heart of the model.

where $\hat{x}$ are foreign reserves scaled by Home output, subject to the initial condition $\hat{a}(0) = 0$. The laws of motion of domestic output and inflation depend on the price setting assumption. If prices are sticky, then domestic output follows

$$dy(t) = (i(t) - \rho - \pi_H(t)) dt - \alpha d\lambda(t) \quad (15)$$

This is the log-linearized IS curve of the model. PPI inflation evolves according to

$$d\pi_H(t) = [\rho\pi_H(t) - \kappa(1 + \varphi)y(t) - \alpha\kappa\lambda(t)] dt \quad (16)$$

where $\kappa \equiv \theta(\rho + \theta)$. This is the log-linearized New Keynesian Phillips Curve (NKPC) of the model. Notice that both the IS curve and the NKPC include an additional term that captures the effect of the consumption wedge on output and inflation. When prices are completely flexible domestic output is proportional to the consumption wedge and it is given by $y(t) = -\frac{\alpha}{1+\varphi}\lambda(t)$. Finally, when prices are completely rigid output is determined by $y(t) = e(t) + (1 - \alpha)\lambda(t)$, where e is the log of the nominal exchange rate.

The loss function for the Home policymaker is derived from a second order approximation to the domestic households utility, given in equation (1), and a second order approximation to the Home country budget constraint, given by equation (12). Details of the derivation are provided in the appendix. The policymaker wishes to minimize

$$\mathbb{L} = \frac{1}{2} \int_0^\infty e^{-\rho t} \left(\phi_x \hat{x}(t)^2 + \phi_\lambda \lambda(t)^2 + \phi_\pi \pi_H(t)^2 + \phi_y y(t)^2 \right) dt \quad (17)$$

where

$$\phi_x = 2\gamma(2 - \alpha)(1 - \beta) \quad \phi_\lambda = \alpha(1 - \alpha)^2 \quad \phi_\pi = \frac{\epsilon}{\kappa}(1 - \alpha) \quad \phi_y = (1 - \alpha)(1 + \varphi)$$

There are four sources of welfare loss. The last two terms in the loss function are common in the New Keynesian literature. The last term captures the welfare cost of deviating from the efficient level of output, while the third term captures the welfare losses induced by inflation through its effect on prices dispersion and domestic total factor productivity. The first two terms are new and arise as a consequence of the imperfection in the international asset market. The second term captures the welfare cost caused by an inefficient intertemporal allocation of consumption relative to the path of the real exchange rate.¹⁹ The log-linearized consumption wedge is $\lambda = c - q$. A positive consumption wedge is associated with an appreciation of the real exchange rate relative to domestic consumption, while a negative consumption wedge is associated with a depreciation of the real exchange rate relative to domestic consumption. By iterating the budget constraint forward and using initial and terminal conditions I obtain $\int_0^\infty e^{-\rho t} \lambda(t) dt = 0$. Thus, periods with a positive consumption wedge must be followed or preceded by periods with a negative consumption wedge. These fluctuations are inefficient since they reduce the present discounted value of the domestic consumption stream. Domestic consumption is too high when the domestic consumption bundle is relatively more

¹⁹An additional term related to imperfect international risk-sharing appears in the quadratic loss function of many models with incomplete markets. See for example De Paoli (2009) for a small open economy model and Corsetti et al. (2010) for two-country model. In all these models, however, this additional term disappears when the elasticity of intertemporal substitution and the elasticity of substitution between domestic and foreign goods are unitary.

expensive, that is the real exchange rate is appreciated, and too low when it is relatively cheaper. A reallocation of consumption from periods with positive consumption wedge to periods with negative consumption wedge, would raise its present discounted value. Finally, the first term in the loss function comes from a second order approximation of the Home budget constraint and captures the monetary cost of holding reserves. This cost arises from the fact that, by intervening in the foreign exchange market, the central bank moves the return of domestic bonds against its own position. For example, a foreign exchange purchase increases foreign reserves and reduces the central bank holdings of Home bonds. By increasing the supply of Home-currency bonds, however, the central bank depreciates the exchange rate and increases their return relative to foreign bonds, therefore increasing the cost of funding reserves. Vice versa, a foreign exchange sale increases the central bank holding of Home-currency assets and appreciates the exchange rate, reducing their return. Since financiers must take the opposite position in for markets to clear, returns adjust in their favor and against the central bank. Notice that the share of revenue from financial intermediation that accrue to domestic households, captured by β , does not affect the dynamic of the system but only determines the monetary cost of using foreign exchange intervention. A higher β reduces the cost of intervening in the foreign exchange market, since part of the losses incurred by the central bank are recovered by domestic households through their own financial intermediaries.

The loss function is derived without specific assumptions about how prices are set (indeed whether prices are sticky or not). This highlights the fact that the loss in welfare arises not specifically from price stickiness but from prices that do not deliver the efficient allocation. In the rest of the paper, I study the effects of and characterize the optimal policy response to a time-zero shock to the foreign demand for Home assets, $\hat{f}_H^*(0) = \varepsilon$, which evolves according to

$$d\hat{f}_H^*(t) = -\varrho \hat{f}_H^*(t) \quad (18)$$

where $\varrho \in [0, \infty]$. I will characterize optimal policies by solving the Home planner's problem under different price-settings assumption, starting with flexible price.

4 Optimal Foreign Exchange Intervention with Flexible Prices

In this section, I characterize the optimal use of foreign exchange intervention when prices are flexible and the only friction in the model is imperfect financial intermediation. I will start by solving for the natural allocation, i.e. the allocation that arises without intervention. This allows me to study the sources of welfare loss induced by capital flow shocks. I will then solve for the optimal foreign exchange intervention policy and derive the optimal foreign exchange intervention policy rule. Finally, using Swiss data I will provide a quantitative assessment of the model and estimate the financial friction at its core.

4.1 The Natural Allocation

When prices are flexible and the central bank does not intervene in the foreign exchange market, the allocation solves the system of first order differential equations given by (13), (14), and (18), subject to the initial

conditions $\hat{f}_H^*(0) = \varepsilon$ and $\hat{a}(0) = 0$. With flexible prices, output is proportional to the consumption wedge, $y = -\frac{\alpha}{1+\varphi}\lambda$. Therefore, the welfare loss caused by a capital flow shock is given by

$$\mathbb{L}^N = \frac{1}{2} \int_0^\infty e^{-\rho t} \phi_F \lambda(t)^2$$

where $\phi_F = \phi_\lambda + \phi_y \left(\frac{\alpha}{1+\varphi} \right)^2$.

Proposition 2. *Under the natural allocation, the consumption wedge is*

$$\lambda(t) = \begin{bmatrix} \frac{\gamma}{\nu} \\ \frac{\gamma}{\rho + \varrho + \nu} \end{bmatrix}^\top \mathbf{s}(t)$$

with $\nu = \frac{-\rho + \sqrt{\rho^2 + 4\alpha\gamma}}{2}$, where the states $\mathbf{s} = \begin{bmatrix} \hat{a} & \hat{f}_H^* \end{bmatrix}^\top$ evolve as $d\mathbf{s}(t) = -\mathbf{M}\mathbf{s}(t) dt$, with

$$\mathbf{M} = \begin{bmatrix} \nu & \frac{\alpha\gamma}{\rho + \varrho + \nu} \\ 0 & \varrho \end{bmatrix}$$

and $\mathbf{s}(0) = \begin{bmatrix} 0 & \varepsilon \end{bmatrix}^\top$. For a given $\bar{\varepsilon} \equiv \int_0^\infty \varepsilon e^{-\rho t} dt$, we have

$$\frac{\partial \mathbb{L}^N}{\partial \gamma} > 0$$

and $\mathbb{L}^N$ is strictly quasiconcave in α and ϱ over $(0, 1) \times (0, \infty)$, while it is zero at the boundary of the region.

To describe the effects of portfolio flow shocks, I will focus on the case of an increase in the foreign demand for domestic assets, that is $\varepsilon > 0$. Since the solution is linear, the effects of a decrease in the demand for domestic assets are symmetric. At $t = 0$, in response to an increase in the foreign demand for domestic assets, the consumption wedge turns positive

$$\lambda(0) = \frac{\gamma\varepsilon}{\rho + \varrho + \nu} \quad (19)$$

and decreasing, $d\lambda(0) = -\gamma\varepsilon dt$. As foreign households want to increase their holdings of Home bonds, the excess demand for Home-currency assets generates an imbalance in the asset market that financiers have to absorb. The nominal exchange rate is given by $e = -\lambda$.²⁰ Therefore, the Home currency appreciates on the spot and depreciates in expectation, such that the price of domestic assets increases and their foreign-currency return falls. The excess return of foreign assets vis-à-vis domestic assets induces financiers to short Home bonds until the imbalance is absorbed. Similarly, the real exchange rate, given

²⁰For simplicity, I assume that the domestic nominal interest rate remains constant $i = \rho$. Obviously, this assumption only affect the path of the nominal exchange rate, not the return of financial intermediaries, which is determined by the excess demand for domestic assets, nor other real variables.

by $q = -(1 - \alpha) \frac{1+\varphi-\varphi\alpha}{1+\varphi} \lambda$, appreciates on impact and is expected to depreciate. The expected real depreciation reduces the real borrowing cost for Home households, inducing them to accumulate debt. Net foreign assets held by Home households turn negative, $d\hat{a}(0) = -\frac{\alpha\gamma\varepsilon}{\rho+\varrho+\nu}$, as foreign capital flows into the domestic economy and finance a consumption boom, $c = \alpha \frac{1+(2-\alpha)\varphi}{1+\varphi} \lambda$. When the consumption wedge turns positive, Home output falls below its efficient level. The reason behind this result lies in the wealth effect on labor supply triggered by the consumption boom. The increase in domestic consumption pushes up real wages and act as a positive mark-up shock for domestic producers.²¹ Therefore, a positive portfolio flow shock generates an immediate consumption boom and a fall in domestic output. As exports fall and imports increases, the domestic terms of trade appreciate, $s = -\frac{1+\varphi-\varphi\alpha}{1+\varphi} \lambda$, and the trade balance turns negative.

This is however not the end of the story, as the boom part of the cycle is followed by a bust. While the capital inflow shock initially generates an excess demand for Home assets, as domestic agents accumulate debt and foreign demand subsides an excess supply of Home bonds arises. That is, $\exists \bar{t} > 0$ such that $-\hat{a} < \hat{f}_H^*$ for $t < \bar{t}$, while $-\hat{a} > \hat{f}_H^*$ for $t > \bar{t}$. For $t > \bar{t}$, the dynamics of the consumption wedge reverse, and λ is negative and increasing. As the imbalance in the asset market changes sign, financial intermediaries must absorb the debt accumulated by the Home country and their portfolio shifts towards Home-currency assets. The exchange rate falls below its steady state value and is expected to appreciate. All dynamics are now reversed. Consumption falls below its steady state level, the trade balance turns positive and output rises above its efficient level. Hence, a positive portfolio flow shock generates an inefficient boom-bust cycle, since the appreciation of the domestic currency, associated with the inflow of capital, is followed by a period of depreciation caused by its outflow.

Capital flow shocks reduce domestic welfare through two channels. First, they reduce the present discounted value of consumption by causing adverse movements in the terms of trade. A positive capital flow shock generates a domestic consumption boom accompanied by an appreciation of the terms of trades. Vice versa, during the bust phase of the cycle, Home consumption falls while the terms of trade depreciate. Hence, domestic households consume more when the cost of their consumption bundle is high, and consume less when the cost is low. These fluctuations reduce the present discounted value of their consumption stream and therefore welfare.²² The second channel through which capital flow shocks reduce domestic welfare, is through their effect on output and employment. The optimal level of employment is chosen by the planner by maximizing (1) subject to the technological constraint $Y(t) = L(t)$, and the market clearing conditions (10), and (11) given the consumption/output possibility set implied by the consumption wedge Λ . In the special case where the elasticities of intertemporal and intratemporal substitutions are equal to one, the optimal allocation implies constant employment $L = (1 - \alpha)^{\frac{1}{1+\varphi}}$. The possibility of altering the domestic terms of trade through foreign exchange intervention does not alter the well-known result, derived by Gali and Monacelli (2005) in a model with complete markets, that under a Cole-Obstfeld parametrization a constant

²¹In the working paper version, domestic producers also produce a non-tradable good. The inclusion of a non-tradable good does not alter this result. Non-tradable output, unlike tradable output, would increase in response to the shock due to the increase in domestic consumption. However, total output and employment would fall, independently on the relative sizes of the two sectors. The underlying reason behind the recessionary impact of the capital inflow is the increase in the real wage which hits both sector equally.

²²This logic is similar to Costinot et al. (2014), where the planner wishes to manipulate the dynamic path of the terms of trade to maximize the present discounted value of a time-varying endowment.

level of employment is optimal. The presence of imperfections in international financial markets, however, affects the ability of the planner to implement the optimal allocation. The labor subsidy that implements the efficient level of employment is

$$\tau^{OPT}(t) = 1 - \frac{\alpha + (1 - \alpha) \Lambda(t)}{\mathcal{M}(1 - \alpha) \Lambda(t)}$$

If markets are frictionless, then Λ is constant, and a constant labor subsidy is enough to render the flexible price allocation efficient. When domestic consumption increases, a depreciation of the real exchange rate reduces the relative price of domestic goods and boosts both domestic and foreign demand. The increase in demand offsets the increase in real wages triggered by the domestic consumption boom and employment remains constant. When international financial markets are imperfect and the exchange rate is determined by financial forces, this expenditure-switching mechanism breaks down. Portfolio flow shocks cause consumption and the exchange rate to move in opposite directions. As a result, employment deviates from its efficient level.

Finally, Proposition 2 shows how the welfare loss induced by capital flow shocks depend on the parameters of the model. The welfare loss is increasing in the magnitude of the financial friction measured by γ . The lower the risk-bearing capacity of the international financial sector, that is, the higher γ , the higher the effect of the capital flow shock on the exchange rate. As γ increases, financial intermediaries require a higher premium to absorb the foreign demand for domestic assets. Hence, the domestic currency appreciates more and depreciates faster. This cause larger swings in λ , consumption, and output, which increase the welfare loss associated with the shock. The relationship between the welfare cost of capital flow shocks and the degree of openness of the economy is non-monotonic. Capital flow shocks cause no welfare loss when the domestic economy is closed ($\alpha = 0$) or completely open ($\alpha = 1$), since in both cases, $\phi = 0$. In the former case, domestic consumption and output are independent from international prices, while in the latter the share of home goods is infinitesimal and it effectively disappears from the model, rendering international prices irrelevant for welfare. Therefore, the welfare cost of capital flow shocks is maximum for an intermediate degree of openness, $\alpha \in (0, 1)$. Finally, when the total size of the shock is kept constant, that is $\varepsilon = \varrho \bar{\varepsilon}$, the welfare cost of the shock is decreasing in its persistence for low values of ϱ . Hence, large and transitory capital flow shocks are more disruptive than small and persistent ones. This result is due to the slow endogenous response of domestic net foreign assets. At time $t = 0$, the increase in the foreign demand for domestic assets is completely absorbed by financial intermediaries, since $\hat{a}(0) = 0$. Only the consequent reduction in the real interest rate induces domestic households to borrow, reducing the imbalance in the international asset market and its effect on the exchange rate. When the shock is small but persistent, most of it will be absorbed by domestic households with little effect on the exchange rate. Vice versa, when the shock is large and transitory, financial intermediaries will absorb a large fraction of it since households do not react fast enough, causing larger swings in the exchange rates and the endogenous variables. However, when the mean-reversion of the shock is very fast, its effect on domestic welfare becomes negligible as fluctuations in λ dissipate quickly.

4.2 Optimal Foreign Exchange Intervention

In this section, I characterize the optimal foreign exchange intervention policy under flexible prices. When prices are flexible, the objective function of the planner simplifies to

$$\mathbb{L} = \frac{1}{2} \int_0^\infty e^{-\rho t} \left(\phi_x \hat{x}(t)^2 + \phi_F \lambda(t)^2 \right) dt$$

The next proposition characterizes the solution to the planner's problem under flexible prices, when the central bank can credibly commit at time $t = 0$ to the entire path of foreign exchange intervention.

Proposition 3. *The solution to the problem*

$$\min_{\hat{x}} \frac{1}{2} \int_0^\infty e^{-\rho t} \left(\hat{x}(t)^2 + \phi \lambda(t)^2 \right) dt$$

subject to (13), (14), and (18), with $\hat{a}(0) = 0$ and $\hat{f}_H^*(0) = \varepsilon$, is

$$\lambda(t) = \begin{bmatrix} -\frac{\gamma}{\rho + \underline{\nu} + \bar{\nu}} \\ \frac{\xi}{\bar{\nu}} \gamma + \frac{1-\xi}{\underline{\nu}} \gamma \\ \frac{\xi \gamma}{\rho + \varrho + \bar{\nu}} + \frac{(1-\xi)\gamma}{\rho + \varrho + \underline{\nu}} \end{bmatrix}^\top \mathbf{s}(t)$$

where the states $\mathbf{s} = \begin{bmatrix} \hat{x} & \hat{a} & \hat{f}_H^* \end{bmatrix}^\top$ evolve as $d\mathbf{s}(t) = -\mathbf{M}\mathbf{s}(t) dt$, with

$$\mathbf{M} = \begin{bmatrix} \xi \bar{\nu} + (1-\xi) \underline{\nu} & -\frac{\alpha \gamma}{\rho + \underline{\nu} + \bar{\nu}} & 0 \\ -\frac{\phi \gamma^2}{\rho + \underline{\nu} + \bar{\nu}} & \xi \underline{\nu} + (1-\xi) \bar{\nu} & 0 \\ \frac{-\phi \gamma^2}{\rho + \varrho + \bar{\nu}} \frac{\rho + \varrho + \frac{\nu \bar{\nu}}{\rho + \underline{\nu} + \bar{\nu}}}{\rho + \varrho + \underline{\nu}} & \frac{\alpha \gamma \xi}{\rho + \varrho + \bar{\nu}} + \frac{\alpha \gamma (1-\xi)}{\rho + \varrho + \underline{\nu}} & \varrho \end{bmatrix}^\top$$

and $\mathbf{s}(0) = \begin{bmatrix} 0 & 0 & \varepsilon \end{bmatrix}^\top$. The parameters $\bar{\nu} > \nu > \underline{\nu} > 0$, and $\xi \in (0, 1)$ are defined in the appendix.

The optimal foreign exchange intervention “leans against the wind”. Following an increase in the demand for domestic assets, the Home central bank increases their net supply and accumulates foreign reserves:

$$d\hat{x}(0) = \frac{\phi \gamma^2}{\rho + \varrho + \bar{\nu}} \frac{\rho + \varrho + \frac{\nu \bar{\nu}}{\rho + \underline{\nu} + \bar{\nu}}}{\rho + \varrho + \underline{\nu}} > 0$$

The intervention stabilizes the path of the consumption wedge and reduces its fluctuations. At time $t = 0$ the consumption wedge is given by

$$\lambda(0) = \gamma \varepsilon \left(\frac{\xi}{\rho + \varrho + \bar{\nu}} + \frac{1-\xi}{\rho + \varrho + \underline{\nu}} \right) \quad (20)$$

and is smaller than (19).²³ By supplying domestic assets and accumulating foreign reserves, the central bank absorbs part of the excess demand for Home assets generated by the foreign demand shock. This, in

²³It is easy to check that, if $\phi = 0$ then $\bar{\nu} = \underline{\nu} = \nu$, and (20) is decreasing in ϕ . See details reported in the appendix.

turns, reduces the imbalance in the international asset market absorbed by financial intermediaries. Since their required premium is proportional to the size of their balance sheet, the intervention reduces the initial appreciation of the domestic currency and moderates its expected depreciation. This increases the domestic real interest rate, compared to the natural allocation, and mitigates the initial consumption boom, reducing the welfare loss associated with it. Since the fall in output is caused by the wealth effect on labor supply triggered by the increase in consumption, mitigating its boom also reduces the negative output gap. The foreign exchange intervention not only mitigates the boom phase of the cycle, but it also stabilizes its bust phase. By stabilizing the real interest rate, the central bank slows down the net inflow of capital in the domestic economy and reduces debt buildup by domestic households. This reduces the size of the subsequent outflow and mitigates the severity of the bust phase of the cycle. By leaning against the wind, the central bank reduces macroeconomic fluctuations and stabilizes the domestic economy.

Despite the inefficiencies induced by fluctuations in λ , it is never optimal for the central bank to fully stabilize it. The reason is that, by creating a wedge between the return of domestic and foreign assets, portfolio flow shocks are akin to positive wealth shocks for the Home country. Although domestic households cannot take advantage of the arbitrage opportunity provided by deviations from UIP, they can still exploit the lower real cost of borrowing induced by a positive portfolio flow shock. By leaning against the wind the central bank mitigates the effects of the shock on the path of the exchange rate and on the real interest rate available to domestic households. In determining the optimal amount of stabilization the central bank trades off the benefits that arise from a cheaper cost of borrowing and the inefficiencies described in the previous section. The next Lemma characterizes the optimal foreign exchange intervention policy rule.

Corollary 1. *Let*

$$\{\hat{x}(t)\}_{t \geq 0} = \arg \min_{\hat{x}} \frac{1}{2} \int_0^{\infty} e^{-\rho t} \left(\hat{x}(t)^2 + \phi \lambda(t)^2 \right) dt$$

subject to (13), (14), and (18). Then $\hat{x}$ satisfies the following differential equation

$$d\hat{x}(t) = \psi_{\lambda} \lambda(t) dt + \psi_a \hat{a}(t) dt + \psi_x \hat{x}(t) dt$$

where

$$\begin{aligned} \psi_{\lambda} &= \psi \phi \gamma \left(\rho + \varrho + \frac{\bar{\nu} \underline{\nu}}{\rho + \underline{\nu} + \bar{\nu}} \right) > 0 \\ \psi_a &= -\psi \phi \gamma \frac{\rho + \varrho}{\alpha} \frac{(\rho + \underline{\nu})(\rho + \bar{\nu})}{\rho + \bar{\nu} + \underline{\nu}} < 0 \\ \psi_x &= -\psi \underline{\nu} \xi (\rho + \varrho + \underline{\nu}) - \psi \bar{\nu} (1 - \xi) (\rho + \varrho + \bar{\nu}) < 0 \end{aligned}$$

with $\psi = (\rho + \varrho + \xi \underline{\nu} + (1 - \xi) \bar{\nu})^{-1}$. Furthermore

$$\begin{aligned} \frac{\partial |\psi_{\lambda}|}{\partial \gamma} &< 0 \quad \frac{\partial |\psi_{\lambda}|}{\partial \varrho} > 0 \quad \frac{\partial |\psi_{\lambda}|}{\partial \phi} > 0 \\ \frac{\partial |\psi_a|}{\partial \gamma} &> 0 \quad \frac{\partial |\psi_a|}{\partial \varrho} > 0 \quad \frac{\partial |\psi_a|}{\partial \phi} > 0 \end{aligned}$$

while $\frac{\partial|\psi_x|}{\partial\gamma} > 0$, $\frac{\partial|\psi_x|}{\partial\varrho} < 0$, and $\frac{\partial|\psi_x|}{\partial\phi}$ is positive if ϱ is small enough and is negative if ϱ is large.

Corollary 1 shows how the central bank can implement the optimal foreign exchange intervention using three implicit targets: the consumption wedge λ , Home net foreign assets $\hat{a}$, and foreign reserves $\hat{x}$. The first target of the intervention, the consumption wedge, captures the goal of macroeconomic stabilization pursued by the central bank. Since $\psi_\lambda > 0$, the central bank commits to accumulate (decumulate) reserves and increase (decrease) the net supply of domestic bonds when the consumption wedge is positive (negative). The stabilizing effect of such policy can be understood by looking at equation (13). By iterating it forward we obtain

$$\lambda(t) = \gamma \int_t^\infty \left(\hat{a}(u) - \hat{x}(u) + \hat{f}_H^*(u) \right) du \quad (21)$$

The term in parenthesis is the size of the time- u imbalance in the Home assets market that is absorbed by the financial sector. The consumption wedge at time t is equal to the present discounted value of all future imbalances, where the discount term is constant and given by γ . If $\lambda(t) > 0$ then the present discounted value of future demands for domestic bonds exceeds the present discounted value of future supplies and financial intermediaries expect to be, on average, short Home assets. The exchange rate is below its steady state value and is expected to depreciate to compensate them. The fall in the return of Home assets reduces the borrowing cost for domestic households and a positive consumption wedge arises. By committing to increase the supply of domestic bonds the central bank reduces the present discounted value of future imbalances and reduces the consumption wedge. Vice versa when a negative consumption wedge arise, the Home central bank stabilizes the return of domestic assets by committing to decrease their supply. Notice that, when monetary policy is idle, the nominal exchange rate and the consumption wedge are isomorphic $e = -\lambda$. The intervention rule can be directly written in terms of the exchange rate. The isomorphism, however, breaks down when monetary policy is active. In this situation targeting the nominal exchange rate would result in a suboptimal amount of foreign exchange intervention and in an inefficient amount of stabilization.

The second target of the optimal foreign exchange intervention, net foreign assets, captures the interaction between public and private flows. Since $\psi_a < 0$, the central bank commits to accumulate (decumulate) reserves and increase (decrease) the net supply of domestic assets when the country is a net debtor (creditor). By doing so the central bank moves the return of domestic assets against the position held by domestic agents. Ceteris paribus, an increase in the net supply of domestic assets increases their return making borrowing more expensive for Home households. Vice versa, a decrease in the net supply reduces their return and the incentive to save. By committing to increase the cost (reduce the profitability) of their positions, the central bank discourages domestic households from accumulating assets. Once the portfolio flow subsides, these assets will have to be absorbed by financial intermediaries causing the same types of inefficiencies generated by the initial phase of the cycle. Thus, by discouraging a buildup in imbalances the central bank mitigates the severity of the bust phase of the cycle.

Finally, the third target of the intervention, the level of foreign reserves, captures the monetary cost of the intervention. Since $\psi_x < 0$, the path of foreign reserves is mean reverting. That is, the central bank commits to eventually reverse the intervention and bring the level of reserves back to their steady state level.

The monetary cost of the intervention arises from the fact that, when the central bank intervenes in the asset market it moves returns against its own positions. Notice also that, since holding reserves is costly, the central bank optimally smooths out the intervention rather than intervening aggressively at the time of the shock. At $t = 0$, following a positive portfolio flow shock, the central bank keeps reserves at their steady state level, $\hat{x}(0) = 0$, and increases the supply of domestic bonds only gradually over time.

The comparative statics results reported in Corollary 1 shed some light on the determinants of the optimal foreign exchange intervention policy. When international financial markets are distressed, that is when γ is high, the cost of foreign exchange intervention rises. In response, the central bank reacts less strongly to fluctuations in the consumption wedge and reduces the duration of the intervention. Notice, however, that a higher γ causes larger fluctuations in λ , as shown in the previous section. Hence, the overall size of the intervention is increasing in γ . This is consistent with the observation that foreign exchange intervention has been heavily used during the recent global financial crisis. On the other hand, the central bank commits to react more strongly to deviations in the net foreign assets position of the country. By doing so, it discourages households from accumulating assets and reduces the size of the intervention necessary during the second phase of the cycle. The central bank intervenes more aggressively against a transitory shock while it is more accommodative when the shock is persistent. The weights on the consumption wedge and net foreign assets are increasing in ϱ , meaning that, keeping constant the effects of the shock on λ and $\hat{a}$, the size of the intervention is increasing in the speed at which the shock subsides. Not only the intervention is larger in size, but it also relatively more persistent. In fact, the speed at which reserves revert to their steady state level is a decreasing function of ϱ . Finally, the size of the optimal intervention is increasing in the desire for stabilization, measured by ϕ . Its effect on the duration of the intervention, however, depends on the persistence of the shock.

4.3 Empirical Evidence

The goal of this section is to test empirically the implications of the model. Using quarterly data for Switzerland over the period 1999:1 to 2011:3, I provide evidence of the effects of portfolio flow shocks to domestic variables. The choice of using Swiss data is driven by two main considerations. First, the status of the Swiss franc as a safe haven currency simplifies the identification of purely exogenous portfolio flow shocks. Demand for franc-denominated assets increases during periods of market distress. Therefore, as a proxy for foreign demand for Swiss assets I use an equity volatility index. Second, being a relatively small but very open advanced economy, Switzerland fits the assumptions underlying the New Keynesian small open economy model, over which this paper builds upon.²⁴

Since the goal of the empirical analysis is to provide evidence of the welfare cost of portfolio flow shocks, and not to estimate the effectiveness of foreign exchange interventions, I stop the data sample in the second quarter of 2011. In September 2011 the Swiss National Bank (SNB) announced that it would “no longer tolerate” an exchange rate below SF1.20 per euro and that it was “prepared to buy foreign

²⁴Indeed, Rudolf and Zurlinden (2014) show that with the addition of few minor modification, the model is able to replicate Swiss data quite well.

currency in unlimited quantities” to maintain this floor.²⁵ The floor has been successfully maintained until January 2015, when the SNB announced that it would no longer hold the Swiss franc pegged to the euro. Even before announcing the peg, between March 2009 and August 2011 the SNB routinely intervened in the foreign exchange market to stem the appreciation of the Swiss franc.²⁶ Despite this, I retain those years in the sample to maximize the degrees of freedoms of the estimation. If effective, foreign exchange intervention should reduce the impact of portfolio flow shocks on the variables of interest and therefore bias the estimation of γ downward. The choice of starting the sample in 1999 is dictated by the availability of capital flow data.

The empirical model takes the form of a reduced form VAR

$$\begin{aligned} Z_t &= B(L) Z_{t-1} + u_t \\ \mathbb{E} u_t u_t^\top &= \Sigma \end{aligned}$$

where $B(L)$ is a p th-ordered polynomial in the lag operator. The structural economic shocks are related to u_t by $u_t = W\varepsilon_t$, with $\mathbb{E}\varepsilon_t\varepsilon_t^\top = I$, where W is a square matrix and I is the identity matrix. The choice of variables to include in the VAR model is guided by the theory developed in Section 2. Absent nominal rigidities, the equilibrium dynamics of the model can be reduced to a system of first order differential equations for net foreign assets, the real exchange rate, and domestic output, given foreign output and foreign demand for domestic assets. Therefore, the vector Z_t is defined as

$$Z_t = \begin{bmatrix} y_t^* & vix_t^* & q_t & d\hat{a}_t & y_t \end{bmatrix}^\top$$

where y^* and y denote real gross foreign and domestic output respectively, vix^* is a measure of foreign investors risk aversion, $d\hat{a}$ denotes net quarterly capital outflows scaled by domestic GDP, and q is the real domestic exchange rate. Foreign variables are measured using data from the European Union. The European Union is the largest trading partner of Switzerland, accounting for more than 60 percent of its total trade. Trade with the United States account for another 20 percent. However, in the baseline estimation I exclude US data as it could confound some of the results, being the dollar a safe haven currency itself. As shown in the appendix, results are robust to the inclusion of US data. Foreign output is measured by real GDP figures published by the Statistical Office of the European Union (EUROSTAT) while global risk aversion is proxied by the EURO STOXX 50 Volatility index (VSTOXX). Numerous papers have shown that the Swiss franc exhibits the characteristics of a safe haven currency because it provides a hedge during periods of global financial distress.²⁷ Two main mechanisms could explain why safe haven currencies appreciate during periods of global turmoil.²⁸ First, periods of high equity volatility might trigger the unwinding of carry trade positions and reduce the supply of low-yielding currencies. Second, periods of global turmoil might increase capital flows toward safer political and economic environments. Both mechanisms hinge

²⁵Swiss National Bank’s Annual Report 2011, p. 38

²⁶A detailed account of Swiss foreign exchange intervention in the aftermath of the global financial crisis is provided by Humpage et al. (2013)

²⁷See for example Baele et al. (2013); Grisse and Nitschka (2015).

²⁸For an alternative view based on heterogeneous exposure to global long-run risk see Colacito et al. (2017)

upon the assumption that currency net demand curves are downward sloping, i.e. they are not infinitely elastic, and are therefore consistent with the model at test. It is immaterial for the goal of this paper to distinguish between them.

For Switzerland, net capital outflows are measured by (the negative of) total net nonofficial capital inflows from the IMF Financial Flows Analytics Database. The real exchange rate is computed using the nominal franc/euro exchange rate and CPI data for Switzerland and the Euro area published by the Swiss Federal Statistical Office (SFSO) and EUROSTAT, respectively. Finally, Swiss output is measured by real GDP figures published by SFSO. All variables are expressed in logs, except for capital outflows which are expressed as percentage of GDP. A smooth HP trend (smoothing parameter 1600) is removed from the log of the real GDP series, while the VSTOXX series is demeaned.

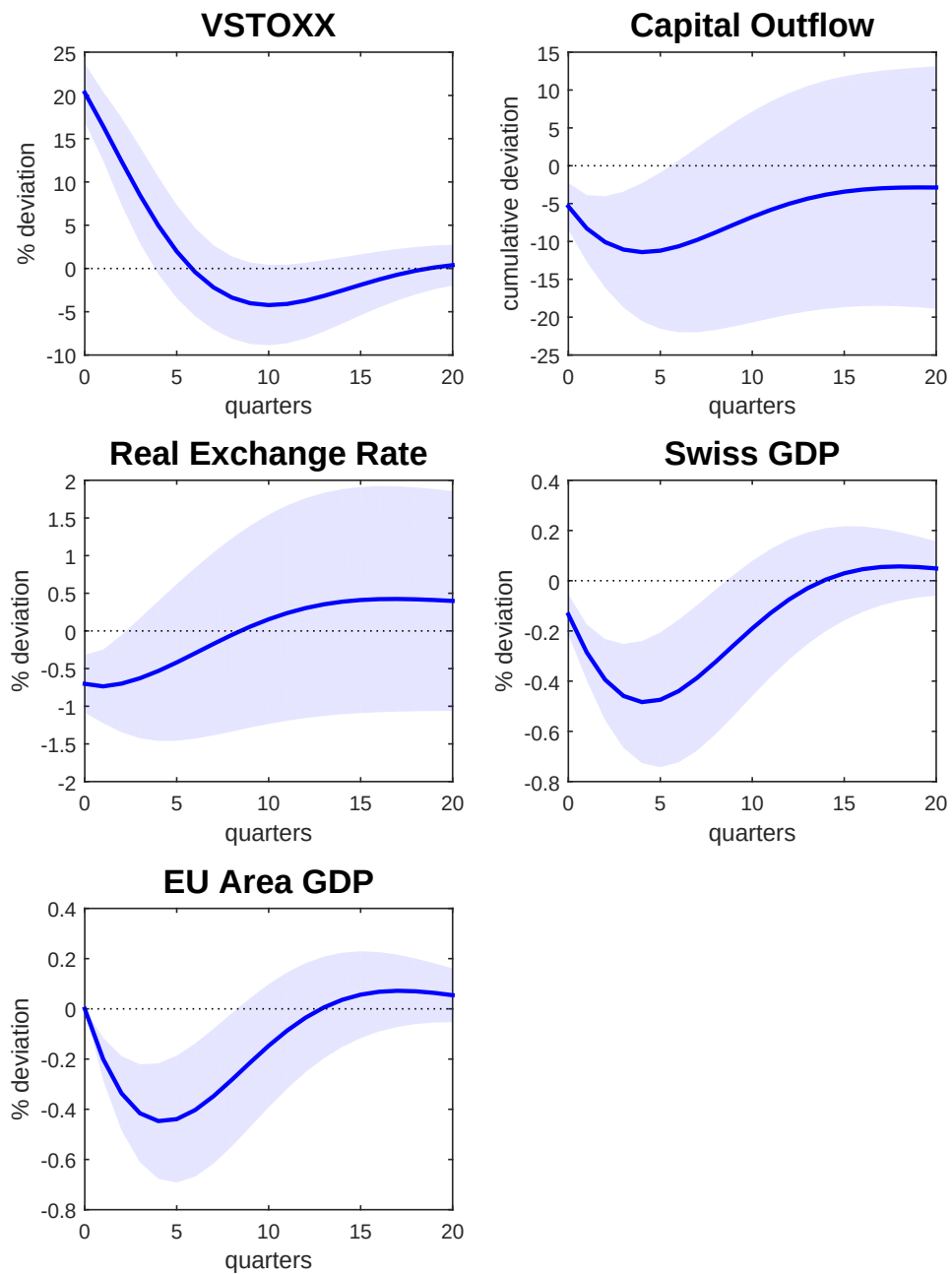
The model is identified recursively by imposing that W^{-1} is lower triangular. The identification strategy presupposes that, consistent with the theoretical model, shocks to global risk aversion affect domestic variables contemporaneously. Note that the relative order of the other variables in the system does not affect the estimates of the impulse responses to a VSTOXX shock. All that matters is the position of each variable relative to $vi x^*$. In the baseline specification, I assume that foreign output responds to innovations in the VSTOXX index only with a lag. In the appendix, I relax this assumption by allowing $vi x^*$ shocks to affect y^* contemporaneously. Results do not change significantly. In estimating the VAR, I also impose the restriction that y^* and $vi x^*$ do not depend on domestic variables. I adopt this assumption because, by construction, disturbances in a small open economy do not affect the rest of the world.

I estimate the VAR with a single lag length. I select $p = 1$ by minimizing the Schwarz's Bayesian information criterion. The Schwarz information criterion is the most accurate lag selection criterion for quarterly data and small sample sizes, as shown by Ivanov and Kilian (2005). A single lag length is also consistent with the theoretical model at test, since the equilibrium dynamics are captured by a system of ODEs of order one. Finally, given the relatively small sample size, the estimated error variance-covariance matrix Σ is corrected using the degrees of freedom of the model.

Figure 1 plots the impulse responses to a one standard deviation shock to $vi x^*$. The estimated shock has a half-life of two quarters and vanishes between four and five quarters. In response to an unanticipated positive shock to the VSTOXX index, capital flows toward Switzerland increase. To facilitate the comparison with the theoretical model, which uses stocks rather than flows, Figure 1 displays the cumulative impulse response function of capital outflows. On impact Switzerland receives a capital inflow equal to 5 percent of GDP which rises to 12 percent within a year from the shock. The inflow of capital lasts on average four quarters and is followed by a very slow outflow. Following an increase in the VSTOXX index, the Swiss real exchange rate appreciates by 0.8 percentage point. The appreciation is relatively small but very persistent, as the real exchange rate remains below its pre-shock level for almost two years. Finally, domestic output contracts by 0.5 percent and the recession achieves its peak four quarters after the onset of the shock. A VSTOXX shock is also associated with a foreign recession that is slightly milder than the domestic one.

The empirical impulse responses show that domestic variables respond significantly to fluctuations in foreign demand for domestic assets, as proxied by the VSTOXX index, and the sign of the response is consistent with the prediction of the theoretical model. A surge in the foreign demand for domestic assets

Figure 1: VAR-based Impulse Responses



VAR-based impulse responses (solid line) and 90 percent confidence intervals (shaded areas). Capital outflow is expressed as cumulative deviation from the unshocked path in percentage of GDP. All other variables are expressed as percentage deviation from their unshocked path.

is associated with an appreciation of the currency and a net capital inflows. This provides evidence of the key friction at work in the model, namely the imperfect intermediation of international capital flows. If γ is equal to zero financial intermediation is frictionless and UIP holds. A surge in foreign demand for domestic assets has no impact on domestic variables as financial intermediaries are willing to take the complementary portfolio, that is supply domestic assets, at no cost. As a result, the exchange rate is unaffected and net capital inflows are zero. In the next section I will try to quantify the magnitude of the friction by minimizing the distance between the empirical and the theoretical impulse response functions.

4.4 Calibration and Estimation

In this section, I calibrate the model and estimate the structural parameter that governs its key friction, γ . I proceed in two steps. First, I develop and calibrate a discrete-time generalized version of the theoretical model presented in Section 2. In order to relate the volatility index to foreign investors' demand for Swiss assets I assume the following functional form

$$\hat{f}_H^* = \chi vix^*$$

where the parameter χ is estimated simultaneously with γ . The equations of the model are reported in the appendix. Table 1 reports details of the calibration. The intertemporal discount factor ρ is set to 0.002 corresponding to an annualized real interest rate of 1.2%, which matches the average realized Swiss real interest over the sample period. The openness parameter α is calibrated to be 0.42 to match the average export and import shares across the sample period. The other preference parameters are calibrated using specifications found in similar studies for Switzerland. The inverted elasticity of labor supply φ is calibrated to 1. In the model, the elasticity of intertemporal substitution in consumption, and the elasticity of substitution between domestic and foreign goods are both equal to 1. These values, chosen for analytical tractability, are also close to the estimates reported in Baurle and Menz (2008), Baurle et al. (2014), and Rudolf and Zurlinden (2014). In the appendix, I relax this assumption and re-estimate the model using more general preferences. The Calvo parameter θ is set to 0.75 to match the average duration of price rigidity in Switzerland which is of four quarters, as reported by Kaufmann et al. (2009). Following Gali and Monacelli (2005) I set ϵ , the elasticity of substitution across varieties, to 6, which implies a steady state markup of 20%. The size of the shock ε and its persistence parameter ϱ are calibrated to their empirical counterparts estimated by the VAR model. I assume that monetary policy follows a simple Taylor rule whose parameters are calibrated using estimates reported by Markov and Nitschka (2013). The policy weight on inflation is set to 1.6 while the weight on output gap is set to 0.4. Finally, the parameter β governs the cost of holding foreign reserves. As such, it does not affect the estimation of γ , but it is key in determining the optimal amount of foreign exchange intervention. I calibrate β to match the estimated ex-ante annual cost of holding reserves reported by Adler and Mano (2016). The authors estimate that the average ex-ante cost for Switzerland of holding reserves over the period 2002-13 is around 0.25 percent of GDP per year. The theoretical cost of holding reserves in the model can be derived using the following second order approximation of the time-zero budget

constraint of the Home country:

$$\int_0^{\infty} e^{-\rho t} \left[\frac{1-\alpha}{2-\alpha} y - \gamma(1-\beta) \hat{x}^2 \right] dt = 0$$

Hence the cost of holding $\hat{x}$ reserves over an interval of time of length dt is $ydt = \gamma(1-\beta) \frac{2-\alpha}{1-\alpha} \hat{x}^2 dt$. The average annual reserves-to-GDP ratio for Switzerland over the period 2002-13 is 0.96. This yields a value for β of 0.995.

Table 1: Calibration

Parameter	Description	Value	Source/Target
ρ	Intertemporal discount factor	0.002	Annual real interest rate
ω	Inverse elasticity of intertemporal substitution	1	Rudolf and Zurlinden (2014)
φ	Inverse elasticity of labor supply	1	Rudolf and Zurlinden (2014)
η	Elasticity of substitution domestic/foreign goods	1	Bäurle and Menz (2008)
α	Weight of foreign goods in total consumption	0.42	Average exp and imp shares
θ	Calvo parameter	0.75	Average price duration
ϵ	Elasticity of substitution across varieties	6	Gali and Monacelli (2005)
ψ_y	Output stabilization weight in Taylor rule	0.4	Markov and Nitschka (2013)
ψ_{π}	Inflation stabilization weight in Taylor rule	1.6	Markov and Nitschka (2013)
ϱ	Shock mean reversion coefficient	0.21	Author's estimation
ε	Shock size	0.16	Author's estimation
β	Fraction of domestic financial intermediaries	0.995	Cost of holding reserves

Note: Parameters on foreign variables are set to the corresponding domestic counterparts.

In the second step, I estimate γ and χ by minimizing the distance between the empirical impulse response functions and those produced by the theoretical model. I consider the first 20 quarters of the impulse responses of 2 variables: the real exchange rate and net capital outflows. I choose not to include domestic output in the baseline estimation since the model lacks investment and capital, and is therefore too simple to properly describe the dynamics of real GDP. Furthermore, the behavior of the exchange rate and net capital outflows is directly impacted by the structure of financial markets and therefore should contain most of the information relevant for the estimation. The estimation consists in choosing 2 parameters to match 82 data points, one for each quarter plus the initial response of each variable. Specifically, let IRF^e denote the 82×1 vector of estimated impulse response functions and $IRF^m(\gamma, \chi)$ the corresponding vector of impulse response functions implied by the theoretical model. Then the estimate of (γ, χ) and the associated distance

between the empirical and the theoretical models, which we denote by Υ , satisfy

$$\Upsilon \equiv \min_{(\gamma, \chi)} [IRF^e - IRF^m(\gamma, \chi)]^\top \Sigma_{IRF^e}^{-1} [IRF^e - IRF^m(\gamma, \chi)]$$

where Σ_{IRF^e} is a diagonal matrix containing the standard errors of the empirical impulse response functions. This matrix penalizes those elements of the estimated impulse response functions associated with large error intervals. The distance Υ is minimized using derivative-free numerical methods and the standard errors of the estimates are computed using the asymptotic delta method. The estimated parameters, reported in Table 2, are $\gamma = 0.194$ and $\chi = 0.764$, and they are both significant at the 5 percent level.

Figure 2 plots the impulse response functions generated by the theoretical model and those produced by the empirical one. The model replicates the data relatively well. All points belonging to the cumulative impulse response function of net outflows lie within or close to the 90 percent confidence interval of its empirical counterpart. Moreover, the model does a good job in replicating the hump-shaped behavior observed in the data. The theoretical impulse response of the real exchange rate captures the key qualitative features of the data: an appreciation followed by a depreciation. The model, however, overestimates the magnitude of the response and underestimates its persistence. A similar argument holds for the impulse response function of output. This is not surprising, though, since the model lacks those components that are necessary to replicate the persistence of real variables such as capital formation with adjustment costs and consumption habits. Magnitude and persistence are two strictly related properties of the impulse response functions, since the budget constraint imposes a restriction on the present discounted values of real variables. Hence, increasing the persistence of the impulse responses of these variables will also help in reducing their magnitudes.

Figure (3) plots the natural allocation of the model, parametrized as described above, together with the allocation implemented by the optimal foreign exchange intervention policy under flexible prices. The magnitude of the estimated parameters implies that a 20 percent increase in the VSTOXX index, equal to a one-standard deviation shock, is associated with a surge in foreign demand for Swiss assets equal to 16 percentage points of GDP, which in turn causes the Swiss franc to appreciate by 6 percentage points, in nominal terms, and around 3 percentage points, in real terms. On impact, consumption rises by more than 3 percent, while output falls by more than 1 percentage point. The optimal foreign exchange intervention leans against the wind and reduces the appreciation by two-thirds, both in nominal and real terms. The central bank accumulates reserves equal to almost 10 percent of GDP in the 1 quarter after the shock, but while the accumulation of reserves is fast, the decumulation is much slower. Within two quarters of the shock, the central bank has absorbed almost entirely the foreign capital flow, while the remaining part is absorbed by domestic households. Hence, financial firms are not intermediating any flow and the exchange rate is completely stabilized.

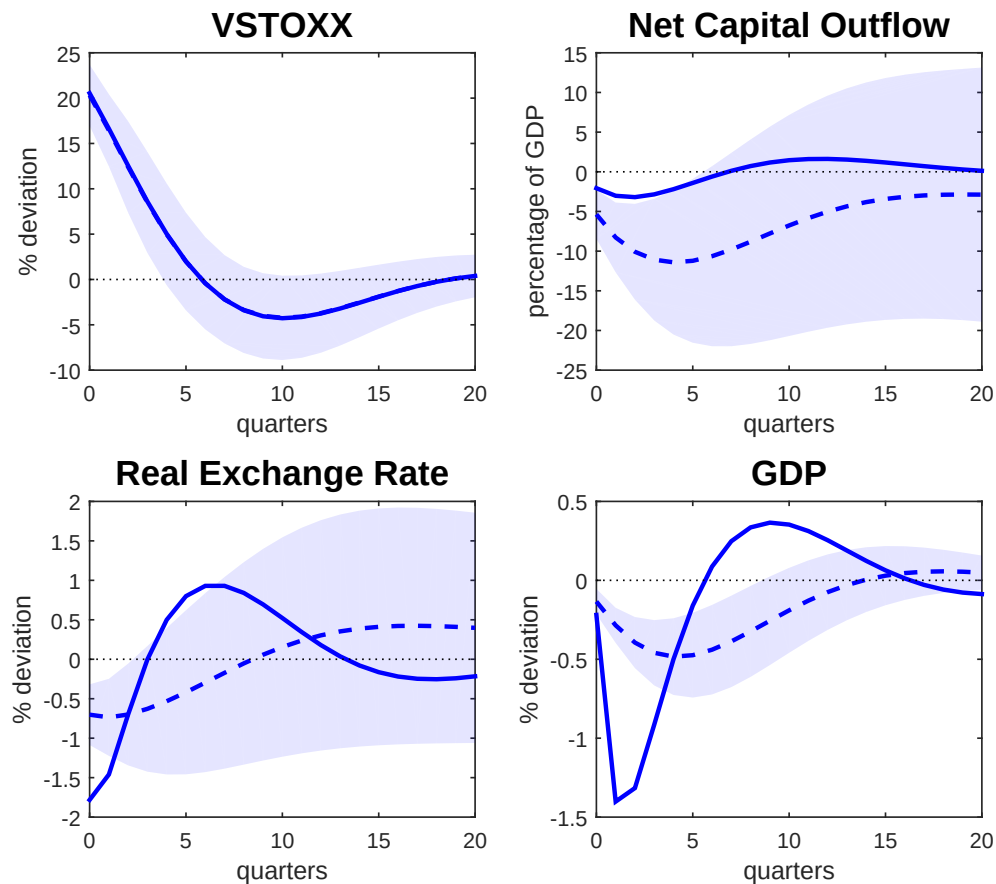
These results show that portfolio flow shocks generate macroeconomic fluctuations that quantitatively relevant, and the size of the optimal foreign exchange intervention is large. I believe these results provide convincing evidence and a reasonable quantification of the inefficiencies caused by portfolio flow shocks and strengthens the case for using foreign exchange intervention as an additional policy tool.

Table 2: Estimation Results

Parameter	Description	Value	S. E.
γ	Financial sector inverse aggregate risk-bearing capacity	0.194	0.082**
χ	Proportionality between vix and foreign demand for domestic assets	0.764	0.340**

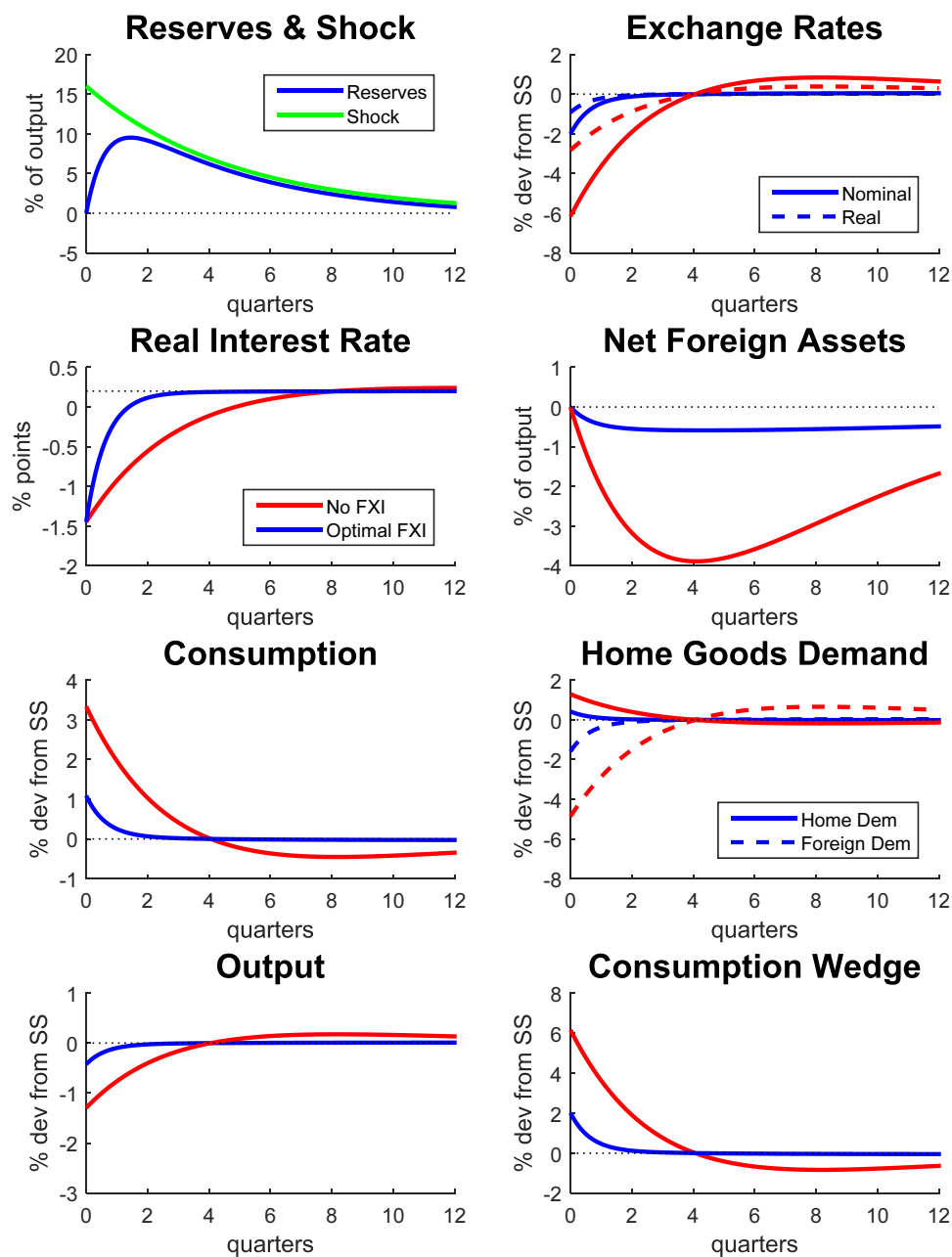
Note: *** indicates significance at the 1 percent level, ** at the 5 percent level, and * at the 10 percent level.

Figure 2: Model and VAR-based Impulse Responses



Model-based impulse responses (solid line), VAR-based impulse responses (dashed line) and 90 percent confidence intervals (shaded areas). Net capital outflow is expressed as cumulative deviation from the unshocked path in percentage of GDP. All other variables are expressed as percentage deviation from their unshocked path.

Figure 3: FX Intervention with Flexible Prices



Impulse responses for the model with flexible prices to a positive shock to foreign demand for Home assets. The blue line represents the allocation implemented by the optimal FX intervention, while the red line represents the allocation without intervention.

5 Optimal Foreign Exchange Intervention with Nominal Rigidities

In this section, I characterize the optimal foreign exchange intervention in the presence of nominal rigidities, and study its interaction with monetary policy. I will start by studying the case of rigid prices, as it allows me to focus on foreign exchange intervention alone. I will then solve for the optimal foreign exchange and monetary policies when prices are sticky.

5.1 Optimal Foreign Exchange Intervention with Rigid Prices

When prices are rigid the domestic terms of trade are given by $s = e$. If monetary policy is idle, that is if the central bank keeps the nominal interest rate at its steady state level, $i = \rho$, then $e = -\lambda$ and the output gap is $y = -\alpha\lambda$.²⁹ Similarly to the flexible price case, the objective function of the planner simplifies to

$$\min_{\hat{x}} \frac{1}{2} \int_0^{\infty} e^{-\rho t} \left(\phi_x \hat{x}(t) + \phi_R \lambda(t)^2 \right) dt$$

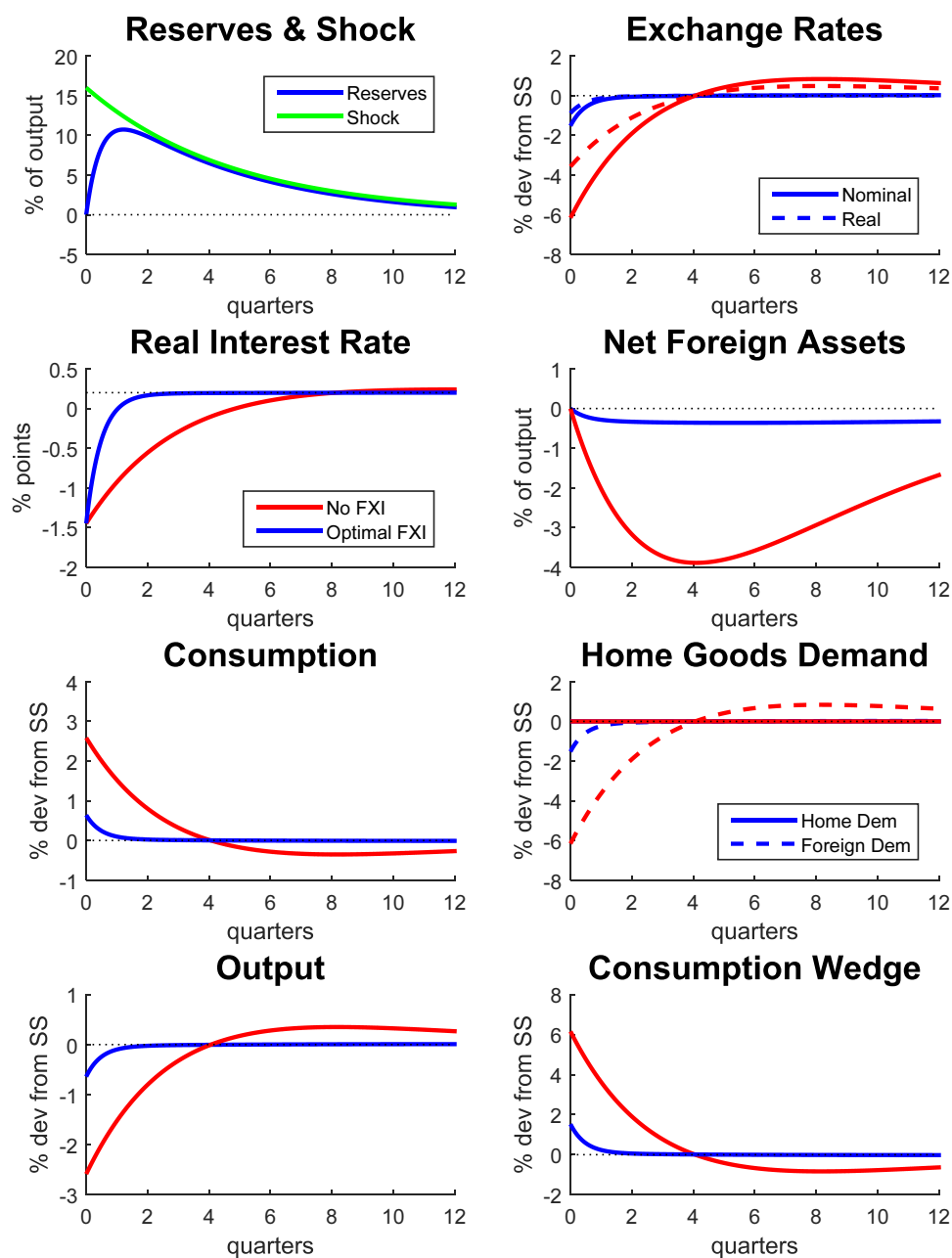
where the relative weight on the consumption wedge is given by $\phi_R \equiv \phi_\lambda + \phi_y \alpha^2$. The planner's problem is similar to the one solved under flexible prices. Hence, its solution has the structure described in Proposition (3) and the optimal foreign exchange intervention rule is the same as in Corollary 1, with $\phi = \phi_R$. The only difference between the flexible and rigid prices problems is the relative weight that the planner attaches to the stabilization of the consumption wedge. When prices are rigid, the welfare loss caused by fluctuations in λ is larger, since $\phi_R > \phi_F$. This is due to the inability of domestic producers to reduce their prices in response to the appreciation of the domestic currency. As the exchange rate falls, the foreign price of Home goods increases, and foreign demand falls. Domestic demand, on the other hand, remains constant as the shift toward foreign goods caused by the appreciation in the terms of trade is compensated by the increase in total demand induced by the fall in the real interest rate.³⁰ Now domestic prices remain too high not only because of the wealth effect on labor supply triggered by the consumption boom, but also because of the nominal friction that prevents domestic producers from adjusting them downwards. Hence, the fall in output is larger. Notice that the presence of nominal rigidities does not directly affect the consumption wedge, nor net foreign assets. Without intervention, the path of both variables is the same under rigid and flexible prices, as can be seen by inspecting (13) and (14). Figure (4) plots the allocation implemented by the optimal foreign exchange intervention with rigid prices, and compares it with the allocation that arises without intervention, both calibrated as described in Section 4.4. Following a portfolio inflow shock, the real exchange rate appreciates by almost 4 percent, compared to 3 percent under flexible prices.

With rigid prices, the central bank responds to an increase in the foreign demand for Home-currency assets by intervening in the foreign exchange market more aggressively, as showed by the comparative static analysis of the optimal foreign exchange intervention rule with respect to ϕ . When prices are flexible, the

²⁹Notice that, with rigid prices, the central bank could perfectly stabilize output at no cost, since producer inflation is zero. However here I want to capture situations in which monetary policy is not available, and foreign exchange intervention is used as a substitute tool.

³⁰The two effects have exactly the same magnitude, and opposite signs, when the elasticities of substitution across goods and across periods are identical and equal to one.

Figure 4: FX Intervention with Rigid Prices



Impulse responses for the model with rigid prices to a positive shock to foreign demand for Home assets. The blue line represents the allocation implemented by the optimal FX intervention, while the red line represents the allocation without intervention. In both cases, monetary policy is idle ($i = \rho$).

main goal of the intervention is to mitigate the expected depreciation of the exchange rate that follows the portfolio flow shock. By doing so it stabilizes domestic consumption and alleviates the inefficiencies described in Section 4.1. Since producers are allowed to adjust their prices freely, the nominal appreciation of the exchange rate *per se* does not generate any welfare loss. With nominal rigidities, however, the nominal appreciation reduces the competitiveness of domestic producers and causes a bigger fall in domestic output. Now the central bank has an additional reason for intervening as it wishes to stabilize not only the dynamic of the exchange rate, but also its level. By fighting the appreciation, the intervention reduces the foreign price of domestic goods and sustains foreign demand. This additional motive is the “monetary” aspect of foreign exchange intervention: the central bank leans against the wind to mitigate the effect of the portfolio flow shock on the competitiveness of domestic exports.

5.2 Optimal Monetary Policy and Foreign Exchange Intervention

When prices are sticky, and the central bank uses both available tools, its problem is to minimize 17 subject to 13, 14, 15, 16, and 18. Unfortunately, this problem does not admit a closed form solution and can be only solved numerically. In order to characterize the optimal foreign exchange intervention rule, I will study a sequential planner’s problem in which the central bank first chooses the optimal path for λ and then, taking it as given, chooses the optimal paths for y and π_H . The solution to the sequential problem approximates very well the solution to joint problem, as shown numerically in the appendix, and has a closed form solution. The optimal foreign exchange intervention policy rule is the same as the one described in Corollary (1).

Proposition 4. *The solution to the sequential problem*

$$\min_i \frac{1}{2} \int_0^\infty e^{-\rho t} \left(\phi_\pi \pi_H(t)^2 + \phi_y y(t)^2 \right) dt$$

subject to 15 and 16, where $\{\lambda(t)\}_{t \geq 0}$ is given by

$$\{\lambda(t)\}_{t \geq 0} = \arg \min_{\hat{x}} \frac{1}{2} \int_0^\infty e^{-\rho t} \left(\hat{x}(t)^2 + \phi \lambda(t)^2 \right) dt$$

subject to 13, 14, and 18, with $\hat{f}_H^*(0) = \varepsilon$ and $\hat{a}(0) = 0$, is

$$\begin{bmatrix} \pi_H(t) \\ \lambda(t) \end{bmatrix} = \begin{bmatrix} \frac{\kappa \frac{1+\varphi}{\rho+\iota}}{(\rho+\iota+\bar{\nu})(\rho+\iota+\bar{\nu})(\rho+\iota+\bar{\nu})} & 0 \\ \frac{-\alpha\gamma\iota\kappa}{(\rho+\iota+\bar{\nu})(\rho+\iota+\bar{\nu})(\rho+\iota+\bar{\nu})} & \frac{\gamma}{\rho+\iota+\bar{\nu}} \\ \kappa\xi \frac{\rho+\bar{\nu}}{\rho+\iota+\bar{\nu}} + \kappa(1-\xi) \frac{\rho+\bar{\nu}}{\rho+\iota+\bar{\nu}} & \frac{\xi}{\bar{\nu}}\gamma + \frac{1-\xi}{\bar{\nu}}\gamma \\ \frac{\alpha\gamma}{\bar{\nu}-\varrho} \frac{\rho\varrho+\varrho^2-\alpha\gamma}{\rho+\varrho+\bar{\nu}} \frac{\kappa}{\bar{\nu}-\varrho} \frac{\rho+\varrho}{\rho+\varrho+\bar{\nu}} + & \frac{\xi\gamma}{\rho+\varrho+\bar{\nu}} + \frac{(1-\xi)\gamma}{\rho+\varrho+\bar{\nu}} \\ \frac{1-\xi}{\bar{\nu}-\varrho} \frac{\alpha\gamma\kappa \frac{\rho+\bar{\nu}}{\rho+\iota+\bar{\nu}}}{\rho+\varrho+\bar{\nu}} + \frac{\xi}{\bar{\nu}-\varrho} \frac{\alpha\gamma\kappa \frac{\rho+\bar{\nu}}{\rho+\iota+\bar{\nu}}}{\rho+\varrho+\bar{\nu}} & \end{bmatrix}^\top \mathbf{s}(t)$$

while the states $\mathbf{s} = \begin{bmatrix} y & x & \hat{a} & \hat{f}_H^* \end{bmatrix}^\top$ evolve as $d\mathbf{s}(t) = -\mathbf{M}\mathbf{s}(t) dt$ where

$$\mathbf{M} = \begin{bmatrix} \iota & 0 & 0 & 0 \\ \frac{-\alpha\gamma\iota^2\frac{\rho+\iota}{1+\varphi}}{(\rho+\underline{\nu}+\bar{\nu})(\rho+\iota+\underline{\nu})(\rho+\iota+\bar{\nu})} & \xi\bar{\nu} + (1-\xi)\underline{\nu} & -\frac{\alpha\gamma}{\rho+\underline{\nu}+\bar{\nu}} & 0 \\ \iota\frac{\rho+\iota}{1+\varphi} \left(\xi\frac{\rho+\underline{\nu}}{\rho+\iota+\underline{\nu}} + (1-\xi)\frac{\rho+\bar{\nu}}{\rho+\iota+\bar{\nu}} \right) & -\frac{\phi\gamma^2}{\rho+\underline{\nu}+\bar{\nu}} & \xi\underline{\nu} + (1-\xi)\bar{\nu} & 0 \\ \frac{\iota\alpha\gamma}{\bar{\nu}-\varrho} \frac{\rho\varrho+\varrho^2-\alpha\gamma}{\rho+\varrho+\bar{\nu}} \frac{1}{\bar{\nu}-\varrho} \frac{\frac{\rho+\varrho}{\rho+\iota+\varrho}}{\rho+\varrho+\underline{\nu}} \frac{\rho+\iota}{1+\varphi} + \iota\alpha\gamma\frac{\rho+\iota}{1+\varphi} \left(\frac{1-\xi}{\bar{\nu}-\varrho} \frac{\rho+\underline{\nu}}{\rho+\iota+\underline{\nu}} + \frac{\xi}{\bar{\nu}-\varrho} \frac{\rho+\bar{\nu}}{\rho+\iota+\bar{\nu}} \right) & \frac{-\phi\gamma^2}{\rho+\varrho+\bar{\nu}} \frac{\rho+\varrho+\frac{\nu\bar{\nu}}{\rho+\underline{\nu}+\bar{\nu}}}{\rho+\varrho+\underline{\nu}} & \frac{\alpha\gamma\xi}{\rho+\varrho+\bar{\nu}} + \frac{\alpha\gamma(1-\xi)}{\rho+\varrho+\underline{\nu}} & \varrho \end{bmatrix}^\top$$

with $\mathbf{s}(0) = \begin{bmatrix} 0 & 0 & 0 & \varepsilon \end{bmatrix}^\top$. The parameter ι is defined in the appendix.

The blue line in Figure (5) depicts the allocation described in Proposition (4), where the relative weight on λ, ϕ , is chosen to minimize (17). When dealing with portfolio inflows, the central bank increases the supply of domestic bonds, and accumulates reserves, while simultaneously reduces the nominal interest rate. To understand the role played by each instrument and their interaction, it is instructive to focus on them separately.

The red line in Figure (5) depicts the allocation implemented by using monetary policy only ($\hat{x} = 0$). The analytical solution to this problem is reported in the appendix. Monetary policy alone cannot deal with the inefficiencies induced by fluctuations in the consumption path. In the previous sections, we have seen that a positive portfolio flow shock appreciates the exchange rate and reduces the domestic real interest rate. The latter effect triggers a consumption boom which reduces output and the present discounted value of the consumption stream. The appreciation of the domestic currency, on the other hand, increases the foreign price of domestic goods which, if prices are sticky, depresses domestic output even further. One might naively guess that, in analogy with the use of foreign exchange intervention, the central bank should tighten monetary policy to mitigate the consumption boom and its negative effects on domestic welfare. This logic however does not take into account the effects of such a policy on the price of domestic assets and their returns. An increase in the nominal interest rate would make domestic assets even more attractive for foreign investors. In order for markets to clear, the exchange rate would have to appreciate even more, depressing foreign demand further. Therefore, by simultaneously reducing domestic consumption and domestic demand, a tighter monetary policy would cause a bigger fall in output. As seen in the previous section, the consumption wedge and Home net foreign assets are independent of the degree of price stickiness and of the output gap. Hence, monetary policy is unable to affect them. By using the domestic and foreign demand schedules we can rewrite the consumption wedge as the gap between domestic and foreign demand for Home goods:

$$\lambda(t) = c_H(t) - c_H^*(t)$$

Monetary policy moves domestic and foreign demand in the same direction, therefore it has limited power in reducing the tension between them generated by the portfolio flow shock. In the special case considered in this paper, where the elasticities of substitution across time and across goods are equal to one, the effects of monetary policy on domestic and foreign demand have exactly the same magnitude. Thus, their ratio is

independent of the path of the nominal interest rate chosen by the central bank. This result has two striking implications. First, monetary policy cannot be used to maximize the present discounted value of the path of domestic consumption as it cannot simultaneously reduce consumption and depreciate the terms of trade. Second, its ability to reduce the output gap is limited by wages inflation triggered by the consumption boom. In response to a positive portfolio flow shock, the central bank reduces the nominal interest rate and reduces the attractiveness of domestic assets. This depreciates the exchange rate and helps domestic producers to reduce the foreign price of domestic goods. The expansionary monetary policy also increases domestic consumption and therefore domestic demand. This reduces the output gap even further but exacerbates wages inflation and eventually leads to positive PPI inflation. Due to their wealth effect on labor supply, portfolio flow shocks shift the Phillips curve into an inefficient position and worsen the output/inflation trade-off for monetary policy.

Foreign exchange intervention, when used in conjunction with monetary policy, plays two distinct roles. First, it manipulates the dynamics of the terms of trade to maximize the present discounted value of domestic consumption. Second, it helps monetary policy to achieve its goals by reducing the shift of the Phillips curve and improving the output/inflation trade-off. Therefore, foreign exchange intervention helps mitigating the recession caused by the capital inflow only indirectly, through its effect on the Phillips curve. This is in sharp contrast with the standard view of foreign exchange intervention which emphasizes their role in managing foreign demand for domestic goods. The monetary aspect of foreign exchange intervention arises only when monetary policy is not available, for example because the zero lower bound is binding. Only then, foreign exchange intervention should also be used to fight the exchange rate appreciation with the goal of sustaining foreign demand, as we have seen in the previous section.

In dealing with the effects of portfolio flow shocks, the central bank optimally uses both instruments. Foreign exchange intervention and monetary policy are complementary tools rather than substitutes. The effectiveness of monetary policy is independent of foreign exchange intervention only in two extreme cases. When $\theta \rightarrow 0$ and prices become fully rigid, the central bank can independently use monetary policy to fully stabilize output, since there is no inflation cost. This is also true when an appropriate state-contingent labor subsidy is in place to offset the wealth effect on labor supply. In this case the divine coincidence holds and monetary policy alone can simultaneously achieve zero inflation and zero output gap.

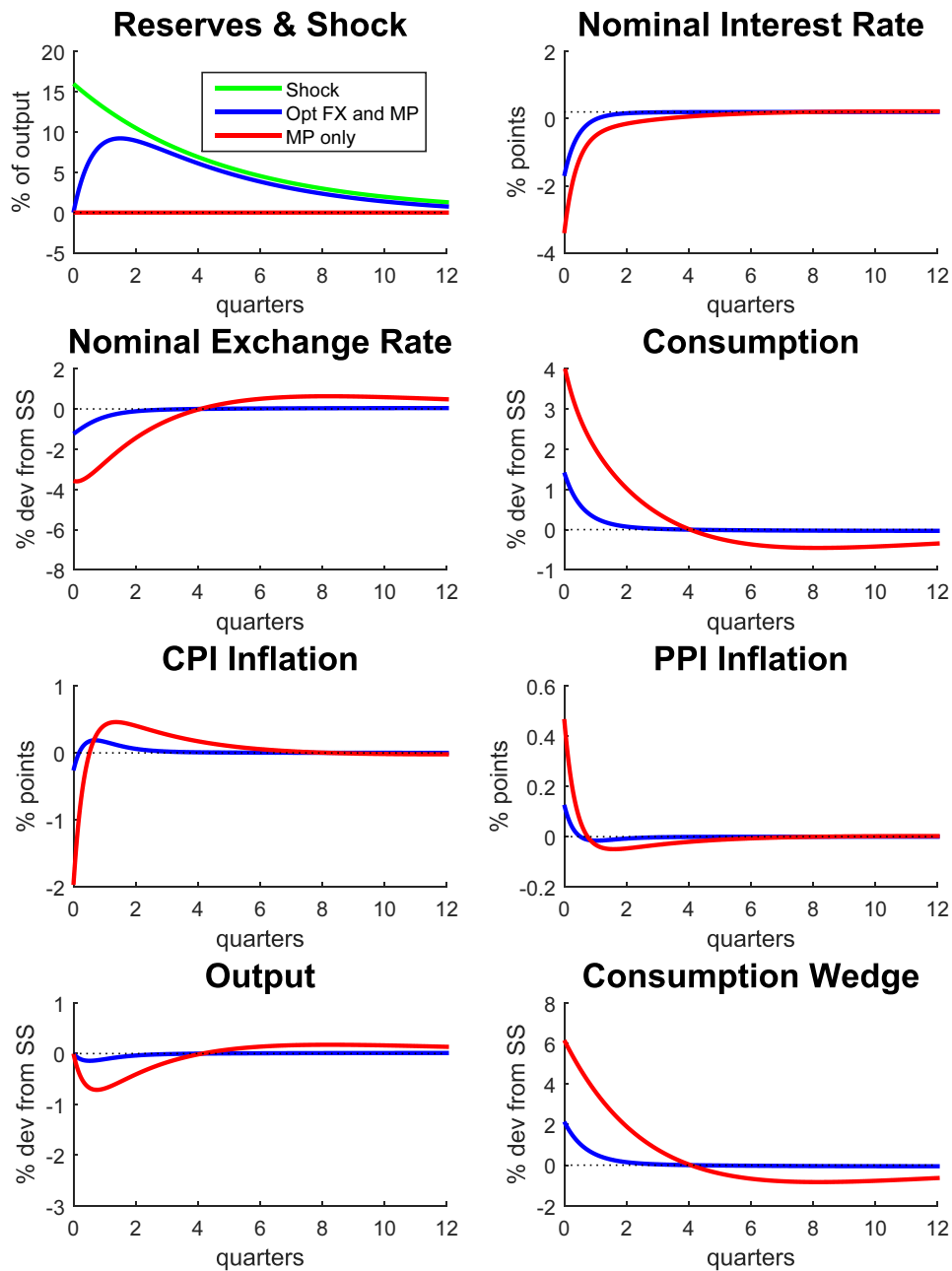
6 Conclusions

The objective of this paper is to study the desirability of foreign exchange intervention in response to portfolio flow shocks. I considered a New Keynesian model of a small open economy where international financial markets are imperfect and the exchange rate is determined by capital flows. I used this framework to study the effects of exchange rate fluctuations driven by portfolio flow shocks and to characterize the optimal foreign exchange intervention policy rule. When financial intermediation across countries is not frictionless, portfolio flow shocks cause inefficient exchange rate fluctuations that trigger boom-bust cycles in the domestic economy. The optimal policy response is to partially stabilize these fluctuations using both foreign exchange intervention and monetary policy. The optimal foreign exchange intervention leans against the

wind and stabilizes the path of the exchange rate: following an increase in the foreign demand for domestic assets, the central bank increases their net supply and accumulates foreign reserves. By doing so, the central bank stabilizes the path of the exchange rate and smooths out fluctuations in domestic consumption. Simultaneously, the central bank reduces the nominal interest rate to reduce the relative price of domestic goods and mitigate the output gap. The optimal foreign exchange intervention policy can be implemented by a simple rule which is a function of three endogenous targets: the consumption wedge, the domestic net foreign asset position, and the level of accumulated foreign reserves. Finally, using Swiss data I provided empirical evidence of the inefficiencies caused by portfolio flow shocks and quantify them by estimating the key parameters of the model. The empirical impulse responses show that domestic variables respond significantly to fluctuations in foreign demand for Swiss assets, and the sign of the response is consistent with the prediction of the theoretical model. Furthermore, the estimated value of the financial friction implies that the inefficiencies associated with these fluctuations are quantitatively relevant and the size of the optimal foreign exchange intervention is large.

These results shed light on the surge in foreign exchange interventions observed since the onset of the global financial crisis, in emerging and advanced economies alike. Foreign exchange intervention is an additional tool for central banks which does not substitute for monetary policy, but rather complements it when shocks in international financial markets spill over and threatens domestic stability.

Figure 5: FX Intervention and Monetary Policy with Sticky Prices



Impulse responses for the model with sticky prices to a positive shock to foreign demand for Home assets. The blue line represents the allocation implemented by the joint use of FX intervention and monetary policy (sequential problem), and the red line represents the allocation implemented by the optimal monetary policy ($\hat{x} = 0$) alone.

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Appendix

A The Household's Problem

The representative Home household maximizes

$$\mathbb{E} \left[\int_0^\infty e^{-\rho t} \left(\frac{C^{1-\omega}}{1-\omega} - \frac{L^{1+\varphi}}{1+\varphi} \right) dt \right]$$

subject to

$$dA = (WL - PC) dt + A(\mu_A dt + \sigma_A dZ)$$

with $A(\mu_A dt + \sigma_A dZ) = A i dt + \mathcal{E}F[(i^* - i + \mu_\mathcal{E}) dt + \sigma_\mathcal{E} dZ] + dM + dR + dT$, where

$$C \equiv \left[(1 - \alpha)^{\frac{1}{\eta}} C_H^{\frac{\eta-1}{\eta}} + \alpha^{\frac{1}{\eta}} C_F^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}$$

and $C_F \equiv \exp \int_0^1 \ln C_i di$. The intratemporal first order conditions yield

$$C_H = (1 - \alpha) \left(\frac{P_H}{P} \right)^{-\eta} C \quad C_F = \alpha \left(\frac{P_F}{P} \right)^{-\eta} C \quad C_i = \left(\frac{P_i}{P_F} \right)^{-1} C_F$$

$\forall i \in [0, 1]$. Home's CPI is defined as $P \equiv \left[(1 - \alpha) P_H^{1-\eta} + \alpha P_F^{1-\eta} \right]^{\frac{1}{1-\eta}}$. The price of imported goods is given by $P_F \equiv \exp \int_0^1 \ln P_i di$. The intertemporal household's problem is:

$$V(A, \mathbf{S}) = \max_{C, L} \mathbb{E} \left[\int_0^\infty e^{-\rho t} \left(\frac{C^{1-\omega}}{1-\omega} - \frac{L^{1+\varphi}}{1+\varphi} \right) dt \right]$$

subject to

$$dA = (WL - PC) dt + A(\mu_A dt + \sigma_A dZ)$$

$$d\mathbf{S} = \mathbf{S} \mu_\mathbf{S} dt + \mathbf{S} \sigma_\mathbf{S} dZ$$

where $\mathbf{S}$ is a generic vector of states and μ_A and σ_A are a function of the states only. The HJB for this problem is

$$\rho V = \sup_{C, L} \left[\frac{C^{1-\omega}}{1-\omega} - \frac{L^{1+\varphi}}{1+\varphi} + \mathcal{A}(V) \right]$$

where

$$\begin{aligned} \mathcal{A}(V) = & V_A (WL - PC + A\mu_A) + \sum_i V_{S_i} S_i \mu_{S_i} \\ & + \frac{1}{2} \left(V_{AA} A^2 \sigma_A^2 + 2 \sum_i A S_i \sigma_A \sigma_{S_i} V_{AS_i} + \sum_i \sum_j V_{S_i S_j} S_i S_j \sigma_{S_i} \sigma_{S_j} \right) \end{aligned}$$

is the infinitesimal generator operator. The first order conditions with respect to C and L are:

$$C^\omega P = \frac{1}{V_A}$$

$$L^\varphi = V_A W$$

Derive both sides of the HJB with respect to A to obtain the law of motion of V_A :

$$\frac{dV_A}{V_A} = (\rho - i) dt + \sigma_{V_A} dZ$$

where $\sigma_{V_A} = \frac{V_{AA}}{V_A} A \sigma_A + \sum_S \frac{V_{AS}}{V_A} S \sigma_S$. Finally we can apply Ito's Lemma to the first order condition to derive the Euler equation

$$dC = \frac{1}{\omega} \left(i - \mu_P - \rho + \sigma_P^2 + \sigma_C \sigma_P + \frac{1 + \omega}{2\omega} \sigma_C^2 \right) dt + \frac{1}{\omega} \sigma_C dZ$$

where $\sigma_C = -\sigma_{V_A} - \sigma_P$.

B Calvo Pricing in Continuous Time

B.1 Producer Price Dynamics

Domestic producer price indexes are defined by

$$P_k(t) \equiv \left[\int_0^1 P_{H,j}(t)^{1-\epsilon} dj \right]^{\frac{1}{1-\epsilon}}$$

In order to derive its law of motion, let's write it down in discrete time first and then take the limit as the length of the time interval goes to zero. The length of a period is $[t, t + dt)$. During a period a fraction $1 - e^{-\theta dt}$ of firms receive the Calvo signal that will allow them to set a new price at time $t + dt$. The remaining fraction $e^{-\theta dt}$ will not be able to change price and will be stuck with the price posted at t . Let $S(t)$ be the set of firms not re-optimizing their posted price at time t . Using the fact that all firms resetting prices will choose an identical price $\bar{P}_H(t)$ we obtain

$$\begin{aligned} P_H(t + dt) &= \left[\int_0^1 P_{H,j}(t + dt)^{1-\epsilon} dj \right]^{\frac{1}{1-\epsilon}} = \left[\int_{S(t)} P_{H,j}(t)^{1-\epsilon} dj + (1 - e^{-\theta dt}) \bar{P}_H(t + dt)^{1-\epsilon} \right]^{\frac{1}{1-\epsilon}} \\ &= \left[e^{-\theta dt} P_H(t)^{1-\epsilon} + (1 - e^{-\theta dt}) \bar{P}_H(t + dt)^{1-\epsilon} \right]^{\frac{1}{1-\epsilon}} \end{aligned}$$

where the last equality follows from the fact that the distribution of prices among firms not adjusting at time $t + dt$ corresponds to the distribution of posted prices at time t , though with a total mass reduced to $e^{-\theta dt}$.

The equation above can be rewritten as

$$P_H(t+dt)^{1-\epsilon} - P_H(t)^{1-\epsilon} = -\left(1 - e^{-\theta dt}\right) P_H(t)^{1-\epsilon} + \left(1 - e^{-\theta dt}\right) \bar{P}_H(t+dt)^{1-\epsilon}$$

Now, we cannot take the the limit as $dt \rightarrow 0$ since the last term is not time t measurable. Therefore add and subtract $(1 - e^{-\theta dt}) \bar{P}_H(t)^{1-\epsilon}$ to obtain

$$P_H(t+dt)^{1-\epsilon} - P_H(t)^{1-\epsilon} = \left(1 - e^{-\theta dt}\right) \left(\bar{P}_H(t)^{1-\epsilon} - P_H(t)^{1-\epsilon}\right) + \left(1 - e^{-\theta \Delta t}\right) \Delta \bar{P}_H(t)^{1-\epsilon}$$

Now Taylor expand $e^{-\theta dt}$ around $dt = 0$ and take the limit for $dt \rightarrow 0$ to obtain

$$\pi_H(t) = \frac{dP_H(t)}{P_H(t)} = \frac{\theta}{1-\epsilon} \left[\left(\frac{\bar{P}_H(t)}{P_H(t)} \right)^{1-\epsilon} - 1 \right] dt$$

-

B.2 Price Dispersion

The aggregate loss of efficiency induced by price dispersion among firms is $\Delta(t) \equiv \int_0^1 \left[\frac{P_{H,j}(t)}{P_H(t)} \right]^{-\epsilon} dj$. Its law of motion can be derived as in the previous section. Let's write the discrete time analogue and appropriately shrink the length of the period

$$\begin{aligned} \Delta(t+dt) &= \int_0^1 \left[\frac{P_{H,j}(t+dt)}{P_H(t+dt)} \right]^{-\epsilon} dj \\ &= P_H(t+dt)^\epsilon \left[\int_{S(t)} P_{H,j}(t)^{-\epsilon} dj + \left(1 - e^{-\theta dt}\right) \bar{P}_{Hk}(t+dt)^{-\epsilon} \right] \\ &= \left(\frac{P_H(t+dt)}{P_H(t)} \right)^\epsilon \int_{S(t)} \left(\frac{P_{H,j}(t)}{P_H(t)} \right)^{-\epsilon} dj + \left(1 - e^{-\theta dt}\right) P_H(t+dt)^\epsilon \bar{P}_H(t+dt)^{-\epsilon} \\ &= \left(\frac{P_H(t+dt)}{P_H(t)} \right)^\epsilon e^{-\theta dt} \Delta(t) + \left(1 - e^{-\theta dt}\right) P_H(t+dt)^\epsilon \bar{P}_H(t+dt)^{-\epsilon} \end{aligned}$$

As before we add and subtract the last term lagged, Taylor expand the exponential terms and take the limit as $dt \rightarrow 0$ to obtain

$$d\Delta(t) = \left[\theta \left(\frac{\bar{P}_H(t)}{P_H(t)} \right)^{-\epsilon} + \Delta(t) (\epsilon \pi_H(t) - \theta) \right] dt$$

or

$$d\Delta(t) = \left[\theta \left(1 - \frac{\epsilon-1}{\theta} \pi_H(t) \right)^{\frac{\epsilon}{\epsilon-1}} + \Delta(t) (\epsilon \pi_H(t) - \theta) \right] dt$$

B.3 Optimal Price Setting

A measure one of monopolistic firms (indexed by $j \in [0, 1]$) engage in infrequent price setting a la Calvo. Each firm re-optimizes its price $P_{H,j}(t)$ only at discrete dates determined by a Poisson process with intensity

θ . The time δ between two re-optimizations is distributed according to the exponential density: $\theta e^{-\theta\delta}$. A firm that is allowed to re-optimize its price at time t maximizes the present discounted value of future profits³¹:

$$\max_{\bar{P}_{H,j}(t)} \mathbb{E} \left[\int_t^\infty \frac{P(t)}{C(t)^{-\omega}} \frac{C(u)^{-\omega}}{P(u)} e^{-(\rho+\theta)(u-t)} \{ \bar{P}_{H,j}(t) Y_j(u|t) - \mathcal{C}_H(Y_j(u|t)) \} du \right]$$

subject to the demand schedule

$$Y_j(u|t) = \left[\frac{\bar{P}_{H,j}(t)}{P_H(u)} \right]^{-\epsilon} Y(u)$$

where $\mathcal{C}(\cdot)$ is the firm's nominal cost function. The first-order condition associated with the problem is

$$\mathbb{E} \left[\int_t^\infty \frac{P(t)}{C(t)^{-\omega}} \frac{C(u)^{-\omega}}{P(u)} e^{-(\rho+\theta)(u-t)} Y_j(u|t) \{ \bar{P}_{H,j}(t) - \mathcal{M}MC(Y_j(u|t)) \} du \right] = 0$$

$$\bar{P}_{H,j}(t) = P_H(t) \frac{\mathbb{E} \left[\int_t^\infty \frac{P(t)}{C(t)^{-\omega}} \frac{C(u)^{-\omega}}{P(u)} e^{-(\rho+\theta)(u-t)} \left[\frac{P_k^*(t)}{P_k^*(u)} \right]^{-\epsilon} Y^*(u) \mathcal{M}MC(u) du \right]}{\mathbb{E} \left[\int_t^\infty \frac{P(t)}{C(t)^{-\omega}} \frac{C(u)^{-\omega}}{P(u)} e^{-(\rho+\theta)(u-t)} \left[\frac{P_k^*(t)}{P_k^*(u)} \right]^{1-\epsilon} P^*(u) \mathcal{E}(u) Y^*(u) du \right]}$$

where MC is the nominal marginal cost function and $\mathcal{M} \equiv \frac{\epsilon}{\epsilon-1}$. Note that in the limiting case of no price rigidities ($\theta \rightarrow \infty$), this condition collapses to the familiar optimal price-setting condition under flexible prices $P_{H,j}(t) = \mathcal{M}MC(Y_j(t))$.

The firm's cost function is $\mathcal{C}(Y_j(u|t)) = Y_j(u|t) (1 - \tau) W(u)$, therefore the nominal marginal cost is

$$MC(Y_j(u|t)) = (1 - \tau) W(u) \equiv MC(u)$$

The FOC can be rewritten as

$$\bar{P}_{H,j}(t) = P_H(t) \frac{\mathbb{E} \left[\int_t^\infty \left\{ e^{-\int_t^u [\rho+\theta-\epsilon\pi_H(s)] ds} \frac{C(u)^{-\omega}}{P(u)} \mathcal{M}MC(u) Y(u) \right\} du \right]}{\mathbb{E} \left[\int_t^\infty \left\{ e^{-\int_t^u [\rho+\theta-(\epsilon-1)\pi_H(s)] ds} \frac{C(u)^{-\omega}}{P(u)} P_H(u) Y_H(u) \right\} du \right]}$$

where I used the result that the dynamics of the price level is locally deterministic. Let

$$\mathcal{U}(t) \equiv \mathbb{E} \left[\int_t^\infty \left\{ e^{-\int_t^u [\rho+\theta-\epsilon\pi_H(s)] ds} \frac{C(u)^{-\omega}}{P(u)} \mathcal{M}MC(u) Y(u) \right\} du \right]$$

$$\mathcal{V}(t) \equiv \mathbb{E} \left[\int_t^\infty \left\{ e^{-\int_t^u [\rho+\theta-(\epsilon-1)\pi_H(s)] ds} \frac{C(u)^{-\omega}}{P(u)} P_H(u) Y_H(u) \right\} du \right]$$

then, the Feynman-Kac representation formula establishes that $\mathcal{U}$ and $\mathcal{V}$ are the unique solutions to the partial

³¹We assume that firms commit to supply whatever quantity demanded at the posted price, even if that implies negative profits

differential equations

$$(\rho + \theta - \epsilon \pi_H(t)) \mathcal{U}(t) = \mathcal{A}(\mathcal{U}) + \frac{C(t)^{-\omega}}{P(t)} \mathcal{M} \mathcal{M} C(t) Y(t)$$

$$[\rho + \theta - (\epsilon - 1) \pi_H(t)] \mathcal{V}(t) = \mathcal{A}(\mathcal{V}) + \frac{C(t)^{-\omega}}{P(t)} P_H(t) Y(t)$$

where the operator $\mathcal{A}$ is the infinitesimal generator of the stochastic process, defined as $\mathcal{A}f = \mu^x \nabla_x f(x) + \frac{1}{2} \text{tr} \left[\sigma^x H(f) (\sigma^x)^T \right]$. Hence, their laws of motion are

$$\frac{d\mathcal{U}(t)}{\mathcal{U}(t)} = \left[\rho + \theta - \epsilon \pi_H(t) - \frac{C(t)^{-\omega}}{P(t)} \frac{\mathcal{M} \mathcal{M} C(t) Y(t)}{\mathcal{U}(t)} \right] dt + \sigma_{\mathcal{U}}(t) dZ(t)$$

$$\frac{d\mathcal{V}(t)}{\mathcal{V}(t)} = \left[\rho + \theta - (\epsilon - 1) \pi_H(t) - \frac{C(t)^{-\omega}}{P(t)} \frac{P_H(t) Y(t)}{\mathcal{V}(t)} \right] dt + \sigma_{\mathcal{V}}(t) dZ(t)$$

Finally, we can use $\pi_H(t) = \frac{\theta}{\epsilon - 1} \left[1 - \left(\frac{\mathcal{U}(t)}{\mathcal{V}(t)} \right)^{1-\epsilon} \right]$ and the laws of motion for $\mathcal{U}$ and $\mathcal{V}$ to derive the law of motion of PPI inflation

$$\begin{aligned} d\pi_H(t) = & [(\epsilon - 1) \pi_H(t) - \theta] \left[\pi_H(t) + \frac{C(t)^{-\omega}}{P(t)} Y(t) \left(\frac{\mathcal{M} \mathcal{M} C(t)}{\mathcal{U}(t)} - \frac{P_H(t)}{\mathcal{V}(t)} \right) \right] dt \\ & + [(\epsilon - 1) \pi_H(t) - \theta] \left[\left(\epsilon \frac{\sigma_{\pi}(t)^2}{2} - \sigma_{\pi}(t) \sigma_{\mathcal{V}}(t) \right) dt + \sigma_{\pi}(t) dZ(t) \right] \end{aligned}$$

where $\sigma_{\pi}(t) = \sigma_{\mathcal{V}}(t) - \sigma_{\mathcal{U}}(t)$.

C Log-linearization

The deterministic laws of motion that describe the equilibrium dynamics are

$$\begin{aligned} d\Lambda &= -\gamma \Lambda \left(\hat{A} + \hat{F}_H^* + \hat{X}_H \right) dt \\ d\hat{A} &= \left[(i - \pi_H) \hat{A} - \gamma \left(\hat{A} + \hat{F}_H^* + \hat{X}_H \right) \left((1 - \beta) \hat{X}_H - \beta \left(\hat{A} + \hat{F}_H^* \right) \right) + 1 - \frac{\mathcal{S} C}{\mathcal{Q} Y} \right] dt - \hat{A} \frac{dY}{Y} \\ dY &= Y \frac{dY^*}{Y^*} + \frac{1}{\omega} [Y - \alpha \mathcal{S}^{\eta} Y^*] \frac{d\Lambda}{\Lambda} + \left[\eta Y + \left(\frac{1}{\omega} - \eta \right) (1 - \alpha) (Y - \alpha \mathcal{S}^{\eta} Y^*) \left(\frac{\mathcal{S}}{\mathcal{Q}} \right)^{\eta-1} \right] \frac{d\mathcal{S}}{\mathcal{S}} \\ d\mathcal{S} &= \mathcal{S} (i - \rho - \pi_H) dt - \mathcal{S} \omega \frac{dY^*}{Y^*} - \mathcal{S} \frac{d\Lambda}{\Lambda} \\ d\pi_H &= [(\epsilon - 1) \pi_H - \theta] \left[\pi_H + \frac{1}{\mathcal{V}} \left(\mathcal{M} (1 - \tau) \Delta^{\varphi} Y^{1+\varphi} \left(1 - \frac{\epsilon - 1}{\theta} \pi_H \right)^{-\frac{1}{1-\epsilon}} - Y C^{-\omega} \frac{\mathcal{Q}}{\mathcal{S}} \right) \right] dt \\ d\mathcal{V} &= \left[(\rho + \theta - (\epsilon - 1) \pi_H) \mathcal{V} - C^{-\omega} \frac{P_H Y}{P} \right] dt \end{aligned}$$

$$\begin{aligned}
d\Delta &= \left[\theta \left(1 - \frac{\epsilon - 1}{\theta} \pi_H \right)^{\frac{\epsilon}{\epsilon-1}} + \Delta (\epsilon \pi_H - \theta) \right] dt \\
dY^* &= \frac{1}{\omega} Y^* (i^* - \rho - \pi^*) dt
\end{aligned}$$

with

$$\begin{aligned}
C &= (\Lambda \mathcal{Q})^{\frac{1}{\omega}} C^* \\
\mathcal{Q} &= [\alpha + (1 - \alpha) \mathcal{S}^{\eta-1}]^{\frac{1}{\eta-1}} \\
\mathcal{S} &= \left(\frac{Y}{C^*} \right)^{\frac{1}{\eta}} \left[(1 - \alpha) \Lambda^{\frac{1}{\omega}} \mathcal{Q}^{\frac{1}{\omega}-\eta} + \alpha \right]^{-\frac{1}{\eta}}
\end{aligned}$$

The log linearized versions are

$$\begin{aligned}
d\lambda &= -\gamma (\hat{a} + \hat{f}_H^* - \hat{x}) dt \\
d\hat{a} &= [\rho \hat{a} - (s - q + c - y)] dt \\
dy &= dy^* + \frac{1-\alpha}{\omega} d\lambda + \left[\eta + \left(\frac{1}{\omega} - \eta \right) (1 - \alpha)^2 \right] ds \\
ds &= (i - \rho - \pi_H) dt - \omega dy^* - d\lambda \\
d\pi_H &= \rho \pi_H - \theta (\rho + \theta) (\varphi y + \omega c + \alpha s) dt \\
dy^* &= \frac{1}{\omega} (i^* - \rho - \pi^*) dt
\end{aligned}$$

and

$$\begin{aligned}
c &= y^* + \frac{1}{\omega} (\lambda + q) \\
q &= (1 - \alpha) s \\
s &= \frac{1}{\eta} (y - y^*) - \frac{1 - \alpha}{\eta} \left(\frac{1}{\omega} \lambda + \left(\frac{1}{\omega} - \eta \right) q \right)
\end{aligned}$$

where we used the first order approximation $\hat{x}_H = -\hat{x}$. Therefore we obtain

$$\begin{aligned}
d\lambda &= -\gamma (\hat{a} + \hat{f}_H^* - \hat{x}) dt \\
d\hat{a} &= \left(\rho \hat{a} - \frac{1}{\omega} \lambda - \frac{1 - \alpha (1 - \omega)}{\omega} s - y^* + y \right) dt \\
dy &= dy^* + \frac{1-\alpha}{\omega} d\lambda + \left[\eta + \left(\frac{1}{\omega} - \eta \right) (1 - \alpha)^2 \right] ds \\
ds &= (i - \rho - \pi_H) dt - \omega dy^* - d\lambda \\
d\pi_H &= \rho \pi_H - \kappa (\varphi y + \omega y^* + \lambda + s) dt \\
dy^* &= \frac{1}{\omega} (i^* - \rho - \pi^*) dt
\end{aligned}$$

where $\kappa \equiv \theta (\rho + \theta)$, with

$$s = \frac{\omega (y - y^*) - (1 - \alpha) \lambda}{\omega \eta + (1 - \omega \eta) (1 - \alpha)^2}$$

Now assume $\omega = \eta = 1$. Then

$$\begin{aligned} d\lambda &= -\gamma \left(\hat{a} + \hat{f}_H^* - \hat{x} \right) dt \\ d\hat{a} &= (\rho \hat{a} - \alpha \lambda) dt \\ dy &= (i - \rho - \pi_H) dt - \alpha d\lambda \\ d\pi_H &= \rho \pi_H - \kappa (1 + \varphi) y - \kappa \alpha \lambda dt \end{aligned}$$

and y^* does not affect equilibrium variables. The welfare function is

$$\mathbb{W} = \int_0^\infty e^{-\rho t} \left[\ln C - \frac{L^{1+\varphi}}{1+\varphi} \right] dt$$

where

$$C = \Lambda \mathcal{S}^{1-\alpha} C^* \quad L = \Delta Y$$

and

$$\mathcal{S} = \frac{Y_H}{[\alpha + (1 - \alpha) \Lambda] C^*}$$

Log-linearize both C and L to obtain

$$\begin{aligned} \ln C &= t.i.p. + \ln \Lambda + (1 - \alpha) \ln \mathcal{S} \\ &= t.i.p. + \ln \Lambda - (1 - \alpha) \ln [\alpha + (1 - \alpha) \Lambda] - (1 - \alpha) (1 - \beta) \ln Y \\ &\simeq t.i.p. + \alpha (2 - \alpha) \lambda - \alpha (1 - \alpha)^2 \frac{1}{2} \lambda^2 + (1 - \alpha) y \end{aligned}$$

$$\frac{L^{1+\varphi}}{1+\varphi} \simeq t.i.p. + (1 - \alpha) (\delta + y) + (1 + \varphi) \frac{1 - \alpha}{2} y^2$$

Thus

$$\mathbb{W} = \int_0^\infty e^{-\rho t} \left[t.i.p. + \alpha (2 - \alpha) \lambda - \frac{\alpha(1-\alpha)^2}{2} \lambda^2 - (1 - \alpha) \frac{\epsilon}{2\kappa} \pi_H^2 - (1 - \alpha) \frac{1+\varphi}{2} y^2 \right] dt$$

Now use the following second order approximation to the budget constraint

$$\int_0^\infty e^{-\rho t} [\alpha \lambda + \gamma (1 - \beta) \hat{x}^2] dt = 0$$

$$y = \gamma (1 - \beta) \frac{2 - \alpha}{1 - \alpha} \hat{x}^2$$

to replace the linear term in $\mathbb{W}$ and obtain

$$\mathbb{W} = \int_0^\infty e^{-\rho t} \left[\begin{array}{c} t.i.p. - \gamma (2 - \alpha) (1 - \beta) \hat{x}^2 - \frac{\alpha(1-\alpha)^2}{2} \lambda^2 \\ - (1 - \alpha) \frac{\epsilon}{2\kappa} \pi_H^2 - (1 - \alpha) \frac{1+\varphi}{2} y^2 \end{array} \right] dt$$

Finally, the loss function is

$$\mathbb{L} = \frac{1}{2} \int_0^\infty e^{-\rho t} (\phi_x \hat{x}^2 + \phi_\lambda \lambda^2 + \phi_\pi \pi_H^2 + \phi_y y^2) dt$$

where

$$\begin{aligned} \phi_x &= 2\gamma (2 - \alpha) (1 - \beta) \\ \phi_\lambda &= \alpha (1 - \alpha)^2 \\ \phi_\pi &= \frac{\epsilon}{\kappa} (1 - \alpha) \\ \phi_y &= (1 - \alpha) (1 + \varphi) \end{aligned}$$

D Proofs

Proof of Proposition 1

Proof. Under flexible prices, the deterministic planner problem is

$$\max_{\hat{X}_H, C, Y, S, Q} \int_0^\infty e^{-\rho t} \left(\frac{C^{1-\omega}}{1-\omega} - \frac{Y^{1+\varphi}}{1+\varphi} \right) dt$$

subject to

$$\begin{aligned} C &= (\Lambda Q)^{\frac{1}{\omega}} C^* \\ Y &= C^* S^\eta \left[(1-\alpha) \Lambda^{\frac{1}{\omega}} Q^{\frac{1}{\omega}-\eta} + \alpha \right] \\ Q &= [\alpha + (1-\alpha) S^{\eta-1}]^{\frac{1}{\eta-1}} \end{aligned}$$

and the laws of motion

$$\begin{aligned} d\Lambda &= -\gamma \Lambda (\hat{A} + \hat{X}_H) dt \\ d\hat{A} &= \hat{A} \left\{ i - \pi_H - \frac{1}{\omega} \left[\omega \eta + (1-\omega \eta) (1-\alpha) \left(1 - \alpha S^\eta \frac{Y^*}{Y} \right) \left(\frac{S}{Q} \right)^{\eta-1} \right] (i - \rho - \pi_H) \right\} dt \\ &\quad \left\{ \left(1 - \frac{S}{Q} \frac{C}{Y} \right) - \gamma (\hat{A} + \hat{X}_H) \left[(1-\beta) \hat{X}_H + \hat{A} (\eta - \beta) \right] \right\} dt \\ &\quad - \frac{\gamma}{\omega} \hat{A} (\hat{A} + \hat{X}_H) \left((1-\omega \eta) (1-\alpha) \left(\frac{S}{Q} \right)^{\eta-1} - 1 \right) \left(1 - \alpha S^\eta \frac{Y^*}{Y} \right) dt \end{aligned}$$

Notice that in steady state we must have $\hat{X}_H = \hat{A}$. This implies that, in steady state, the central bank uses FX intervention only to correct the financial friction and not to permanently alter the terms of trades or the real exchange rate. Therefore, in a symmetric steady state we have $\hat{X}_H = \hat{A} = 0$ and $\Lambda = Q = S = 1$. The Lagrangian is

$$\begin{aligned} \mathcal{L} &= \frac{C^{1-\omega}}{1-\omega} - \frac{Y^{1+\varphi}}{1+\varphi} + \lambda_C \left[C - (\Lambda Q)^{\frac{1}{\omega}} C^* \right] + \lambda_Q \left\{ Q - [\alpha + (1-\alpha) S^{\eta-1}]^{\frac{1}{\eta-1}} \right\} \\ &\quad + \lambda_Y \left[Y - C^* S^\eta \left[(1-\alpha) \Lambda^{\frac{1}{\omega}} Q^{\frac{1}{\omega}-\eta} + \alpha \right] \right] + \lambda_A \mu_A + \lambda_\Lambda \mu_\Lambda \end{aligned}$$

The FOCs evaluated at the symmetric steady state are

$$\begin{aligned} 0 &= Y^{1-\omega} + Y \lambda_C - \lambda_A \\ 0 &= -Y^{1+\varphi} + Y \lambda_Y + \lambda_A \\ 0 &= -Y \lambda_C \frac{1}{\omega} + \lambda_Y (\alpha - 1) Y \left(\frac{1}{\omega} - \eta \right) + \lambda_Q + \lambda_A \\ 0 &= -Y \lambda_Y \eta - \lambda_Q (1-\alpha) - \lambda_A \\ 0 &= -\lambda_\Lambda \gamma \Lambda \end{aligned}$$

and

$$\begin{aligned} d\lambda_A &= \left\{ \lambda_A \left\{ \left[i - \pi_H - \frac{1}{\omega} \left[\omega\eta + (1 - \omega\eta)(1 - \alpha)^2 \right] (i - \rho - \pi_H) \right] \right\} - \gamma\Lambda\lambda_A \right\} dt \\ d\lambda_\Lambda &= \left\{ \rho\lambda_\Lambda + \lambda_C \frac{1}{\omega} Y + \lambda_Y Y (1 - \alpha) \frac{1}{\omega} \right\} dt \end{aligned}$$

Hence we obtain

$$Y^{\omega+\varphi} = \frac{(2 - \alpha)\eta - 1}{(2 - \alpha)\eta - 1 + \alpha}$$

When prices are flexible

$$P_H = (1 - \tau) \frac{\epsilon}{\epsilon - 1} W = (1 - \tau) \frac{\epsilon}{\epsilon - 1} Y^{\omega+\varphi} P$$

Therefore the optimal labor subsidy is

$$\tau = 1 - \frac{\epsilon - 1}{\epsilon} \frac{(2 - \alpha)\eta - 1 + \alpha}{(2 - \alpha)\eta - 1}$$

□

Proof of Proposition (2)

Proof. When prices are flexible and the central bank does not intervene in the asset market, the allocation solves

$$\begin{aligned} d\lambda &= -\gamma (\hat{a} + \hat{f}_H^*) dt \\ d\hat{a} &= (\rho\hat{a} - \alpha\lambda) dt \\ d\hat{f}_H^* &= -\varrho\hat{f}_H^* \end{aligned}$$

given the initial conditions $\hat{f}_H^*(0) = \varepsilon$, $\hat{a}(0) = 0$ and the terminal condition $\lim_{t \rightarrow \infty} e^{-\rho t} \hat{a}(t) = 0$.

Let $\mathbf{z}^\top = \begin{bmatrix} \lambda & \hat{a} & \hat{f}_H^* \end{bmatrix}$. The natural allocation solves the system of differential equations $d\mathbf{z} = \mathbf{A}\mathbf{z}dt$, where

$$\mathbf{A} = \begin{bmatrix} 0 & -\gamma & -\gamma \\ -\alpha & \rho & 0 \\ 0 & 0 & -\varrho \end{bmatrix}$$

The eigenvalues of $\mathbf{A}$ are $\begin{bmatrix} -\varrho & -\nu & \rho + \nu \end{bmatrix}$, where $\nu = \frac{-\rho + \sqrt{\rho^2 + 4\alpha\gamma}}{2} > 0$. The first two eigenvectors of $\mathbf{A}$ are

$$\mathbf{v}_1 = \begin{bmatrix} -\frac{\gamma(\rho+\varrho)}{(\nu-\varrho)(\rho+\varrho+\nu)} \\ -\frac{\nu(\rho+\nu)}{(\nu-\varrho)(\rho+\varrho+\nu)} \\ 1 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} \frac{\rho+\nu}{\alpha} \\ 1 \\ 0 \end{bmatrix}$$

The solution is

$$\begin{bmatrix} \lambda \\ \hat{a} \\ \hat{f}_H^* \end{bmatrix} = \zeta_1 \begin{bmatrix} -\frac{\gamma(\rho+\varrho)}{(\nu-\varrho)(\rho+\varrho+\nu)} \\ -\frac{\nu(\rho+\nu)}{(\nu-\varrho)(\rho+\varrho+\nu)} \\ 1 \end{bmatrix} e^{-\varrho t} + \zeta_2 \begin{bmatrix} \frac{\rho+\nu}{\alpha} \\ 1 \\ 0 \end{bmatrix} e^{-\nu t}$$

where the parameters ζ_1 and ζ_2 are determined by the initial conditions $\hat{a}(0) = 0$ and $\hat{f}_H^*(0) = \varepsilon$. Thus we obtain

$$\lambda = \begin{bmatrix} \frac{\gamma}{\nu} \\ \frac{\gamma}{\rho+\varrho+\nu} \end{bmatrix}^\top \begin{bmatrix} \hat{a} \\ \hat{f}_H^* \end{bmatrix}$$

where the states evolve as follows

$$\begin{bmatrix} d\hat{a} \\ d\hat{f}_H^* \end{bmatrix} = \begin{bmatrix} -\nu & -\nu\frac{\rho+\nu}{\rho+\varrho+\nu} \\ 0 & -\varrho \end{bmatrix} \begin{bmatrix} \hat{a} \\ \hat{f}_H^* \end{bmatrix} dt$$

with $\begin{bmatrix} \hat{a}(0) & \hat{f}_H^*(0) \end{bmatrix} = \begin{bmatrix} 0 & \varepsilon \end{bmatrix}$. We can rewrite λ as

$$\lambda = \gamma\varepsilon \frac{(\rho+\nu)e^{-\nu t} - (\rho+\varrho)e^{-\varrho t}}{(\nu-\varrho)(\rho+\varrho+\nu)}$$

and thus obtain

$$\begin{aligned} \mathbb{L} &= \frac{\phi(\gamma\varepsilon)^2}{2\rho+2\varrho} \frac{\rho(\rho+\nu+\varrho)+2\varrho\nu}{(\rho+\varrho+\nu)^3(\rho+2\nu)} \\ &\quad \frac{\phi(\gamma\varepsilon\varrho)^2}{2\rho+2\varrho} \frac{\rho(\rho+\nu+\varrho)+2\varrho\nu}{(\rho+\varrho+\nu)^3(\rho+2\nu)} \end{aligned}$$

where $\phi = \phi_F = \alpha(1-\alpha)^{\frac{1+\varphi-\alpha\varphi}{1+\varphi}}$. Finally, we can take derivatives to obtain

$$\begin{aligned} \frac{\partial \mathbb{L}}{\partial \gamma} &= \frac{2\mathbb{L}}{\gamma} \frac{2\varrho(\nu^3+5\nu^2\rho+4\nu\rho^2+\rho^3)(\nu+\rho)+\varrho^2(2\nu+\rho)^3+\rho(\nu+\rho)^4}{(\rho+\varrho+\nu)(\rho+2\nu)^2(\rho(\rho+\nu+\varrho)+2\varrho\nu)} > 0 \\ \frac{\partial \mathbb{L}}{\partial \varrho} \frac{\varrho}{\mathbb{L}} &= 2 \frac{\rho^2(\nu+\rho)^2 - \varrho^3(2\nu+\rho) + 4\nu\varrho^2(\nu+\rho) + 2\rho\varrho(\nu+\rho)(2\nu+\rho)}{[\rho(\rho+\nu+\varrho)+2\varrho\nu](\rho+\varrho+\nu)(\rho+2\varrho)} \\ \frac{\partial \mathbb{L}}{\partial \alpha} \frac{\alpha}{\mathbb{L}} &= \frac{1-2\alpha+(1-3\alpha)(1-\alpha)\varphi}{(1-\alpha)(1+\varphi-\alpha\varphi)} - 2\nu(\rho+\nu) \frac{3\nu^2(\rho+2\varrho)+\nu\rho(5\rho+6\varrho)+2\rho^2(\rho+\varrho)}{[\rho(\rho+\nu+\varrho)+2\varrho\nu](\rho+\varrho+\nu)(\rho+2\nu)^2} \end{aligned}$$

It is easy to show that

$$\begin{aligned} \lim_{\varrho \downarrow 0} \mathbb{L} &= 0 & \lim_{\varrho \downarrow 0} \frac{\partial \mathbb{L}}{\partial \varrho} \frac{\varrho}{\mathbb{L}} &> 0 \\ \lim_{\varrho \uparrow \infty} \mathbb{L} &= 0 & \lim_{\varrho \uparrow \infty} \frac{\partial \mathbb{L}}{\partial \varrho} \frac{\varrho}{\mathbb{L}} &< 0 \end{aligned}$$

and the discriminant of the cubic polynomial at the numerator of $\frac{\partial \mathbb{L}}{\partial \varrho}$ is negative. Hence, $\exists! \tilde{\varrho} \in (0, \infty)$ such

that $\frac{\partial \mathbb{L}}{\partial \varrho} > 0$ for $\varrho < \tilde{\varrho}$, and $\frac{\partial \mathbb{L}}{\partial \varrho} < 0$ for $\varrho > \tilde{\varrho}$. Similarly

$$\begin{aligned} \lim_{\alpha \downarrow 0} \mathbb{L} &= 0 & \lim_{\alpha \downarrow 0} \frac{\partial \mathbb{L}}{\partial \alpha} \alpha &> 0 \\ \lim_{\alpha \uparrow 1} \mathbb{L} &= 0 & \lim_{\alpha \uparrow 1} \frac{\partial \mathbb{L}}{\partial \alpha} \alpha &< 0 \end{aligned}$$

The first term in the expression for $\frac{\partial \mathbb{L}}{\partial \alpha} \alpha$ is decreasing in α , going from 1 to $-\infty$, while the second is increasing, going from 0 to a positive number. Hence, $\exists! \tilde{\alpha} \in (0, 1)$ such that $\frac{\partial \mathbb{L}}{\partial \alpha} > 0$ for $\alpha < \tilde{\alpha}$, and $\frac{\partial \mathbb{L}}{\partial \alpha} < 0$ for $\alpha > \tilde{\alpha}$. $\square$

Proof of Propositions 3

Proof. The planner's problem is

$$\min_{\hat{x}} \frac{1}{2} \int_0^\infty e^{-\rho t} (\hat{x} + \phi \lambda^2) dt$$

subject to

$$\begin{aligned} d\lambda &= -\gamma (\hat{a} + \hat{f}_H^* - \hat{x}) dt \\ d\hat{a} &= (\rho \hat{a} - \alpha \lambda) dt \\ d\hat{f}_H^* &= -\varrho \hat{f}_H^* dt \end{aligned}$$

given the initial conditions $\hat{f}_H^*(0) = \varepsilon$, $\hat{a}(0) = 0$ and the terminal condition $\lim_{t \rightarrow \infty} e^{-\rho t} \hat{a}(t) = 0$. The Hamiltonian associated with this problem is

$$H = \frac{\phi}{2} \lambda^2 + \frac{1}{2} \hat{x}^2 + \mu^\lambda (-\gamma \hat{a} + \gamma \hat{x} - \gamma \hat{f}_H^*) + \mu^a (\rho \hat{a} - \alpha \lambda)$$

The FOC is $\hat{x} = -\gamma \mu^\lambda$, while the laws of motion of the costates are

$$\begin{aligned} d\mu^\lambda &= \rho \mu^\lambda - \phi \lambda + \alpha \mu^a \\ d\mu^a &= \gamma \mu^\lambda \end{aligned}$$

subject to the initial condition $\mu^\lambda(0) = 0$. We can use the first order conditions to replace μ^λ and $d\mu^a$ to obtain a second order differential equation that the optimal foreign exchange intervention must satisfy

$$dd\hat{x} = \rho d\hat{x} + (\alpha \gamma + \phi \gamma^2) \hat{x} - \phi \gamma^2 (\hat{a} + \hat{f}_H^*)$$

with $\hat{x}(0) = 0$. Let $\mathbf{z}^\top = \begin{bmatrix} d\hat{x} & \hat{x} & \lambda & \hat{a} & \hat{f}_H^* \end{bmatrix}$, then the system of differential equations that must be

solved is $dz = \mathbf{A}zdt$, where

$$\mathbf{A} = \begin{bmatrix} \rho & \alpha\gamma + \phi\gamma^2 & 0 & -\phi\gamma^2 & -\phi\gamma^2 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & \gamma & 0 & -\gamma & -\gamma \\ 0 & 0 & -\alpha & \rho & 0 \\ 0 & 0 & 0 & 0 & -\varrho \end{bmatrix}$$

The eigenvalues of $\mathbf{A}$ are $\begin{bmatrix} -\varrho & -\underline{\nu} & -\bar{\nu} & \rho + \underline{\nu} & \rho + \bar{\nu} \end{bmatrix}$, where

$$\underline{\nu} \equiv \frac{-\rho + \sqrt{\rho^2 + 2\gamma \left(2\alpha + \phi\gamma - \sqrt{\phi\gamma(4\alpha + \phi\gamma)} \right)}}{2}$$

$$\bar{\nu} \equiv \frac{-\rho + \sqrt{\rho^2 + 2\gamma \left(2\alpha + \phi\gamma + \sqrt{\phi\gamma(4\alpha + \phi\gamma)} \right)}}{2}$$

The solution can be written as

$$\begin{bmatrix} d\hat{x} \\ \hat{x} \\ \lambda \\ \hat{a} \\ \hat{f}_H^* \end{bmatrix} = \zeta_1 \begin{bmatrix} \frac{\varrho^2(\rho+\varrho)\phi\gamma^2}{(\bar{\nu}-\varrho)(\rho+\varrho+\bar{\nu})(\underline{\nu}-\varrho)(\rho+\varrho+\underline{\nu})} \\ -\frac{\varrho(\rho+\varrho)\phi\gamma^2}{(\bar{\nu}-\varrho)(\rho+\varrho+\bar{\nu})(\underline{\nu}-\varrho)(\rho+\varrho+\underline{\nu})} \\ \frac{\gamma(\rho+\varrho)(\varrho(\rho+\varrho)-\alpha\gamma)}{(\bar{\nu}-\varrho)(\rho+\varrho+\bar{\nu})(\underline{\nu}-\varrho)(\rho+\varrho+\underline{\nu})} \\ \frac{\alpha\gamma(\varrho(\rho+\varrho)-\alpha\gamma)}{(\bar{\nu}-\varrho)(\rho+\varrho+\bar{\nu})(\underline{\nu}-\varrho)(\rho+\varrho+\underline{\nu})} \\ 1 \end{bmatrix} e^{-\varrho t} + \zeta_2 \begin{bmatrix} \frac{\underline{\nu}(\rho+\underline{\nu})-\alpha\gamma}{\alpha\gamma} \underline{\nu} \\ -\frac{\underline{\nu}(\rho+\underline{\nu})-\alpha\gamma}{\alpha\gamma} \\ \frac{\rho+\underline{\nu}}{\alpha} \\ 1 \\ 0 \end{bmatrix} e^{-\underline{\nu}t} + \zeta_3 \begin{bmatrix} \frac{\bar{\nu}(\rho+\bar{\nu})-\alpha\gamma}{\alpha\gamma} \bar{\nu} \\ -\frac{\bar{\nu}(\rho+\bar{\nu})-\alpha\gamma}{\alpha\gamma} \\ \frac{\rho+\bar{\nu}}{\alpha} \\ 1 \\ 0 \end{bmatrix} e^{-\bar{\nu}t}$$

where the coefficients $\begin{bmatrix} \zeta_1 & \zeta_2 & \zeta_3 \end{bmatrix}$ are determined using the initial conditions $\hat{a}(0) = 0$, $\hat{x}(0) = 0$ and $\hat{f}_H^*(0) = \varepsilon$. Thus we obtain

$$\lambda = \begin{bmatrix} -\frac{\gamma}{\rho+\underline{\nu}+\bar{\nu}} \\ \frac{\xi}{\bar{\nu}}\gamma + \frac{1-\xi}{\underline{\nu}}\gamma \\ \frac{\xi\gamma}{\rho+\varrho+\bar{\nu}} + \frac{(1-\xi)\gamma}{\rho+\varrho+\underline{\nu}} \end{bmatrix}^\top \begin{bmatrix} \hat{x} \\ \hat{a} \\ \hat{f}_H^* \end{bmatrix}$$

with $\xi \equiv \frac{\alpha\gamma}{\alpha\gamma+\underline{\nu}(\rho+\underline{\nu})}$. The states $\begin{bmatrix} \hat{x} & \hat{a} & \hat{f}_H^* \end{bmatrix}$ evolve as

$$\begin{bmatrix} d\hat{x} \\ d\hat{a} \\ d\hat{f}_H^* \end{bmatrix} = \begin{bmatrix} -\xi\bar{\nu} - (1-\xi)\underline{\nu} & \frac{\alpha\gamma}{\rho+\underline{\nu}+\bar{\nu}} & 0 \\ \frac{\phi\gamma^2}{\rho+\underline{\nu}+\bar{\nu}} & -\xi\underline{\nu} - (1-\xi)\bar{\nu} & 0 \\ \phi\gamma^2 \frac{\rho+\varrho+\frac{\underline{\nu}\bar{\nu}}{(\rho+\varrho+\bar{\nu})(\rho+\varrho+\underline{\nu})}}{(\rho+\varrho+\bar{\nu})(\rho+\varrho+\underline{\nu})} & -\frac{\alpha\gamma\xi}{\rho+\varrho+\bar{\nu}} - \frac{\alpha\gamma(1-\xi)}{\rho+\varrho+\underline{\nu}} & -\varrho \end{bmatrix}^\top \begin{bmatrix} \hat{x} \\ \hat{a} \\ \hat{f}_H^* \end{bmatrix}$$

with $\begin{bmatrix} \hat{x}(0) & \hat{a}(0) & \hat{f}_H^*(0) \end{bmatrix} = \begin{bmatrix} 0 & 0 & \varepsilon \end{bmatrix}$. □

Proof of Lemma 1

Proof. The optimal intervention rule can be derived by manipulating the closed form solutions of Proposi-

tion (3). The explicit expressions of the derivatives with respect to ϱ are

$$\begin{aligned}\frac{\partial |\psi_\lambda|}{\partial \varrho} &= \frac{\phi\gamma}{\rho + \bar{\nu} + \underline{\nu}} \frac{\xi \underline{\nu} (\rho + \underline{\nu}) + (1 - \xi) \bar{\nu} (\rho + \bar{\nu})}{[\rho + \varrho + \xi \underline{\nu} + (1 - \xi) \bar{\nu}]^2} > 0 \\ \frac{\partial |\psi_a|}{\partial \varrho} &= \frac{\phi\gamma}{\alpha} \left(\rho + \frac{\bar{\nu} \underline{\nu}}{\rho + \bar{\nu} + \underline{\nu}} \right) \frac{\xi \underline{\nu} + (1 - \xi) \bar{\nu}}{[\rho + \varrho + \xi \underline{\nu} + (1 - \xi) \bar{\nu}]^2} > 0 \\ \frac{\partial |\psi_x|}{\partial \varrho} &= -\frac{(1 - \xi) \xi (\bar{\nu} - \underline{\nu})^2}{[\rho + \varrho + \xi \underline{\nu} + (1 - \xi) \bar{\nu}]^2} < 0\end{aligned}$$

The explicit expressions of the derivatives with respect to ϕ and γ have been derived using the symbolic toolkit available in Wolfram Mathematica. Their expressions are too big to be reported here and are available upon request. $\square$

Optimal Joint FX and Monetary Policies The planner's problem is

$$\min_{\hat{x}, i} \frac{1}{2} \int_0^\infty e^{-\rho t} [\hat{x} + \phi_\lambda \lambda^2 + \phi_\pi \pi_H^2 + \phi_y y^2] dt$$

subject to

$$\begin{aligned}d\lambda &= -\gamma (\hat{a} + \hat{f}_H^* - \hat{x}) dt \\ d\hat{a} &= (\rho \hat{a} - \alpha \lambda) dt \\ dy &= [i - \rho - \pi_H + \alpha \gamma (\hat{a} + \hat{f}_H^* - \hat{x})] dt \\ d\pi_H &= [\rho \pi_H - \kappa (1 + \varphi) y - \alpha \kappa \lambda] dt \\ d\hat{f}_H^* &= -\varrho \hat{f}_H^*\end{aligned}$$

given the initial conditions $\hat{f}_H^*(0) = \varepsilon$, $\hat{a}(0) = 0$ and the terminal condition $\lim_{t \rightarrow \infty} e^{-\rho t} \hat{a}(t) = 0$. The Hamiltonian associated with this problem is

$$\begin{aligned}H &= \frac{1}{2} \hat{x}^2 + \frac{\phi_\lambda}{2} \lambda^2 + \frac{\phi_\pi}{2} \pi_H^2 + \frac{\phi_y}{2} y^2 + \mu^\lambda (-\gamma \hat{a} - \gamma \hat{f}_H^* + \gamma \hat{x}) + \mu^a (\rho \hat{a} - \alpha \lambda) \\ &\quad + \mu^y [i - \rho - \pi_H + \alpha \gamma (\hat{a} + \hat{f}_H^* - \hat{x})] + \mu^{\pi_H} [\rho \pi_H - \kappa (1 + \varphi) y - \alpha \kappa \lambda]\end{aligned}$$

The FOCs are $\mu^y = 0$ and $\hat{x} = \alpha \gamma \mu^y - \gamma \mu^\lambda$, while the laws of motion of the costates are

$$\begin{aligned}d\mu^\lambda &= \rho \mu^\lambda - \phi_\lambda \lambda + \alpha \mu^a + \alpha \kappa \mu^{\pi_H} \\ d\mu^a &= \gamma \mu^\lambda - \alpha \gamma \mu^y \\ d\mu^y &= \rho \mu^y - \phi_y y + \kappa (1 + \varphi) \mu^{\pi_H} \\ d\mu^{\pi_H} &= -\phi_\pi \pi_H + \mu^y\end{aligned}$$

The optimal monetary policy is

$$i - \rho = \left[1 - \kappa (1 + \varphi) \frac{\phi_\pi}{\phi_y} \right] \pi_H dt - \alpha \gamma (\hat{a} + \hat{f}_H^* - \hat{x})$$

while the optimal foreign intervention satisfies

$$d\hat{x} = \rho d\hat{x} + (\alpha \gamma + \phi_\lambda \gamma^2) \hat{x} - \phi_\lambda \gamma^2 \hat{a} - \phi_\lambda \gamma^2 \hat{f}_H^* + \gamma \alpha \kappa \phi_\pi \pi_H$$

Let $\mathbf{z}^\top = \left[y \quad \pi_H \quad d\hat{x} \quad \hat{x} \quad \lambda \quad \hat{a} \quad \hat{f}_H^* \right]$, then the system of differential equations that must be solved is $d\mathbf{z} = \mathbf{A}\mathbf{z}dt$ where

$$\mathbf{A} = \begin{bmatrix} 0 & -\kappa(1+\varphi)\frac{\phi_\pi}{\phi_y} & 0 & 0 & 0 & 0 & 0 \\ -\kappa(1+\varphi) & \rho & 0 & 0 & -\alpha\kappa & 0 & 0 \\ 0 & \kappa\alpha\gamma\phi_\pi & \rho & \alpha\gamma + \gamma^2\phi_\lambda & 0 & -\phi_\lambda\gamma^2 & -\phi_\lambda\gamma^2 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \gamma & 0 & -\gamma & -\gamma \\ 0 & 0 & 0 & 0 & -\alpha & \rho & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\varrho \end{bmatrix}$$

The solution has the following form

$$\mathbf{z} = \zeta_1 \mathbf{v}_1 e^{\nu_1 t} + \zeta_2 \mathbf{v}_2 e^{\nu_2 t} + \zeta_3 \mathbf{v}_3 e^{\nu_3 t} + \zeta_4 \mathbf{v}_4 e^{\nu_4 t} + \zeta_5 \mathbf{v}_5 e^{\nu_5 t} + \zeta_6 \mathbf{v}_6 e^{\nu_6 t} + \zeta_7 \mathbf{v}_7 e^{\nu_7 t}$$

where ν_j and $\mathbf{v}_j$ are the eigenvalues and associated eigenvectors of $\mathbf{A}$. Unfortunately, the eigenvalues of $\mathbf{A}$ cannot be derived analytically, therefore the system can only be solved numerically.

Optimal Monetary Policy

Proof. The problem is similar to the one solved before, but this time $\hat{x} = 0$. Let $\mathbf{z}^\top = \left[y \quad \pi_H \quad \lambda \quad \hat{a} \quad \hat{f}_H^* \right]$, then the system of differential equations that must be solved is $d\mathbf{z} = \mathbf{A}\mathbf{z}dt$ where

$$\mathbf{A} = \begin{bmatrix} 0 & -\kappa(1+\varphi)\frac{\phi_\pi}{\phi_y} & 0 & 0 & 0 \\ -\kappa(1+\varphi) & \rho & -\alpha\kappa & 0 & 0 \\ 0 & 0 & 0 & -\gamma & -\gamma \\ 0 & 0 & -\alpha & \rho & 0 \\ 0 & 0 & 0 & 0 & -\varrho \end{bmatrix}$$

given the initial conditions $\hat{f}_H^*(0) = \varepsilon$, $\hat{a}(0) = y_H(0) = 0$ and the terminal condition $\lim_{t \rightarrow \infty} e^{-\rho t} \hat{a}(t) = 0$. The eigenvalues of $\mathbf{A}$ are $\left[-\varrho \quad -\nu \quad -\iota \quad \rho + \nu \quad \rho + \iota \right]$ where $\iota = \frac{-\rho + \sqrt{\rho^2 + 4\kappa^2(1+\varphi)^2 \frac{\phi_\pi}{\phi_y}}}{2}$. The solu-

tion is

$$\begin{bmatrix} y \\ \pi_H \\ \lambda \\ \hat{a} \\ \hat{f}_H^* \end{bmatrix} = \zeta_1 \begin{bmatrix} -\frac{\nu(\rho+\nu)(\rho+\varrho)\iota(\rho+\iota)}{(\varrho-\iota)(\rho+\varrho+\iota)} \\ -\kappa \frac{\nu(\rho+\nu)\varrho(\rho+\varrho)}{(\varrho-\iota)(\rho+\varrho+\iota)} \\ -\gamma(\rho+\varrho) \\ -\nu(\rho+\nu) \\ (\nu-\varrho)(\rho+\varrho+\nu) \end{bmatrix} e^{-\varrho t} + \zeta_2 \begin{bmatrix} \frac{(\rho+\nu)\iota(\rho+\iota)}{(\nu-\iota)(\rho+\nu+\iota)} \\ \kappa \frac{\nu(\rho+\nu)}{(\nu-\iota)(\rho+\nu+\iota)} \\ \frac{\gamma}{\nu} \\ 1 \\ 0 \end{bmatrix} e^{-\nu t} + \zeta_3 \begin{bmatrix} \frac{\rho+\iota}{\kappa} \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} e^{-\iota t}$$

Thus we obtain

$$\begin{bmatrix} \pi_H \\ \lambda \end{bmatrix} = \begin{bmatrix} \kappa \frac{1+\varphi}{\rho+\iota} & 0 \\ \kappa \frac{\rho+\nu}{\rho+\nu+\iota} & \frac{\gamma}{\nu} \\ \frac{\kappa\alpha\gamma(1+\frac{\rho+\nu}{\varrho-\iota})}{(\rho+\varrho+\iota)(\rho+\varrho+\nu)} & \frac{\gamma}{\rho+\varrho+\nu} \end{bmatrix}^\top \begin{bmatrix} y \\ \hat{a} \\ \hat{f}_H^* \end{bmatrix}$$

$$\pi_H = \frac{\kappa\alpha\gamma(1+\frac{\rho+\nu}{\varrho-\iota})}{(\rho+\varrho+\iota)(\rho+\varrho+\nu)} \hat{f}_H^* + \kappa \frac{\rho+\nu}{\rho+\nu+\iota} \hat{a} + \kappa \frac{(1+\varphi)}{\rho+\iota} y$$

with

$$\begin{bmatrix} dy \\ d\hat{a} \\ d\hat{f}_H^* \end{bmatrix} = \begin{bmatrix} -\iota & 0 & 0 \\ \frac{-\iota(\rho+\iota)(\rho+\nu)}{(1+\varphi)(\rho+\nu+\iota)} & -\nu & 0 \\ \frac{-\iota^2(\rho+\iota)\nu(\rho+\nu)}{(1+\varphi)(\rho+\varrho+\nu)(\rho+\varrho+\iota)(\rho+\nu+\iota)} & -\nu \frac{\rho+\nu}{\rho+\varrho+\nu} & -\varrho \end{bmatrix}^\top \begin{bmatrix} y \\ \hat{a} \\ \hat{f}_H^* \end{bmatrix}$$

and the initial conditions $\begin{bmatrix} y(0) & \hat{a}(0) & \hat{f}_H^*(0) \end{bmatrix} = \begin{bmatrix} 0 & 0 & \varepsilon \end{bmatrix}$. □

Proof of Proposition 4

Proof. The planner solves

$$\min_i \frac{1}{2} \int_0^\infty e^{-\rho t} (\phi_\pi \pi_H^2 + \phi_y y^2) dt$$

subject to

$$\begin{aligned} dy &= \left[i - \rho - \pi_H + \alpha\gamma (\hat{a} + \hat{f}_H^* - \hat{x}) \right] dt \\ d\pi_H &= (\rho\pi_H - \kappa y - \alpha\kappa\lambda) dt \\ d\hat{f}_H^* &= -\varrho \hat{f}_H^* \end{aligned}$$

The optimal monetary policy is still

$$i - \rho = \left(1 - \kappa \frac{\phi_\pi}{\phi_y} \right) \pi_H dt + \alpha d\lambda$$

while the foreign exchange intervention satisfies

$$d\hat{x} = \rho \hat{x} + (\alpha\gamma + \phi\gamma^2) \hat{x} - \phi\gamma^2 (\hat{a} + \hat{f}_H^*)$$

Let $\mathbf{z}^\top = \begin{bmatrix} y & \pi_H & d\hat{x} & \hat{x} & \lambda & \hat{a} & \hat{f}_H^* \end{bmatrix}$, then the system of differential equations that must be solved is $d\mathbf{z} = \mathbf{A}\mathbf{z}dt$ where

$$\mathbf{A} = \begin{bmatrix} 0 & -\kappa(1+\varphi)\frac{\phi\pi}{\phi_y} & 0 & 0 & 0 & 0 & 0 \\ -\kappa(1+\varphi) & \rho & 0 & 0 & -\alpha\kappa & 0 & 0 \\ 0 & 0 & \rho & \alpha\gamma + \phi\gamma^2 & 0 & -\phi\gamma^2 & -\phi\gamma^2 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \gamma & 0 & -\gamma & -\gamma \\ 0 & 0 & 0 & 0 & -\alpha & \rho & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\varrho \end{bmatrix}$$

given the initial conditions $\hat{f}_H^*(0) = \varepsilon$, $\hat{a}(0) = \hat{x}(0) = y(0) = 0$ and the terminal condition $\lim_{t \rightarrow \infty} e^{-\rho t} \hat{a}(t) = 0$. The eigenvalues of $\mathbf{A}$ are $\begin{bmatrix} -\varrho & -\underline{\nu} & -\bar{\nu} & -\iota & \rho + \underline{\nu} & \rho + \bar{\nu} & \rho + \iota \end{bmatrix}$. Using the same approach as before we obtain

$$\begin{bmatrix} \pi_H \\ \lambda \end{bmatrix} = \begin{bmatrix} \kappa \frac{1+\varphi}{\rho+\iota} & 0 \\ -\frac{\alpha\gamma\iota\kappa}{(\rho+\underline{\nu}+\bar{\nu})(\rho+\iota+\underline{\nu})(\rho+\iota+\bar{\nu})} & \frac{\gamma}{\rho+\underline{\nu}+\bar{\nu}} \\ \kappa\xi \frac{\rho+\underline{\nu}}{\rho+\iota+\underline{\nu}} + \kappa(1-\xi) \frac{\rho+\bar{\nu}}{\rho+\iota+\bar{\nu}} & \frac{\xi}{\bar{\nu}}\gamma + \frac{1-\xi}{\underline{\nu}}\gamma \\ \frac{\alpha\gamma}{\bar{\nu}-\varrho} \frac{\rho\varrho+\varrho^2-\alpha\gamma}{\rho+\varrho+\bar{\nu}} + \frac{\kappa}{\bar{\nu}-\varrho} \frac{\rho+\varrho}{\rho+\varrho+\underline{\nu}} + \frac{1-\xi}{\bar{\nu}-\varrho} \frac{\alpha\gamma\kappa}{\rho+\iota+\underline{\nu}} + \frac{\xi}{\bar{\nu}-\varrho} \frac{\alpha\gamma\kappa}{\rho+\iota+\bar{\nu}} & \frac{\xi\gamma}{\rho+\varrho+\bar{\nu}} + \frac{(1-\xi)\gamma}{\rho+\varrho+\underline{\nu}} \end{bmatrix}^\top \begin{bmatrix} y \\ \hat{x} \\ \hat{a} \\ \hat{f}_H^* \end{bmatrix}$$

while the states $\mathbf{s} = \begin{bmatrix} y & \hat{x} & \hat{a} & \hat{f}_H^* \end{bmatrix}^\top$ evolve as $d\mathbf{s} = -\mathbf{M}\mathbf{s}dt$ where

$$\mathbf{M} = \begin{bmatrix} \iota & 0 & 0 & 0 \\ -\frac{\alpha\gamma\iota^2(\rho+\iota)}{(1+\varphi)(\rho+\underline{\nu}+\bar{\nu})(\rho+\iota+\underline{\nu})(\rho+\iota+\bar{\nu})} & \xi\bar{\nu} + (1-\xi)\underline{\nu} & -\frac{\alpha\gamma}{\rho+\underline{\nu}+\bar{\nu}} & 0 \\ \iota \frac{\rho+\iota}{1+\varphi} \left[\xi \frac{\rho+\underline{\nu}}{\rho+\iota+\underline{\nu}} + (1-\xi) \frac{\rho+\bar{\nu}}{\rho+\iota+\bar{\nu}} \right] & -\frac{\phi\gamma^2}{\rho+\underline{\nu}+\bar{\nu}} & \xi\underline{\nu} + (1-\xi)\bar{\nu} & 0 \\ \frac{\iota\alpha\gamma}{\bar{\nu}-\varrho} \frac{\rho\varrho+\varrho^2-\alpha\gamma}{\rho+\varrho+\bar{\nu}} + \frac{\rho+\iota}{\bar{\nu}-\varrho} \frac{\rho+\varrho}{\rho+\varrho+\underline{\nu}} + \frac{\rho+\iota}{\bar{\nu}-\varrho} \frac{\iota\alpha\gamma(1-\xi)}{(1+\varphi)(\rho+\varrho+\underline{\nu})} + \frac{\rho+\iota}{\bar{\nu}-\varrho} \frac{\iota\alpha\gamma\xi}{(1+\varphi)(\rho+\varrho+\bar{\nu})} & \frac{-\phi\gamma^2}{\rho+\varrho+\bar{\nu}} \frac{\rho+\varrho+\frac{\underline{\nu}\bar{\nu}}{\rho+\varrho+\underline{\nu}}}{\rho+\varrho+\underline{\nu}} & \frac{\alpha\gamma\xi}{\rho+\varrho+\bar{\nu}} + \frac{\alpha\gamma(1-\xi)}{\rho+\varrho+\underline{\nu}} & \varrho \end{bmatrix}^\top$$

with $\begin{bmatrix} y(0) & \hat{x}(0) & \hat{a}(0) & \hat{f}_H^*(0) \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & \varepsilon \end{bmatrix}$. □

Figure (D.1) plots the allocations implemented by the solution of the sequential problem, blue line, and the solution of the joint problem, red line. The two allocations are almost indistinguishable from each other. In light of the discussion developed in the main body of the paper on the relationship between the two tools, this is not a surprising result. The planner wants to stabilize the path of λ for two reasons. First, because it enters directly into the loss function with weight ϕ_λ . Second, because it shifts the Phillips curve and improve the trade-off between the output gap and domestic inflation. The sequential problem does not ignore the second channel, but rather it collapses both channels into the choice of a single parameter, ϕ . In fact, the optimal ϕ that minimizes total welfare in Proposition (4) is strictly greater than ϕ_λ . The remaining

difference between the two solutions is only due to the relative dynamics of λ , y , and π_H , and is therefore negligible.

E Details of the Empirical Analysis and Robustness Checks

Discrete-Time Model

The equations of the model used to estimate the parameters γ and χ are the discrete-time versions of the log-linearized equations in Section C. The complete system is

$$\begin{aligned}\Delta q_{t+1} &= (1 - \alpha) [i_t - \pi_{H,t+1} - \gamma \Delta y_{t+1}^* + \gamma (\hat{a}_t + \chi vix_t^* - \hat{x}_t)] \\ \Delta a_t &= \rho a_{t-1} + \frac{\alpha}{1 - \alpha} \left[y_t^* - y_t + \left(\frac{2 - \alpha}{1 - \alpha} \eta - 1 \right) q_t \right] \\ \Delta y_{t+1} &= \alpha \Delta y_{t+1}^* + \frac{1 - \alpha}{\omega} (i_t - \pi_{H,t+1}) + \alpha \left(\frac{2 - \alpha}{1 - \alpha} \eta - \frac{1}{\omega} \right) \Delta q_{t+1} \\ \Delta \pi_{H,t+1} &= \rho \pi_{H,t} - \kappa \frac{\varphi(1 - \alpha) + \omega}{1 - \alpha} y_t + \kappa \frac{\alpha \omega}{1 - \alpha} y_t^* - \kappa \frac{\alpha}{1 - \alpha} \left(1 - \omega \eta \frac{2 - \alpha}{1 - \alpha} \right) q_t \\ i_t &= \psi_\pi \left(\pi_{H,t} + \frac{\alpha}{1 - \alpha} \Delta q_t \right) + \psi_y y_t\end{aligned}$$

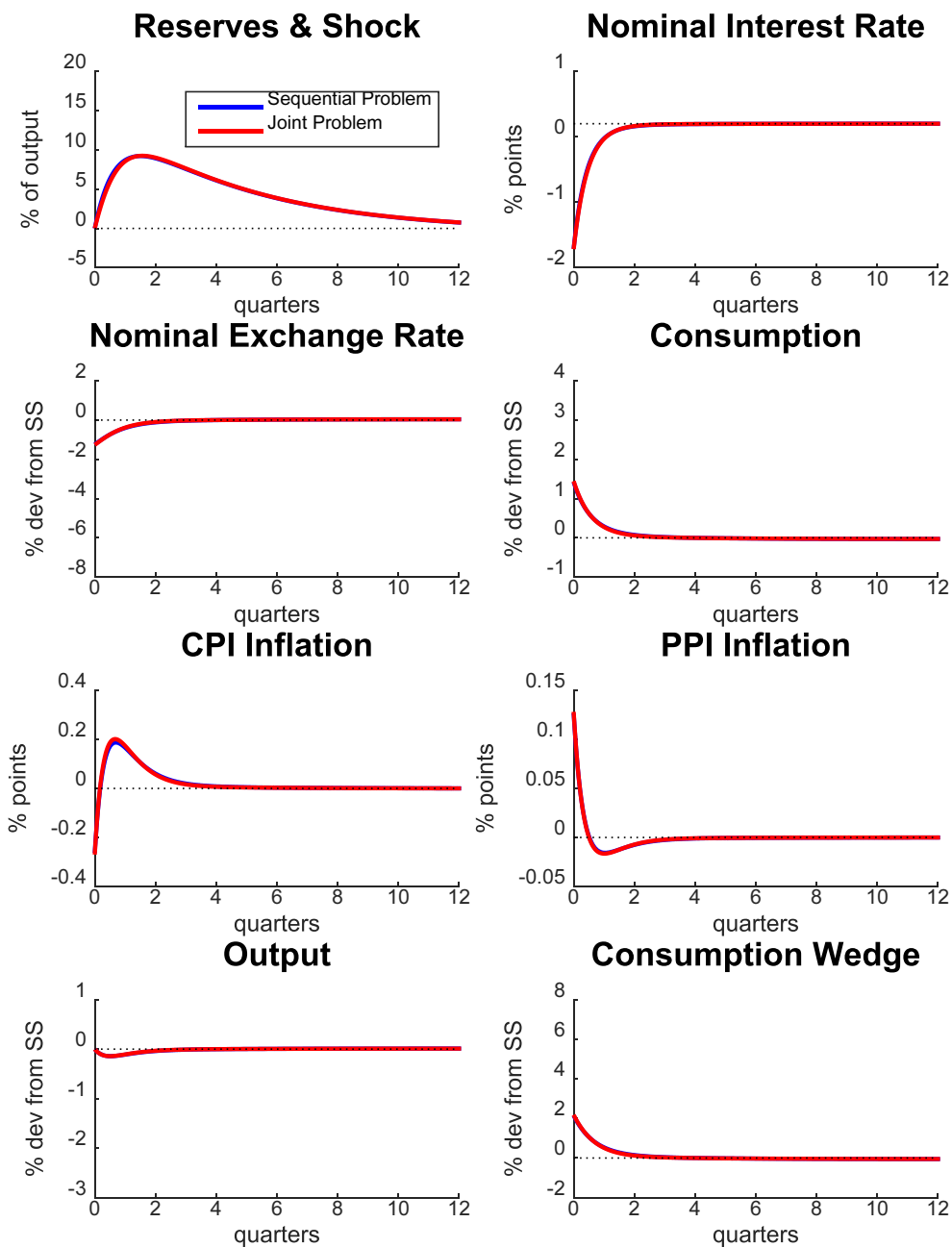
where $\kappa = \frac{1 - \theta}{\theta} (1 + \rho - \theta)$.

Robustness Checks

To gauge the robustness of the empirical analysis I perform three robustness exercises. First, I re-estimate the VAR and the model using both EU and US data. Foreign output and global risk aversion are obtained as a weighted average of EU and US data, with weights of 0.7 and 0.3 respectively. US output is measured by real GDP figures published by the U.S. Bureau of Economic Analysis, while US global risk aversion is proxied by the CBOE Volatility Index (VIX). Results of the estimation are reported in Table E.1 and Figure E.1. The empirical impulse responses are very similar to those obtained with the baseline specification. The main difference is the smaller appreciation of the Swiss franc real exchange rate, which is also reflected in the slightly lower estimate obtained for γ . Both, however, are still statistically significant. This is unsurprising and is due to the fact that the dollar is itself a reserve currency and tend to appreciate in times of global uncertainty.

The second robustness check involves the identification of the global risk-aversion shock. I re-estimate the VAR and the model by ordering vix^* first, to allow foreign output to respond contemporaneously to innovation in the VSTOXX. Results are reported in Table E.2 and Figure E.2. Again, the empirical impulse responses are quite similar to the baseline specification. Interestingly enough, however, the estimation of the structural parameters shifts some weight from χ to γ . The estimates for γ is 50 percent higher than in the baseline. By inspecting the empirical impulse responses it becomes apparent that this result is driven by the larger response of capital outflow.

Figure D.1: Sequential Problem vs Joint Problem



Impulse responses for the solution to the joint problem (blue line) and to the sequential problem (red line).

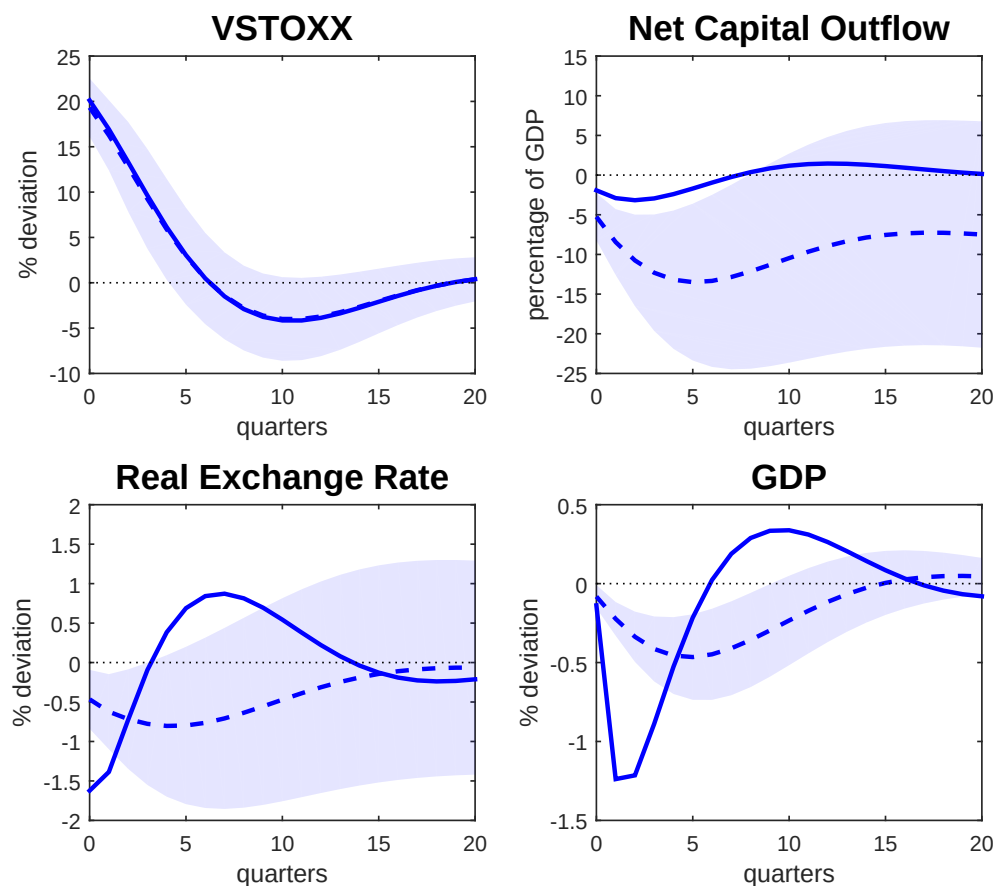
Finally, in the third robustness check I relax the assumption of unitary elasticities of intertemporal and intratemporal substitution. Following Baurle and Menz (2008) and Baurle et al. (2014) I set ω equal to 1.2 and η equal to 1.5. Results are reported in Table E.3 and Figure E.4. The estimate for γ is remarkably close to the baseline estimate, while the estimate for χ is higher. With this calibration, however, the model does a better job at matching the empirical impulse responses. While the theoretical impulse response for the real exchange rate is barely affected, the impulse response for net capital outflow is closer to its empirical counterpart and lies entirely within the 90 percent confidence interval.

Table E.1: Estimation Results (EU and US Data)

Parameter	Description	Value	S. E.
γ	Financial sector inverse aggregate risk-bearing capacity	0.171	0.062**
χ	Proportionality between vix and foreign demand for domestic assets	0.756	0.313**

Note: *** indicates significance at the 1 percent level, ** at the 5 percent level, and * at the 10 percent level.

Figure E.1: Model and VAR-based Impulse Responses (EU and US Data)



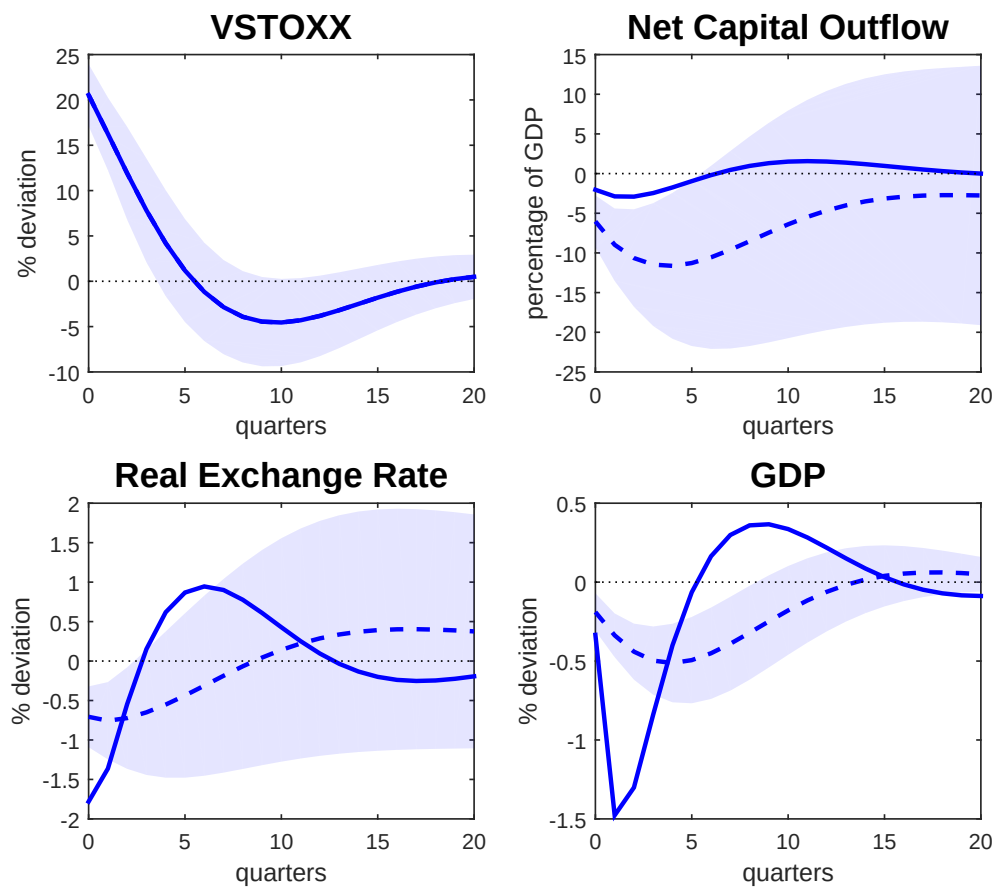
Model-based impulse responses (solid line), VAR-based impulse responses (dashed line) and 90 percent confidence intervals (shaded areas). Net capital outflow is expressed as cumulative deviation from the unshocked path in percentage of GDP. All other variables are expressed as percentage deviation from their unshocked path.

Table E.2: Estimation Results (Alternative Identification)

Parameter	Description	Value	S. E.
γ	Financial sector inverse aggregate risk-bearing capacity	0.296	0.125**
χ	Proportionality between vix and foreign demand for domestic assets	0.573	0.245**

Note: *** indicates significance at the 1 percent level, ** at the 5 percent level, and * at the 10 percent level.

Figure E.2: Model and VAR-based Impulse Responses (Alternative Identification)



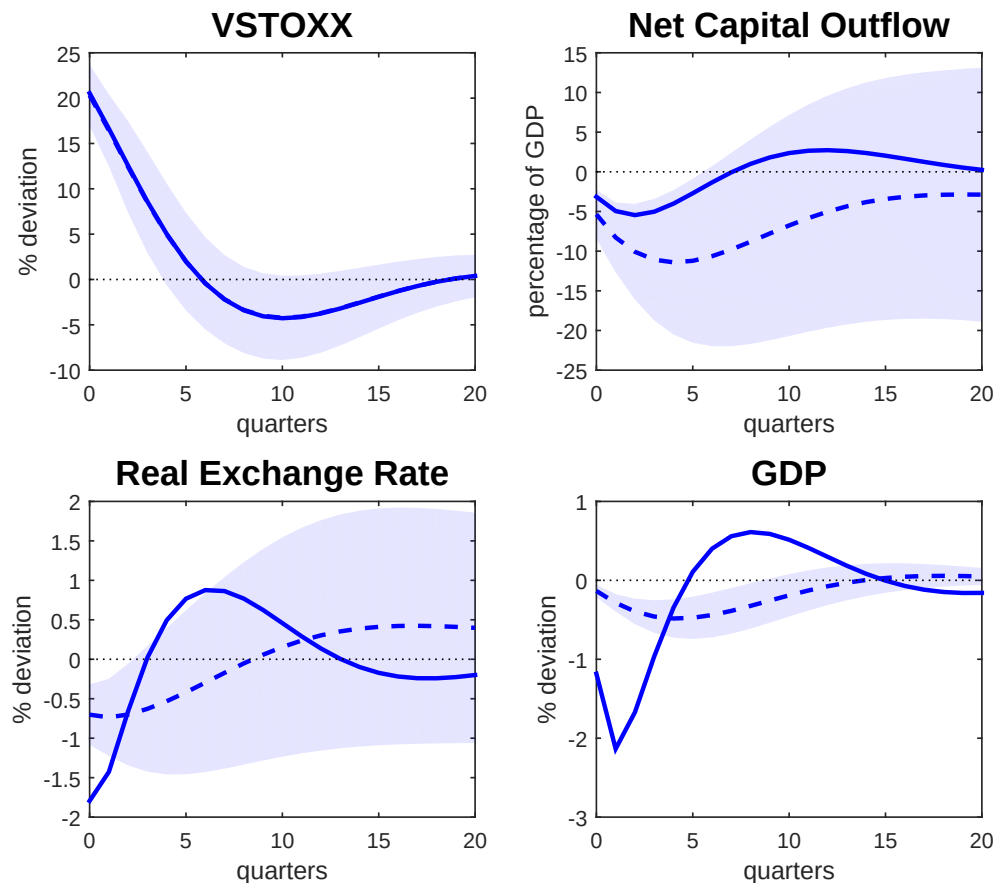
Model-based impulse responses (solid line), VAR-based impulse responses (dashed line) and 90 percent confidence intervals (shaded areas). Net capital outflow is expressed as cumulative deviation from the unshocked path in percentage of GDP. All other variables are expressed as percentage deviation from their unshocked path.

Figure E.3: Estimation Results (Alternative Calibration)

Parameter	Description	Value	S. E.
γ	Financial sector inverse aggregate risk-bearing capacity	0.145	0.072**
χ	Proportionality between vix and foreign demand for domestic assets	1.236	0.613**

Note: *** indicates significance at the 1 percent level, ** at the 5 percent level, and * at the 10 percent level.

Figure E.4: Model and VAR-based Impulse Responses (Alternative Calibration)



Model-based impulse responses (solid line), VAR-based impulse responses (dashed line) and 90 percent confidence intervals (shaded areas). Net capital outflow is expressed as cumulative deviation from the unshocked path in percentage of GDP. All other variables are expressed as percentage deviation from their unshocked path.

